

CONORMAL SOURCE MODULES, RECOVERY, AND OBSTRUCTIONS FOR FINITE-HORIZON DISPERSING BILLIARDS WITH MOVING SINGULARITIES

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ABSTRACT. We develop a collision-map conormal source framework for compact families of finite-horizon dispersing billiards with moving singularities. The conditional finite-order calculus uses a period-one response chart with a genuine no-boundary-pair-birth collar, finite physical Ward data, and a separate triangular strong-primitive certificate for assembled sources. One-face currents and clean two-face atoms are placed in a label-resolved quotient Banach module rather than the ordinary Demers–Zhang weak completion. Scalar susceptibilities are asserted only after complete source recovery; periodic point atoms, moving-face jumps, and a fourth-order grazing atom give explicit obstructions to an eventwise all-seed theory.

For a genuinely nonconjugate radial inflation of one circular scatterer, the complete first- and second-order invariant assemblies recover to centered strong densities, uniformly after summing all insertion positions at every fixed positive waiting length. Historical terminal labels need not close: any surviving time-labelled terminal germ accumulates linearly. An assembly-first reset nevertheless yields finite and countable scalar susceptibilities under summable primitive and reset bounds. Within each incident label, physical regrouping removes the artificial 2^b homogeneity-strip entropy. On the explicitly defined level-three uniform block-flux jet graph domain, the first radial letter satisfies the one-event estimate $2^{-(1-\epsilon_*)^b}$. This graph domain embeds in the formal $\mathfrak{D}_U^{(3)}$ level, but transfer invariance and next-event closure are not proved; the estimate is therefore not an instance of (S1). It does not extend to the isolated second letter: the circular impact angle has $\partial_s^2 \alpha \asymp u^{-6}$, leaving the nonintegrable flux coefficient u^{-3} . Higher radial response must therefore use specified complete assemblies. An explicit exact-current benchmark is supplied by every transported collision observable a_s : the exact current $(I - \mathcal{L}_s)[(a_s - \langle \rho_s, a_s \rangle) \rho_s]$ passes all represented Poisson recovery tests through order J and has a telescoping susceptibility. A nonzero differentiated-invariance radial benchmark satisfies the total-current q_1, q_2 identities and has a twice differentiable infinite susceptibility. These identities do not certify the label-resolved terminal reset and strong remainder in the general Poisson criterion. Generic local correlation sources remain subject to a closed proper moving-face gluing condition.

For continuous time we retain the corrected fixed-table BDL-v3 input and three obstructions only. Flow-box oscillations rule out transverse resolvent smoothing, and generic deformation letters have a necessary u^{-2} grazing multiplier. Although $|v \cdot n| = u^2$ is reflection-seam invariant, the ratio between consecutive collision weights is unbounded near tangency; a global theory therefore needs a dynamical branch/preimage weight, not scalar pointwise conjugation. No moving-family graph-domain or symbol theorem is claimed; those constructions are left as future work.

1. INTRODUCTION

This paper develops conormal source modules and a conditional finite-order response calculus for finite-horizon dispersing billiards whose singularity sets move with a compact mechanical parameter. The central problem is to combine the singular collision geometry with billiard-specific anisotropic transfer scales and a moving-boundary conormal calculus. The BDL suspension problem is treated separately because a generic table deformation produces generator derivatives transverse to the flow direction. The resulting package can later be used in deterministic homogenization, but no macroscopic equation or probabilistic representation is used as an input here.

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Scope and separation of hypotheses. The primitive inputs are the fixed-energy dispersing billiard cell, smooth scatterers with uniform curvature and separation bounds, uniform finite horizon, specular reflection, and compact finite-response parameter data. The period-one simple stratification, the nonconjugate operator Ward certificate, and the triangular strong-primitive certificate are additional analytic hypotheses, stated explicitly in Assumptions 2.6 and 4.8. They are not inferred from graph connectedness or compactness alone. At no point is a singular weak density declared strong merely because a later iterate is regular. Parameter-uniform corrected BDL-v3 graph and symbol estimates are not part of the collision-map theorem; the retained continuous-time content is the fixed-table input and obstruction in Propositions 2.14 and 2.15.

Main contribution and limitation. The central contribution is the collision-map moving-singularity source calculus: physical trace currents, decorated clean two-wall atoms, repeated-wall normal derivatives, and source-specific scalar insertions on top of uniform finite-horizon billiard transfer estimates. The abstract construction is finite order, as every response theorem with finite primitive smoothness must be, but its application beyond first order is source-specific. The dynamical target is the reduced collision-map susceptibility on a certified recovered response tree; static invariant averages are differentiated directly from the explicit flux measure. Construction of that certificate for a genuinely nonconjugate higher-order all-branch response tree remains open. Radial inflation supplies the source-specific invariant blocks and an exact observable-current benchmark, but not the generic S1/S3 tree. The continuous-time obstruction identifies, rather than hides, the extra graph-domain and symbol estimates still needed. The no-go mechanisms show that the missing theorem cannot be obtained by extending the present certificates to all ambient sources: it must exploit the special algebra of a complete physical response tree. At base response level we prove a useful converse: a centered ambient source has a strong primitive exactly when its represented complete assembly belongs to the centered strong density space. Recovery is therefore reduced to cancellation of every surviving face and point current, followed by the ordinary reduced density resolvent; separate terminal bookkeeping still needs either eventwise cancellation or assembly-first scalar reset.

Sharp boundary of the positive theorem. Flux conjugacy proves the finite operator Ward identities for a nonempty symmetry class, including common rigid table translations, but Proposition 2.8 shows that it is not a nonconjugate dynamical example. For a general independent scatterer deformation the Ward identities are a finite certificate, not a consequence of primitive geometry. Likewise, the ambient source scale propagates decorated point currents for finite time but supplies no decay: a regular periodic orbit gives a direct counterexample. Infinite dynamical words are therefore stated only for a certified recovered tree. The generic re-insertion theorem and the moving-flow symbol estimate remain separate new problems rather than hidden consequences of density mixing or compactness. In particular, the periodic-atom and fourth-order endpoint obstructions rule out a uniform certificate on all point sources or all localized smooth seeds. The recent small-hole response theorem [4] uses a different, genuinely positive coupling mechanism: its vertical-line source is pushed to a regular unstable standard family. A physical moving-scatterer grazing trace instead has a one-sided image on the grazing seam and is terminal in the present atlas. Thus the small-hole coupling theorem is an important model but does not verify (S1) or (S3) for the sources considered here. The radial-inflation example below is genuinely nonconjugate (a period-two multiplier changes) and verifies this singular-sector cancellation for the complete invariant source through order two. Its fixed-length block recovery allows arbitrary positive gaps after all insertion positions are summed; it does not verify eventwise re-insertion for a general dynamical susceptibility.

Informal result map. For every compact finite-response family of finite-horizon dispersing billiard tables satisfying the displayed geometric and analytic compatibility assumptions below, the paper constructs:

- (i) an a.e. singular hyperbolic collision map with uniform homogeneity-strip, cone, distortion, and growth estimates;
- (ii) strong/weak anisotropic transfer scales, a spectral gap, and an ambient decorated source scale for one-face and point-current outputs;

(iii) moving-singularity trace currents in transported Banach fibers, with finite-order observable-side response only along a certified recovered tree;

(iv) a scalar calculus for certified recovered dynamical susceptibilities, together with the exact geometric reduction of static invariant averages and an obstruction to generic point-source decay. The source-specific billiard statement is Theorem 5.1; the general certificate-based implication is Theorem 5.2; the continuous-time boundary is Proposition 2.15.

1.1. Five-result spine. The proof is organised around the following five load-bearing groups; all other named assertions are local lemmas or bookkeeping consequences.

1. Theorems 2.21 and 2.24 give the uniform collision geometry and the theorem-numbered Demers–Zhang spectral input.
2. Propositions 3.5 and 4.3 and Theorem 4.14 construct the label-resolved source module and identify recovery with centered strong assembled membership.
3. Theorems 3.6, 3.7 and 3.12 and Proposition 3.10 prove the nonconjugate radial invariant-source and assembly-first scalar results.
4. Propositions 3.16 and 3.17 and Theorems 3.15, 3.18 and 4.6 separate the general Poisson test, its obstruction, exact benchmarks, and the one-event radial graph bound from any unproved recursive certificate.
5. Theorem 5.2 and Proposition 2.15 state the conditional finite-order implication and the continuous-time boundary. Neither is used to enlarge the actual radial theorem.

The collision-map argument follows this order: geometry and fixed-table spectral input; the conormal source module; recovery and the radial model; the abstract certificate; and exported boundaries. Continuous time is confined to a separate fixed-table-background/obstruction subsection and is not used in that implication.

1.2. Known theory versus new contribution.

Component	Role in this paper	Submission treatment
Sinai billiard hyperbolicity	Foundational geometric mechanism	State as classical background and cite precisely.
Homogeneity strips and growth lemma	Singular-map regularization needed for uniform transfer estimates	Present the uniform finite-response version used by the response theorem.
Anisotropic transfer spaces	Functional setting for spectral gap and observable resolvent	Separate standard construction from any moving-family strengthening.
Dolgopyat suspension estimates	Fixed-table background and obstruction	Retain the corrected BDL input; do not state a moving-family criterion.
Moving-singularity response	Load-bearing finite-order package	Present the transport, resolved conormal currents, recovery tests, and certified scalar calculus.

1.3. Relation to known billiard response theory. The paper uses classical dispersing-billiard and anisotropic-transfer theory in a layered way. The hyperbolicity, homogeneity-strip, growth-lemma, and standard-family mechanisms come from singular billiards [16, 15, 6, 7]; the strong/weak transfer architecture for billiards follows [10, 11], with the general compactness mechanism from [14, 1]; and the suspension estimates use the Dolgopyat finite-horizon billiard-flow route [12, 2]. What is local to this paper is the finite-response stability, renormalized grazing-block scale, resolved moving-boundary currents, and their observable-side scale pairing. Detailed citation and dependency conventions are recorded in Appendix A.

1.4. Moving singularities and response obstructions. For systems with moving discontinuities, differentiability of statistical averages is not a formal consequence of smooth parameter dependence. In piecewise expanding unimodal maps, for example, linear response can fail unless the perturbation is compatible with the relevant topological class [3]. This paper avoids that obstruction by not differentiating a pointwise invariant density across a moving partition. Instead the first parameter

derivative is taken in transported anisotropic fibers, moving boundaries are represented by trace functionals, and a recovered source is paired only through a source-specific scalar functional. The theorem is scoped to the finite conormal and decorated ambient source scales and their renormalized physical grazing-block identity, not to arbitrary moving discontinuities.

Two nearby billiard perturbation settings should be distinguished. The moving-scatterer model of Stenlund–Young–Zhang [17] allows the table to change between collisions and proves uniform loss of memory for nonstationary compositions; here the table is stationary for each parameter and the problem is differentiability of invariant statistics inside one transported Banach bundle. The small-force works of Chernov and Chernov–Zhang–Zhang [5, 8] perturb the flight dynamics and study transport/current response. In contrast, the present perturbation moves the scatterer boundaries and therefore creates moving singular curves; the new load-bearing object here is the first physical trace pairing.

2. UNIFORM COLLISION AND SPECTRAL INPUT

Definition (Billiard cell, collision section, and flux). For each parameter $\theta \in \Theta$, the billiard cell is

$$(1) \quad Q_\theta = \mathbb{T}^2 \setminus \bigcup_{i=1}^N K_i(\theta), \quad \Gamma_\theta = \partial Q_\theta.$$

The velocity shell is the unit circle. The collision section is the outgoing boundary phase space

$$(2) \quad \Sigma_\theta = \{(q, v) : q \in \Gamma_\theta, |v| = 1, \langle v, n(q, \theta) \rangle \geq 0\},$$

with local coordinates $z = (i, s, \varphi)$, where s is boundary arclength and $\varphi \in (-\pi/2, \pi/2)$ is the angle with the inward normal. The reference flux is

$$(3) \quad d\nu_\theta = Z_\theta^{-1} \cos \varphi \, ds \, d\varphi,$$

where Z_θ is the normalizing flux constant. The symbol $\mathfrak{c}_\theta$ is reserved below for crossing speeds. Throughout, ν_θ denotes the collision-section flux reference measure, whereas μ_θ denotes the invariant probability measure or its suspension lift after the invariant density has been constructed. The billiard map $\mathcal{P}_\theta$ follows the free flight to the next regular impact and then applies specular reflection; the roof function is the flight time τ_θ . Grazing, multiple-impact, and branch-change states form the one-step singular set $\mathcal{S}_\theta$.

Remark (Invariant-density convention). In the moving collision fiber $(\Sigma_\theta, \nu_\theta)$, the billiard map preserves ν_θ , so the collision invariant density is $\rho_\theta \equiv 1$. When all objects are pulled back to a fixed reference fiber by $\mathfrak{T}_{\theta_0 \leftarrow \theta}^{\text{den}}$, this constant density becomes the transported density including the transport Jacobian. Thus symbols such as $\nabla_s \rho_\theta|_0$ and $\nabla_s \mathcal{L}_\theta|_0 \rho_{\theta_0}$ are always interpreted in the fixed transported fiber; the derivative of $Z_\theta^{-1} \cos \varphi$ and of the density-transport Jacobian is included in the transfer derivative letters.

Proposition 2.1 (Direct geometric derivative of static collision averages). *Let $\Phi_\theta : \Sigma_{\theta_0} \rightarrow \Sigma_\theta$ be the chosen collision-section transport and write*

$$J_\theta = \frac{d(\Phi_\theta^* \nu_\theta)}{d\nu_{\theta_0}}.$$

For a C^J transported observable f_θ , the static invariant collision average is

$$I(\theta) = \int_{\Sigma_{\theta_0}} (f_\theta \circ \Phi_\theta) J_\theta \, d\nu_{\theta_0},$$

and therefore

$$(4) \quad \partial_\theta^j I(\theta_0) = \sum_{\ell=0}^j \binom{j}{\ell} \int \partial_\theta^\ell (f_\theta \circ \Phi_\theta)|_{\theta_0} \partial_\theta^{j-\ell} J_\theta|_{\theta_0} \, d\nu_{\theta_0}.$$

In particular a static invariant average needs no dynamical resolvent. Any transfer/resolvent representation of its derivative must equal (4); this equality is used below as a consistency identity, not as the source of static differentiability.

Proof. The invariant collision density is 1 relative to ν_θ . Pull the integral back by Φ_θ and differentiate the product $(f_\theta \circ \Phi_\theta)J_\theta$ j times. $\square$

Definition 2.2 (Finite-response parameter data). The admissible parameter set Θ is a compact C^{r+3} manifold with finite atlas, or a compact subset of such a manifold after restricting to one atlas chart. Locally, every scatterer boundary is moved by a compact normal graph

$$(5) \quad q_i(s, \theta) = q_i(s, \theta_0) + a_i(s, \theta)n_i(s, \theta_0), \quad \|a_i\|_{C^{r+3}} \leq C_{\text{par}}.$$

The map $\theta \mapsto a_i(\cdot, \theta)$ is C^{r+3} into a bounded subset of $C^{r+3}(\partial K_i)$. Any microscopic drift, action density, or observable read-out used by a response functional is specified as a transported section of the branchwise observable space and is C^J in θ after the homogeneous-branch trivialization. Thus the admissible input data are:

- (P1) the compact parameter set and finite atlas;
- (P2) the normal graph fields a_i with uniform C^{r+3} bounds;
- (P3) the induced table transports, roof/read-out sections, and branchwise parameter derivatives through order J .

No effective coefficient, homogenized equation, Green–Kubo tensor, or value function is part of this finite-response datum.

Assumption 2.3 (Primitive finite-horizon dispersing billiard family). For every parameter $\theta \in \Theta$, the billiard cell is a fixed-energy two-dimensional periodic dispersing billiard whose scatterers satisfy:

- (G1) each scatterer boundary is a closed C^{r+3} curve depending C^{r+3} on θ in local normal coordinates;
- (G2) the dispersing curvature is uniformly bounded above and below by positive constants;
- (G3) physical scatterer boundaries are corner-free and remain uniformly separated;
- (G4) the family has uniform finite horizon: there are $0 < \tau_- < \tau_+ < \infty$ such that every regular free flight satisfies $\tau_- \leq \tau_\theta(z) \leq \tau_+$;

- (G5) the parameter family is a compact C^{r+3} normal-graph family generated by the finite-response parameter data of Definition 2.2, with uniform curvature, separation, and finite-horizon constants;
- (G6) the free region and the collision graph are connected.

No spectral gap, trace summability, response formula, effective coefficient, or homogenized equation is included in this primitive input layer.

Remark (Corners). The physical scatterers in Assumption 2.3 have no boundary corners. Later references to “partition corners” or “multiple-impact corners” refer only to artificial endpoints of refined homogeneous charts or to lower-dimensional multiple-impact strata inserted into the singular partition; they are not geometric corners of Γ_θ .

Assumption 2.4 (Finite response order and primitive reserve). Fix an integer response order $J \geq 1$ and take $r \geq J + 4$. Scatterer charts and normal-graph transports are C^{r+3} , and the finite-response observables are branchwise C^r . The proof constructs parameter derivatives through order J . The map response uses the four displayed reserve levels below. No moving-family continuous-time response order or derivative-loss budget is asserted in this paper.

Definition 2.5 (Checkable flux-conjugate deformation class). A C^J normal-graph family is *flux-conjugate through order J* if there is a C^J family of phase diffeomorphisms $\Psi_\theta : \Sigma_{\theta_0} \rightarrow \Sigma_\theta$ which preserves the grazing boundary, the simple face labels, and the normalized collision flux, and for which

$$\mathcal{P}_\theta \circ \Psi_\theta = \Psi_\theta \circ \mathcal{P}_{\theta_0}, \quad \mathcal{L}_\theta = \mathcal{C}_\theta \mathcal{L}_{\theta_0} \mathcal{C}_\theta^{-1}$$

through parameter order J , where $\mathcal{C}_\theta$ is the induced density transport. These identities are checked before differentiating branch indicators. Common rigid translations of all scatterers on the flat torus, with $\Psi_\theta(q, v) = (q + \theta w, v)$, and tangential boundary reparametrizations are examples. A generic independent normal deformation of one scatterer is not declared flux-conjugate.

Assumption 2.6 (Finite one-step transversality and physical Ward data). Fix a base parameter θ_0 . The following conditions are imposed only on the finitely many labelled one-step branch types and physical grazing germs, represented on the countable homogeneous-strip atlas. Long times will subsequently be handled by operator iterates; no all-iterate singularity isotopy is assumed.

(R1) The collision map is restricted to a fixed period-one mixing component. Period one is not inferred from connectedness of the undirected scatterer graph.

(R2) Each active one-step face has a C^J defining function $g_{a,\theta}$ on a compact rescaled chart. There are a parameter neighbourhood $U_0 \ni \theta_0$, a grazing collar $0 \leq u < u_J$, and constants $\delta_J, c_J > 0$ with the following properties for every $\theta \in U_0$. The labelled stratification is *simple*: every point belongs to at most two active physical faces and no triple-labelled vertex occurs. Distinct physical faces have no common zero in the grazing collar,

$$(6) \quad \{g_{a,\theta} = g_{b,\theta} = 0\} \cap \{0 \leq u < u_J\} = \emptyset \quad (a \neq b).$$

On the central region $u \geq u_J$, every admitted meeting pair is rank two with

$$(7) \quad |\det(Dg_{a,\theta}, Dg_{b,\theta})| \geq 2c_J,$$

and every nonmeeting labelled pair has distance at least $2\delta_J$. Repeated generators are recorded as normal derivatives of one face. A boundary pair birth/death, a distinct-face tangency, or a triple meeting requires a separate unfolding model and lies outside this response chart.

(R3) The operator Ward mechanism is supplied in one of two checkable ways. Either the family is flux-conjugate through order J in the sense of Definition 2.5, or, after differentiating the complete branch identity, cancelling every artificial face, and integrating all normal delta derivatives by parts, let Q_j be the finite set of nonnegative orders of residual endpoint currents which actually occur, let $W_{j,q}$ be the coefficient of $\delta^{(q)}(u)$, and let F_j^{reg} be the remaining regular coefficient. The operator-level physical Ward conditions are

$$(8) \quad W_{j,q} = 0 \quad (q \in Q_j), \quad F_j^{\text{reg}}(0, s; \rho, \psi) = 0, \quad 1 \leq j \leq J,$$

as identities on the smooth flux-seed/test core. In the flux-conjugate case these identities follow from Proposition 2.7; in the general case they are an additional finite Ward certificate. The first u -derivative of F_j^{reg} , with the required unstable-holonomy differences, is bounded by the order- $j+2$ flux-seed and order- $j+3$ trace-test norms.

Conditions (R2)–(R3) are finite, base-table conditions. They do not quantify over singularity preimages of arbitrary length. They are not consequences of the unweighted growth lemma or of the fixed cubic flux zero. The exact invariant source satisfies a global Ward identity proved below; (R3) is the strictly stronger operator identity needed when derivative letters act on an arbitrary response seed. When $J \geq 4$, the arbitrary-seed alternative is not generic: Proposition 2.10 shows that a physical grazing face with nonzero first parameter velocity violates it already at order four. Thus a nonconjugate theorem must use cancellation of a specified complete assembled source, rather than postulating (R3) on every seed.

Proposition 2.7 (Flux conjugacy implies the operator Ward identities). *Every family in Definition 2.5 satisfies (8) through order J for every smooth flux seed. Hence common rigid table translations and tangential reparametrizations give a nonempty normal-graph class for which (R3) is a theorem rather than an endpoint assumption. This proposition does not prove the separate all-branch response-scale bound (S1).*

Proof. Transport by Ψ_θ fixes the labelled branch partition, including the grazing face, and preserves the complete normalized branch flux. On the fixed fiber the transfer family is therefore the conjugate family $\mathcal{C}_\theta \mathcal{L}_0 \mathcal{C}_\theta^{-1}$, with no moving indicator. Differentiating this equality $j \leq J$ times gives a finite sum of commutators of the smooth transport generators with $\mathcal{L}_0$. Writing the same equality in the resolved branch atlas and applying Stokes to each branch, internal faces cancel in oppositely oriented pairs. Because Ψ_θ preserves $u = 0$ and its flux, the sum has no distribution supported only at the endpoint. Thus every $W_{j,q}$ and $F_j^{\text{reg}}(0)$ in (R3) is zero. The chain rule for the face-preserving C^J diffeomorphism Ψ_θ preserves the resulting finite distributional identities. Bounds on the completed countable branch scale are not inferred from this finite Stokes calculation; they remain the distinct analytic statement (S1).

A common torus translation is a phase conjugacy preserving $ds \cos \varphi d\varphi$; a tangential reparametrization changes only the boundary chart. Both examples satisfy the definition directly. $\square$

Proposition 2.8 (Symmetry limitation of the flux-conjugate class). *For a flux-conjugate family, every finite-time collision correlation is the corresponding reference correlation with transported observables. Consequently the class in Definition 2.5 verifies the Ward mechanism but does not constitute a nonconjugate deformation of the billiard dynamics. In particular it cannot, by itself, verify (S2)–(S3) for the conormal response tree of an independently moving scatterer.*

Proof. Insert $\mathcal{P}_\theta \Psi_\theta = \Psi_\theta \mathcal{P}_0$, transport the invariant flux by Ψ_θ , and change variables in each finite-time correlation. Only the transported observables remain parameter dependent. The last statement follows because an independent scatterer deformation is not covered by the conjugacy identity. $\square$

Theorem 2.9 (Local clean atlas and finite-order flux factorization). *Under Assumptions 2.3 and 2.6, after shrinking to a compact parameter neighbourhood $U \ni \theta_0$:*

- (i) *the labelled one-step branch domains form a parameter-uniform simple C^J stratified isotopy with constant face incidence and no triple vertices; distinct-face intersections occur only in the central region $u \geq u_J$, while the grazing collar contains only one-face strata;*
- (ii) *every admitted distinct-face intersection satisfies*

$$|\det(Dg_{a,\theta}, Dg_{b,\theta})| \geq c_J;$$

- (iii) *the complete order- j grazing coefficient has the exact factorization*

$$(9) \quad F_{j,m}(u, s; \rho, \psi) = u \tilde{F}_{j,m}(u, s; \rho, \psi), \quad 1 \leq j \leq J,$$

with the rescaled coefficient and holonomy differences bounded uniformly in the dyadic block. Consequently the geometry needed by a derivative letter is constructed once at one step. The arbitrary times in a resolvent word are powers of the undifferentiated transfer operator and do not require an isotopy of all iterated singularity preimages.

Proof. There are finitely many scatterer labels and, by finite horizon, finitely many one-step lift labels in a fundamental cell. In the compact blown-up atlas the functions $g_{a,\theta}$ are C^J . Condition (6) removes the boundary-fold mechanism before the implicit-function theorem is used: no distinct pair can enter from or leave through $u = 0$. On the compact central region, the uniform rank-two bound (7) gives C^J labelled pair-intersection charts, while the uniform distance bound preserves every nonincidence. Simplicity and exclusion of triple-labelled vertices then give constant incidence. The one-face charts in the grazing collar are transported separately and never promoted to two-face corners. This proves (i)–(ii) with one genuine determinant lower bound and without an infinite union of preimages or determinant sublevel shells.

Finally, the distribution identity $u^3 \delta^{(n)}(u) = -n(n-1)(n-2) \delta^{(n-3)}(u)$ shows that the complete list of residual endpoint currents through order J is the finite set Q_j in (8). Those coefficients vanish by (R3), so only F_j^{reg} remains. The fundamental theorem of calculus gives

$$F_j^{\text{reg}}(u, s) = u \int_0^1 \partial_u F_j^{\text{reg}}(tu, s) dt.$$

The derivative bound in (R3), applied also to matched holonomy differences, is exactly the asserted uniform bound for $\tilde{F}_{j,m}$. Artificial faces have already cancelled before this argument, so no boundary term is discarded. $\square$

Proposition 2.10 (Sharpness of the clean and Ward data). *The primitive assumptions alone do not imply the conclusions of Theorem 2.9. In local coordinates the boundary-fold family*

$$g_{1,s}(x, y) = y, \quad g_{2,s}(x, y) = y - x^2 + s$$

has no intersection for one sign of s , a tangency at $s = 0$, and two intersections for the other sign, with determinant $2|x| = 2\sqrt{|s|}$ at the intersections. On the half-chart $x = u \geq 0$ it is exactly a pair born from the grazing boundary and therefore violates (6). Likewise the local moving-cutoff model $u^3 H(u - s)$ satisfies

$$\partial_s^4 [u^3 H(u - s)]|_{s=0} = u^3 \delta^{(3)}(u) = -6\delta(u)$$

up to the fixed orientation sign. Thus neither stratified nonbifurcation nor finite-order endpoint cancellation follows from smooth parameter dependence, uniform curvature, separation, finite horizon, or the cubic flux zero. More precisely, for

$$c_s(u, y)u^3H(u - \eta_s(y)), \quad c_0(0, y_0) \neq 0, \quad \dot{\eta}_0(y_0) \neq 0,$$

the coefficient of $\delta(u)$ in the fourth derivative at y_0 is

$$(10) \quad -6\dot{\eta}_0(y_0)^4 c_0(0, y_0).$$

Consequently an operator Ward identity required for every localized smooth seed/test cannot hold through order four on such a moving germ. In particular, a first-order nonconjugate deformation which moves a physical grazing face cannot satisfy the arbitrary-seed version of (R3) for $J \geq 4$ unless an additional exact identity removes that germ before the localization is made. Moreover, pairwise transversality alone does not imply constant incidence:

$$g_{1,s}(x, y) = x, \quad g_{2,s}(x, y) = y, \quad g_{3,s}(x, y) = x + y - s$$

has three pairwise transverse faces with determinants ± 1 . They meet at one triple-labelled point when $s = 0$, but for $s \neq 0$ the three pairwise intersections are distinct. This is why simplicity/no-triple-unfolding is an actual hypothesis rather than the vacuous statement that three independent normals cannot occur in dimension two.

Proof. The fold and triple-incidence statements are the displayed direct calculations. For the Ward obstruction, differentiate $H(u - \eta_s)$ four times. The unique highest normal derivative is $\dot{\eta}_0^4 \delta^{(3)}(u)$; every term containing a parameter derivative of c_s or a higher parameter jet of η_s carries at most $\delta^{(2)}(u)$. Multiplication by u^3 kills the latter terms, while $u^3 \delta^{(3)} = -6\delta$, proving (10). Localizing the seed and test near one physical germ prevents cancellation by a disjoint branch. These calculations show that replacing (R2)–(R3) by compactness or by pairwise rank alone would make the main theorem false rather than stronger. $\square$

2.1. Fixed-table flow input and obstructions.

Assumption 2.11 (Fixed-table corrected BDL input). For each table, the suspension flow is placed on the corrected BDL-v3 strong/weak anisotropic spaces and corrected generator graph domain of [2]. The finite-horizon, contact, central-pair, roof, and fake-foliation constants are assumed uniform on the compact parameter set after a finite normal-graph cover. This supplies the *fixed-table* high-frequency generator-resolvent theorem with common constants.

It does not supply a common parameter graph domain. In particular, no parameter difference quotient, branch/preimage-weighted completion, weighted Lasota–Yorke inequality, or zero-waiting trace/lift/seam cancellation is included in this assumption.

Definition 2.12 (Collision/flow regularity separation). Collision-map response uses the finite source/test scale through order $J + 4$. A future moving-flow theorem would additionally require a specified common graph-domain scale, one transverse level for each parameter letter, and a negative-order high-frequency symbol estimate strong enough for the inverse-Laplace integral. No fixed numerical flow derivative loss is assigned before those two mapping properties have been constructed.

Proposition 2.13 (Primitive-to-transfer finite-time bounds). *The flight-time, singularity, cone, finite-time distortion, and growth constants used by the collision map are deterministic functions of the primitive record in Assumption 2.3. They are uniform on a compact normal-graph chart. This finite-time conclusion produces neither a moving-flow graph domain nor a weighted flow resolvent.*

Proof. Finite horizon and separation bound the roof; the impact equations and homogeneity subdivision give the singularity charts; the optical curvature recurrence gives the cones; differentiated branch Jacobians give distortion; and the standard descendant induction gives the growth constant. All these operations involve a fixed finite time and vary continuously in the normal-graph topology. A finite cover makes their constants uniform. $\square$

Proposition 2.14 (Fixed-table BDL correspondence and its uniform geometric data). *Under Assumption 2.11, the fixed-table BDL-v3 argument is invoked with the following exact division of labour:*

BDL item	input used here
Theorem 6.2	fixed-table fake foliations and the four-point temporal holonomy on central flow boxes;
Lemma 8.1	the contact/non-joint-integrability pair and the resulting local Dolgopyat amplitude loss, including the corrected v3 statement;
Proposition 9.1	iteration of that loss to the high-frequency generator-resolvent contraction;
Theorems 1.2 and 1.4	the fixed-table finite-horizon flow consequences on the corrected flow space.

The contact identity on a selected regular rectangle is

$$\Delta_{\mathcal{R}}(u, s) = d\alpha_x(u, s) + O(|u||s|(|u| + |s|)), \quad |\partial_u \partial_s \Delta_{\mathcal{R}}| \geq \Delta_c > 0.$$

The common finite-time margins follow from a finite cover of the regular core; terminal descendants are controlled by the ordinary growth tail. Uniformity of the completed BDL spaces and high-frequency operators is precisely the conditional input in Assumption 2.11, not a consequence of this compactness argument.

Proof. For one table, specular reflection preserves the contact form on the cross-section, and Stokes' formula gives the displayed four-point action identity. Stable and unstable lines are transverse in the contact plane, so a selected central pair has a nonzero $d\alpha$ margin. Delete a terminal strip tail, cover the remaining compact regular core by finitely many such rectangles, and then use normal-graph continuity and a finite parameter cover. This verifies only the finite-time geometric rows. The passage to BDL Lemma 8.1 and Proposition 9.1 uses the corrected fixed-table functional setting assumed above. $\square$

Proposition 2.15 (Three obstructions to a naive moving-flow extension). *A moving-family flow theorem cannot be obtained from the fixed-table resolvent by any of the following same-space arguments.*

(i) *In a regular flow box with coordinates (t, z_1, z_2) and $Y = \partial_t$, the resolvent*

$$(a - Y)^{-1} \{ \chi(t) e^{iNz_1} \} = e^{iNz_1} \int_0^\infty e^{-ar} \chi(t+r) dr$$

preserves the transverse frequency N . Hence it gives no transverse smoothing.

(ii) *For a generic moving reflection, the transported generator difference contains a nonzero multiplier of order $|v \cdot n|^{-1} = u^{-2}$. An unweighted same-level graph norm therefore cannot absorb one parameter letter.*

(iii) *The seam-compatible pointwise weight $u^2 = |v \cdot n|$ is preserved by one reflection, but*

$$\frac{u(Tx)^2}{u(x)^2}$$

is unbounded along regular orbits approaching a tangency at the next collision. Pointwise conjugation by u^2 is therefore not a globally invariant flow scale.

Moreover, a polynomial bound for the base resolvent at $a + ib$ does not by itself create the negative symbol order needed after parameter differentiation. Any positive construction must use a dynamical branch/preimage weight, prove its weighted Lasota–Yorke and common-resolvent invariance, and cancel every forbidden zero-waiting endpoint distribution.

Proof. The first assertion is the displayed explicit flow-box calculation. The second is the differentiated specular-reflection formula in the incoming/ outgoing frame. For the third, choose incoming collisions bounded away from grazing whose next images approach the grazing seam; the numerator tends to zero while the denominator stays bounded below, or reverse time for the reciprocal ratio. Finally, the differentiated resolvent identity only alternates base resolvents with parameter letters; without an independently proved symbol gain it multiplies, rather than improves, their high-frequency bounds. $\square$

Continuous-time boundary. The preceding proposition is the complete continuous-time content retained in this paper. A future moving-flow theorem would have to construct a common branch/preimage-weighted corrected BDL completion $\mathbb{E}_P$ and prove, for each parameter letter A_θ ,

$$A_\theta : \mathbb{E}_{P+1}^{[q]} \longrightarrow \mathbb{E}_P^{[q+1]}, \quad (a - Y_\theta)^{-1} : \mathbb{E}_P^{[q]} \longrightarrow \mathbb{E}_P^{[q]},$$

together with a genuinely negative-order dyadic scalar bound for every differentiated inverse-Laplace word. Positive-flight time kernels can guide that estimate, but zero-waiting trace/lift/roof/seam clusters require separate endpoint Ward cancellation. None of these mappings is claimed here; the detailed flow programme is removed from the theorem package.

2.2. Singular collision geometry.

Definition 2.16 (Homogeneity strips, cones, and standard curves). For $k \geq k_0$ define the grazing strips

$$(11) \quad \mathbb{H}_k^\pm = \left\{ (s, \varphi) : \frac{1}{(k+1)^2} < \frac{\pi}{2} \mp \varphi < \frac{1}{k^2} \right\},$$

and let $\mathbb{H}_0$ be the complement of the union of these boundary strips. A homogeneous component is a connected component of the complement of singularity curves and strip boundaries. For a nonvertical tangent vector $u = (ds, d\varphi)$ write

$$(12) \quad |u|_p = \cos \varphi |ds|, \quad \mathcal{B}(u) = \frac{d\varphi}{ds} + \kappa(s, \theta), \quad \mathcal{K}(u) = \frac{\mathcal{B}(u)}{\cos \varphi}.$$

Here $\mathcal{B}$ is the collision-coordinate numerator, while $\mathcal{K}$ is the physical wavefront curvature measured in transverse Jacobi arclength. Only $\mathcal{K}$ obeys the free-flight Riccati recurrence used in the appendix. The unstable cone is $\mathcal{C}_\theta^u(z) = \{u : ds \neq 0, \mathcal{B}(u) \geq a_-\}$ with strip truncation $\mathcal{B}(u) \leq a_+ k(z)^2$; stable cones are defined by time reversal. A stable or unstable admissible curve is a connected C^2 curve inside one homogeneous component, tangent to the corresponding cone, with uniform length and curvature bounds in the rescaled strip coordinates. These are the standard quadratic billiard strips used in the growth lemma and the Demers–Zhang anisotropic norms. No later proof is allowed to replace their $O(k^{-3})$ width by a freely chosen exponent.

Definition 2.17 (Renormalized grazing blocks). Let

$$\mathcal{G}_m^\pm = \bigcup_{2^m \leq k < 2^{m+1}} \mathbb{H}_k^\pm.$$

For an order- j physical response current, first sum all homogeneous pieces inside $\mathcal{G}_m^\pm$, cancel the two orientations of every artificial strip face, and combine the interior, Jacobian, flux, and physical-face terms. The resulting distribution is denoted $\mathcal{T}_{j,m}^{\text{phys}}$. It is defined from the resolved physical branch current before any absolute stripwise estimate.

Definition 2.18 (Flux-seed norms). Fix $0 < \epsilon_* < 1/4$. In a resolved grazing chart write $u = (\cos \varphi)^{1/2}$, and write the transported branch flux as $u^3 a_{h,\theta}(u, s) \rho_{h,\theta}(u, s) du ds$, where $a_{h,\theta}$ is the smooth chart and branch-Jacobian factor. For a smooth finite-strip core density set

$$(13) \quad \begin{aligned} \mathcal{N}_q^{\text{flux}}(\rho) := & \sup_{\theta \in U} \|\rho_\theta\|_{\mathcal{B}_\theta} \\ & + \sum_{\ell+|\beta| \leq q} \sup_{\theta, m} 2^{\epsilon_* m} \sum_{h: D_h \cap \mathcal{G}_m \neq \emptyset} \int_{D_h \cap \mathcal{G}_m} \left| u^\ell \partial_u^\ell \partial_s^\beta (u^3 a_{h,\theta} \rho_{h,\theta}) \right| du ds. \end{aligned}$$

Add the analogous total-variation terms on the labelled one-face and clean two-face restrictions of the resolved branch charts, with the physical coarea Jacobian included. These are coefficient norms for the undifferentiated physical flux; they do not contain $\mathcal{T}_{j,m}^{\text{phys}}$. For a resolved trace test define

$$\|\psi\|_{\mathcal{C}_{\text{tr}}^q} = \max_{\ell+|\beta| \leq q} \sup_h \|u^\ell \partial_u^\ell \partial_s^\beta \psi_h\|_\infty,$$

including the corresponding one-sided face and corner values. Common refinement merely subdivides the integrals and total variations, so these norms are refinement invariant on the core.

Lemma 2.19 (Renormalized grazing-block summation). *For each $0 \leq j \leq J$, there is $C_j < \infty$ such that*

$$(14) \quad |\langle \mathcal{T}_{j,m}^{\text{phys}} \rho, \psi \rangle| \leq C_j 2^{-m} \mathcal{N}_{j+2}^{\text{flux}}(\rho) \|\psi\|_{C_{\text{tr}}^{j+3}}.$$

Consequently $\sum_m \mathcal{T}_{j,m}^{\text{phys}}$ converges in the adjoint trace topology and extends to the corresponding flux-seed completions. The lemma does not assert the generally false absolute moment $\sum_k k^{M_j} m_W(E_{j,k}) < \infty$.

Proof. Use the physical grazing coordinate $u = (\cos \varphi)^{1/2}$. The flux is $d\nu = 2u^3 a(u, s) du ds$, with a smooth and bounded above and below. Differentiate the complete flux-preserving branch identity before separating interior and face terms. Artificial faces cancel between adjacent homogeneity cells. The cubic flux zero alone is insufficient at high order: for $n \geq 3$,

$$u^3 \delta^{(n)}(u) = -n(n-1)(n-2) \delta^{(n-3)}(u).$$

Thus no endpoint is discarded merely because a factor u^3 is present.

Apply Theorem 2.9 to the already assembled physical branch identity. It gives the proved exact factorization $F_{j,m} = u \tilde{F}_{j,m}$, after all residual boundary jets in the preceding distribution identity have cancelled. On the dyadic block $u \asymp 2^{-m}$, the rescaled coefficient bound in that theorem, the flux-seed norm, and the trace-test norm therefore give (14). Matched unstable-holonomy transport changes the bound by a fixed Jacobian; unmatched descendants remain in the same dyadic block and are controlled by the weak terminal component. The geometric series in 2^{-m} proves convergence. $\square$

Lemma (Impact-function trace differentiation). *Fix a finite collision word and a homogeneous strip itinerary. The system of impact equations defining the successive flight times has, on each regular homogeneous component, flight-time Jacobian*

$$(15) \quad \prod_{m=1}^j \langle v_m^-, n(q_m, \theta) \rangle.$$

After the grazing change of variable $u = (\cos \varphi)^{1/2}$, all branch maps, finite-word singular curves, and their parameter velocities satisfy polynomial strip bounds. Consequently every moving singular curve Γ generated by a finite word defines a trace functional on the smooth branchwise core, in the stable-leaf pairing sense formalized in Equation (33),

$$(16) \quad \rho \mapsto (V_\Gamma \cdot n_\Gamma) \rho \delta_\Gamma$$

and, for each finite word length n and finite strip cutoff K ,

$$(17) \quad \sum_{\substack{\Gamma \in \mathcal{S}_n \\ k(\Gamma) \leq K}} \|(V_\Gamma \cdot n_\Gamma) \rho \delta_\Gamma\|_{\mathcal{B}_\theta^w} \leq C_{n,K} \|\rho\|_{C_{\text{br}}^1}.$$

This finite-cutoff statement does not take point values of a general element of $\mathcal{B}_\theta$. The first-order completed trace is defined only on the graph domain below, and its grazing contribution is completed by Lemma 2.19.

Proof. On a regular branch, the next impact with a scatterer is determined by an equation $G(q + tv, \theta) = 0$. Differentiating in t gives the normal velocity factor $\langle v, n \rangle$. Near grazing this factor is singular in φ , but in the blown-up coordinate $u = (\cos \varphi)^{1/2}$ every derivative loses only a polynomial power of the strip index. Branch-change curves are handled by the same computation applied to differences of impact functions. Transversality to admissible leaves is a consequence of the dispersing curvature recurrence: failure of transversality is assigned to a refined multiple-impact, branch-change, separation, infinite-flight, or unblown grazing stratum. Intersections with stable leaves are therefore polynomially bounded in high strips. The unweighted growth lemma does not sum the resulting polynomial trace loss absolutely; the physical terms are instead assembled by Lemma 2.19. $\square$

Theorem 2.20 (Growth lemma). *Let $r_n(x)$ be the distance, measured along $\mathcal{P}_\theta^n W$, from $\mathcal{P}_\theta^n x$ to the nearest endpoint of its homogeneous component. There exist $C > 0$ and $\rho_g \in (0, 1)$, uniform in*

the finite-response parameter, such that for every admissible unstable curve W and every $\varepsilon > 0$,

$$(18) \quad m_W\{x \in W : r_n(x) < \varepsilon\} \leq C\varepsilon + C\rho_g^n m_W(W).$$

For response currents, the terminal children are first grouped into the dyadic physical blocks of Definition 2.17; their combined pairings obey

$$\sum_{m \geq M} |\langle \mathcal{T}_{j,m}^{\text{phys}} \rho, \psi \rangle| \leq C_j 2^{-M} \|\rho\|_{\mathfrak{D}_U^{(j+2)}} \|\psi\|_{\mathfrak{A}_U^{(j+3)}}.$$

Proof. Construct the standard family tree for images of W . At each adapted return time, descendants above the reference length continue as long children, while short descendants cut by singularities or high strips are recorded in the boundary layer. Cone expansion and the one-step expansion estimate make the total inverse expansion of long children strictly smaller than one. Bounded distortion converts pointwise inverse expansion into componentwise length control. At the last generation, preimages of points within ε of a fresh endpoint have total length $O(\varepsilon)$. For response currents, sum all children in one dyadic grazing block before estimating them. Lemma 2.19 gives the factor 2^{-m} ; the adapted contraction propagates this terminal block charge down the tree. Iterating and absorbing the finitely many remainder iterates gives both estimates. This is the classical billiard growth-lemma mechanism of [6, 7]; the present proof keeps all constants uniform over the compact finite-response chart. $\square$

Theorem 2.21 (Hyperbolic collision map with singularities). *The collision map is defined globally modulo a zero-measure singular set, is piecewise smooth on homogeneous components, preserves the billiard flux measure, and satisfies uniform cone invariance, distortion, one-step expansion, and growth-lemma estimates. In homogeneous coordinates the map is C^r on every branch, singularities are countable unions of C^r curves after the square-root grazing blow-up, and stable/unstable cones are invariant with uniform expansion. The constants are deterministic functions of Assumption 2.3.*

Proof. Piecewise smoothness is obtained from the implicit impact equation on each homogeneous component. Cone invariance and expansion follow from the optical curvature recurrence, distortion from the branch Jacobian formula, and complexity from the one-step expansion plus Theorem 2.20. The zero-measure exceptional set is the countable union of preimages of grazing, multiple-collision, and branch-change singularities. Uniformity in θ is inherited from the compact finite-response parameter chart. The local singular-hyperbolic mechanism follows the Sinai billiard framework [16, 15, 6]; the uniform finite-response dependence is the specialization needed here. $\square$

2.3. Anisotropic transfer and fixed-table flow background.

Definition 2.22 (Notation for the billiard anisotropic spaces). For the collision map we fix a billiard-specific strong/weak pair from the Demers–Zhang Lorentz-gas construction [10, 11], transported between nearby tables in normal-graph coordinates. The formulas below record only notation and the types of seminorms; they are not a replacement definition of those spaces. Let $\mathcal{W}_\theta^s$ and $\mathcal{W}_\theta^u$ be the stable and unstable admissible curves from Definition 2.16, cut so that each curve lies in one homogeneous component, has length in a fixed interval $[\ell_-/2, \ell_+]$ after standard subdivision, and has the curvature bounds specified there. Two stable curves $W_1, W_2 \in \mathcal{W}_\theta^s$ are *matched* if they lie in the same homogeneous chart and are connected by the local unstable holonomy $\mathfrak{h}^u : W_1 \rightarrow W_2$ on all but terminal children whose total stable mass is charged to the weak tail. The distance $d(W_1, W_2)$ is the C^1 graph distance between the matched stable curves in that chart plus the stable mass of unmatched terminal pieces. For a branchwise smooth density h , set

$$(19) \quad |h|_{w,\theta} = \sup_{W \in \mathcal{W}_\theta^s} \sup_{\|\psi\|_{C^0(W)} \leq 1} \left| \int_W h \psi \, dm_W \right|,$$

$$(20) \quad \|h\|_{s,\theta} = \sup_{W \in \mathcal{W}_\theta^s} \sup_{\|\psi\|_{C^\alpha(W)} \leq 1} \left| \int_W h \psi \, dm_W \right|.$$

The unstable seminorm is

$$(21) \quad \|h\|_{u,\theta} = \sup_{(W_1, W_2, \mathfrak{h}^u)} \sup_{\|\psi_i\|_{C^\alpha(W_i)} \leq 1} \frac{\left| \int_{W_1} h \psi_1 dm_{W_1} - \int_{W_2} h(\psi_2 \circ \mathfrak{h}^u) J_{\mathfrak{h}^u} dm_{W_2} \right|}{d(W_1, W_2)^\beta},$$

where the first supremum ranges over matched pairs, $J_{\mathfrak{h}^u}$ is the unstable-holonomy Jacobian between nearby stable curves, and $0 < \beta < \alpha < 1$. Unmatched terminal pieces are first cut and estimated by $|\cdot|_{w,\theta}$. With fixed $b_0, c_0 > 0$,

$$(22) \quad \|h\|_{\mathcal{B}_\theta} = \|h\|_{s,\theta} + b_0 \|h\|_{u,\theta} + c_0 |h|_{w,\theta},$$

The short-leaf weights, length scaling, admissible test exponents, and unmatched-piece terms are those of the cited billiard space. We denote its completion by $\mathcal{B}_\theta$. The weak space $\mathcal{B}_\theta^w$ is the cited weak completion, schematically represented by $|\cdot|_{w,\theta} + \|\cdot\|_{s,\theta}^{(\alpha_w)}$ with $\alpha_w < \alpha$ and without the unstable comparison seminorm. The completion elements are distributions defined by their stable-leaf pairings. The observable space $\mathcal{B}_\theta^\#$ is the restricted dual generated by physical branchwise C^r observables with the dual norm induced by stable-leaf pairing; $\mathcal{B}_\theta^{\#,0}$ denotes the quotient by constants. The suspension spaces $\widehat{\mathcal{B}}_\theta$, their weak companions, and their restricted observable duals are the actual corrected BDL-v3 anisotropic flow spaces and generator graph-domain levels transported in Assumption 2.11. They are not a suspension completion of the Demers–Zhang map space. The retained fixed-table cross-section trace and lift maps are the interface between these two functional settings. The map transfer and observable operators are defined by

$$(23) \quad \int f \mathcal{L}_\theta h d\nu_\theta = \int f \circ \mathcal{P}_\theta h d\nu_\theta, \quad \mathcal{U}_\theta f = f \circ \mathcal{P}_\theta.$$

The density projector is $\Pi_\theta^{\text{den}} h = \langle h, 1 \rangle \rho_\theta$, and the dual observable projector is $\Pi_\theta^{\text{obs}} f = \langle \rho_\theta, f \rangle 1$. Write $\Pi_\theta^{\text{den},\perp} = I - \Pi_\theta^{\text{den}}$ and $\Pi_\theta^{\text{obs},\perp} = I - \Pi_\theta^{\text{obs}}$. The reduced observable resolvent is

$$(24) \quad (-\mathcal{A}_\theta|_{\ker \Pi_\theta^{\text{obs}}})^{-1} : \widehat{\mathcal{B}}_\theta^{\#,0} \longrightarrow \widehat{\mathcal{B}}_\theta^{\#,0},$$

constructed as the restricted dual of the zero-mass suspension transfer resolvent after the spectral gap is proved. No claim is made that the raw conservative composition semigroup is exponentially decaying in a natural C^0 or L^p observable norm; the decaying object is the transfer semigroup on the zero-mass anisotropic density space, and the observable resolvent is its quotient dual on $\widehat{\mathcal{B}}_\theta^{\#,0}$. Uniform boundedness of the transports $\mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{obs}}$ and $\mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{den}}$ is included in Proposition 4.12; it is not deduced merely from the schematic displays above.

Definition (Pairing and resolvent notation). Density-observable duality is written

$$\langle g, f \rangle_\theta = \int_{\Sigma_\theta} f g d\nu_\theta.$$

Thus the transfer operator acts on densities and the Koopman operator acts on observables. For a centered source g and a target test f , the notation

$$(25) \quad \langle g, R_\theta^{\text{obs}} f \rangle_\theta := \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \langle g, \mathcal{U}_\theta^n \Pi_\theta^{\text{obs},\perp} f \rangle_\theta$$

is used only after scalar Cauchy convergence has been proved for that source in Theorems 4.11 and 4.16. It does not define an element $R_\theta^{\text{obs}} f$ in the positive trace completion, nor a bounded Koopman resolvent on that completion. The Laplace-weighted collision resolvent on the density space is the renewal factor

$$R_{\theta,z}^{\text{Lap}} = (I - \mathcal{L}_{\theta,z})^{-1} \Pi_\theta^{\text{den},\perp}$$

. It enters the flow argument only between the BDL trace/lift maps in Lemma 2.25; it is not identified with the generator resolvent on the BDL flow space. The reduced suspension observable resolvent is

$$R_\theta^{\text{flow}} = (-\mathcal{A}_\theta|_{\ker \Pi_\theta^{\text{obs}}})^{-1}.$$

It is interpreted by density-observable duality:

$$\langle h, R_\theta^{\text{flow}} f \rangle_\theta = \int_0^\infty \langle \widehat{\mathcal{L}}_{\theta,t} \Pi_\theta^{\text{den},\perp} h, f \rangle_\theta dt$$

for zero-mass suspension densities h and zero-mean observable classes f . Thus R_θ^{flow} is not a statement of operator-norm decay for the raw Koopman composition semigroup on ordinary observables. The response formula uses the scalar functional $\langle g, R_\theta^{\text{obs}} f \rangle_\theta$ from (25). It never applies a density-side map resolvent to a weak trace source.

Definition (Reference convention for anisotropic norms). The symbols in this paper use the following convention. The pair $(\mathcal{B}_\theta, \mathcal{B}_\theta^w)$ is the billiard strong/weak anisotropic pair: the stable component is the stable-curve testing norm, the unstable component compares nearby stable curves by the local unstable holonomy $\mathfrak{h}^u$ inside homogeneous charts, and the weak component is the corresponding lower-regularity stable-curve test. Stable holonomies of the flow are used separately in the Dolgopyat/fake-foliation block; they are not the holonomy used to identify two stable test curves in the unstable seminorm. This is the billiard-specific strong/weak architecture of [10, 11], which modifies the bounded-second-derivative setting of [9] to handle grazing singularities. Compact embedding and quasi-compactness are separated through [14, 1]. The reference to [13] is only for the general anisotropic-Banach-space mechanism; all singular-cut, homogeneity-strip, and billiard trace estimates are the local estimates of this paper. The suspension spaces are instead the corrected BDL-v3 flow spaces and graph domains from Assumption 2.11; they are not identified with the Demers–Zhang map spaces. Their norms are compared only through the separately specified flow-domain transports.

Definition (Finite-horizon BDL route). For the suspension estimate we use the finite-horizon Sinai-flow route of [2, arXiv v3, including the correction note] as a generator-resolvent theorem, not as a one-step weighted collision-operator lemma and not as an operator on the Demers–Zhang map space. The approximate unstable foliation and four-point holonomy estimates are those of BDL Theorem 6.2; BDL Lemma 8.1 is the Dolgopyat high-frequency bound after the geometric and phase-separation constants have been fixed; BDL Proposition 9.1 turns this bound into the required resolvent contraction; and BDL Theorems 1.2 and 1.4 give the finite-horizon flow consequences. The uniform geometric inputs are verified in Proposition 2.14; the uniform flow-space and domain identifications are Assumption 2.11. The table records this separation:

BDL finite-horizon item	discharge in this paper
roof and flow boxes	finite horizon and the branch roof margin in Assumption 2.3 and Definition 2.12.
fake foliation/four-point holonomy	the BDL-v3 operator from Assumption 2.11, with the geometric contact margin from Proposition 2.14.
non-joint-integrability / pair selection	Proposition 2.14.
Dolgopyat bound and high-frequency contraction	Proposition 2.14, using BDL Lemma 8.1 and Proposition 9.1 after the hypotheses are matched.

Proposition 2.23 (BDL high-frequency consequence). *By Assumption 2.11 and Proposition 2.14, the generator resolvent of the finite-horizon billiard flow satisfies the high-frequency estimate and the resolvent contraction supplied by BDL Lemma 8.1 and Proposition 9.1, uniformly over the compact family. It makes no claim that the union of all near-resonant phase windows has measure $O(|b|^{-1})$. In the BDL argument the gain comes from local branch pairing, alternating cutoffs, amplitude loss, and iteration on the flow anisotropic space.*

Theorem 2.24 (Billiard collision-map spectral gap). *Under Assumptions 2.3, 2.4 and 2.6, the singular collision maps admit the billiard-specific Demers–Zhang anisotropic transfer spaces with uniform Lasota–Yorke estimates, compact embedding, and a spectral gap on the zero-mean complement. After quotienting constants, there exist $C > 0$ and $\vartheta \in (0, 1)$, depending only on the primitive and period-one chart constants, such that for every allowed table parameter θ ,*

$$(26) \quad \|\mathcal{L}_\theta^n h\|_{\mathcal{B}_\theta} \leq C \vartheta^n \|h\|_{\mathcal{B}_\theta} \quad (\langle 1, h \rangle = 0).$$

Proof of Theorem 2.24. We use the perturbative Lorentz-gas theorem of [11, Theorems 2.2, 2.3, 2.5, 2.6, and 2.11; Corollary 2.4] with the following theorem-number-level correspondence:

Demers–Zhang result	verification used here
Theorem 2.2	Uniform Lasota–Yorke inequalities and a uniform essential spectral-radius bound follow once (H1)–(H5) hold with common constants. Uniform finite horizon, curvature and separation, the cone/distortion bounds, and Theorem 2.20 give those constants on the transported compact chart.
Theorem 2.5	The unforced finite-horizon dispersing billiard class with fixed bounds satisfies (H1)–(H5). The primitive record in Assumption 2.3 is chosen inside one such class.
Theorem 2.6	Moving and deforming scatterers are small in the Demers–Zhang perturbation distance after normal-graph transport; this is the metric input for the present radial and general compact response charts.
Theorem 2.3 and Corollary 2.4	The transfer operators and spectral projectors vary continuously in the strong-to-weak perturbation topology, and the spectral gap of one reference table persists on a sufficiently small chart.
Theorem 2.11	Along a continuous compact path in the uniform billiard class, the spectral data vary continuously/Hölder continuously; a finite cover makes the spectral separation and the Lasota–Yorke constants uniform.

The compact response set is covered by finitely many normal-graph charts to which this route applies. Compactness is used only after Theorems 2.3 and 2.11 provide spectral continuity; pointwise mixing alone is not used to infer a uniform gap.

The cited construction also gives compact strong-to-weak embedding (see the compactness lemma used in the proof of Demers–Zhang Theorem 2.2), hence quasi-compactness with one common essential spectral-radius bound. The period-one hypothesis in Assumption 2.6 removes all peripheral eigenvalues except the simple eigenvalue 1. The continuous spectral projectors and the finite compact cover then give $C < \infty$ and $\vartheta < 1$ common to the family. This proves the displayed zero-mass estimate. Collision-graph connectivity is not used as a substitute for the period-one assumption. $\square$

Remark (Proof order and non-circular use of later estimates). The proof is read in the following forward order. First, Definition 2.22 and Theorem 2.24 fix the map spectral engine; no moving-response formula is used there. Second, Theorem 3.2 constructs the first trace on the graph domain. Third, Definition 3.21 and Proposition 3.22 place its weak source against the admissible adjoint observable resolvent. Fourth, Theorem 3.20 proves the first response pairing without differentiating a projector or resolvent. Any reference from Section 4 to trace notation is therefore only notational anticipation, not a dependency on the response theorem. The flow graph and contour calculus is logically separate and begins with Assumption 2.11 and Lemma 2.25.

Lemma 2.25 (Suspension Laplace contour and frequency partition). *Let X_θ be the corrected BDL- $v\beta$ flow generator and set*

$$\mathcal{G}_{\theta,z} = (z - X_\theta)^{-1} \Pi_\theta^{\text{den},\perp}, \quad z = \sigma + ib.$$

There exist $\eta > 0$, $b_0 > 1$, and $C > 0$, uniform in the family of Assumption 2.11, with the following frequency decomposition.

(i) *For $|b| \leq b_0$ and $|\sigma| < \eta$, the BDL generator resolvent has only the simple invariant pole at $z = 0$. Inserting $\Pi_\theta^{\text{den},\perp}$ removes it and gives a uniformly bounded operator between the stated BDL graph-domain levels.*

(ii) *For $|b| > b_0$ and $|\sigma| < \eta$, the corrected BDL- $v\beta$ estimate gives*

$$(27) \quad \|\mathcal{G}_{\theta,\sigma+ib}\|_{\widehat{\mathcal{B}}_\theta^{(1)} \rightarrow \widehat{\mathcal{B}}_\theta^w} \leq C|b|^{\chi_{\text{res}}}.$$

The high-frequency line is partitioned into dyadic blocks $2^m b_0 \leq |b| \leq 2^{m+1} b_0$.

(iii) The collision family $\mathcal{L}_{\theta,z} = \mathcal{L}_{\theta}(e^{-z\tau_{\theta}})$ enters only through the bounded trace/lift renewal identity

$$\mathcal{G}_{\theta,z} = \mathcal{H}_{\theta,z} + \mathcal{E}_{\theta,z}(I - \mathcal{L}_{\theta,z})^{-1} \mathcal{J}_{\theta,z}$$

on the fixed-table graph domains of Assumption 2.11. This identity is not an equality of the BDL flow space with the Demers–Zhang map space.

(iv) The inverse-Laplace contour for the zero-mean flow semigroup can be shifted from $\Re z = a > 0$ to the broken contour consisting of the low-frequency segment in $|\Re z| \leq \eta$, the two vertical high-frequency rays $\Re z = -\eta$, and small detours around the removed pole at $z = 0$. On the zero-mean complement the pole contribution vanishes.

Consequently, for every zero-mean observable section f with the roof regularity required by the response domain, the reduced observable resolvent is defined by

$$(28) \quad \langle h, (-\mathcal{A}_{\theta}|_{\ker \Pi_{\theta}^{\text{obs}}})^{-1} f \rangle = \int_0^{\infty} \langle \widehat{\mathcal{L}}_{\theta,t} \Pi_{\theta}^{\text{den},\perp} h, f \rangle dt,$$

which converges for every zero-mass suspension density h in the strong anisotropic density fiber. No finite parameter derivative of this flow resolvent is taken in the present paper.

Proof. The fixed-table meromorphic and high-frequency assertions are the corrected BDL-v3 generator-resolvent theorem. Their common norm and domain constants, and the trace/lift renewal maps, are the fixed-table functional input in Assumption 2.11; the contact and pair-selection margins are Proposition 2.14. The zero-mass projector removes the invariant pole. The corrected BDL bound then justifies the fixed-table contour shift on its own flow space. No parameter derivative of this contour is taken in the present paper. $\square$

3. LABEL-RESOLVED SOURCES, RECOVERY, AND THE RADIAL MODEL

Remark (Physical traces versus artificial cuts). Only moving physical scatterer boundaries produce response trace terms in the final formula. Homogeneity-strip boundaries, partition cutoffs, and artificial refinement edges are introduced to define the anisotropic norms and to localize singularities. Their central contributions cancel between adjacent partition pieces because the partition of unity sums to one; the noncentral pieces are recorded only as terminal weak-tail remainders. In the trace theorems below, “strip boundary” and “finite preimage” terms therefore mean bookkeeping terms for the partition calculus, not additional physical response forces.

Definition 3.1 (Moving Banach bundle, transport maps, and response domain). The suspension phase-space bundle is

$$(29) \quad \widehat{\mathfrak{J}} = \{(\theta, z, u) : \theta \in \Theta, z \in \Sigma_{\theta} \setminus \bigcup_{n \in \mathbb{Z}} \mathcal{P}_{\theta}^{-n} \mathcal{S}_{\theta}, 0 \leq u < \tau_{\theta}(z)\} / \sim,$$

with $(\theta, z, \tau_{\theta}(z)) \sim (\theta, \mathcal{P}_{\theta} z, 0)$. Local trivializations

$$(30) \quad \mathfrak{J}_{\theta \leftarrow \theta_0} : \widehat{\Sigma}_{\theta_0} \rightarrow \widehat{\Sigma}_{\theta}$$

move scatterer boundaries in normal coordinates, preserve the component label and physical velocity, and use the normalized roof height $u/\tau_{\theta_0}(z)$. They carry strip boundaries and singular hypersurfaces to corresponding finite unions, modulo zero flux sets.

The observable and density transition maps are

$$(31) \quad \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{obs}} f = f \circ \mathfrak{J}_{\theta \leftarrow \theta_0},$$

$$(32) \quad \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{den}} \rho = (\mathfrak{J}_{\theta \leftarrow \theta_0})_*(J_{\theta, \theta_0}^{-1} \rho),$$

with the Jacobian chosen so that pairings are preserved. A section is C^k if it becomes a C^k map into one fixed Banach fiber after applying these transport maps.

For a compact parameter ball U about θ_0 , let $\mathcal{C}_{\text{hom}}^r(U)$ be the smooth core of densities whose transported representatives are branchwise C^r on every homogeneous component and have finite strip support before completion. On this core define $\mathcal{T}_{1,m,\theta}^{\text{phys,core}}$ by differentiating the homogeneous branch indicators, roof weights, inverse branches, and Jacobian factors once, summing all strips in the dyadic block $\mathcal{G}_m$, cancelling artificial faces, and retaining the complete physical current. The

finite-word refinement is fixed before completion; the appendix proves that changing to a common refinement gives equivalent seminorms. A first moving singular curve Γ with normal velocity V_Γ defines the core trace by stable-leaf testing:

$$(33) \quad \langle \mathcal{B}_\Gamma^{\text{core}} \rho, \psi \rangle_W = \sum_{y \in W \cap \Gamma} (V_\Gamma \cdot n_\Gamma)(y) \rho|_W(y) [\psi J_h]_\Gamma(y).$$

The notation $\rho|_W(y)$ is used only for smooth core densities. No point-value estimate is asserted from the underlying anisotropic norm alone. The response domain $\mathfrak{D}_U$ is the completion of $\mathcal{C}_{\text{hom}}^r(U)$ for the independent seed norm

$$(34) \quad \|\rho\|_{\mathfrak{D}_U} = \mathcal{N}_3^{\text{flux}}(\rho).$$

It controls coefficients of the undifferentiated physical branch flux and its resolved face/corner restrictions, not the target current itself. Lemma 2.19 therefore proves, rather than defines, continuity of $\mathcal{T}_{1,m,\theta}^{\text{phys,core}}$. The weak codomain $\mathfrak{D}_U^w$ is the corresponding weak completion. After Theorem 3.2 is proved, the combined physical current has a unique continuous extension. No isolated point trace or absolutely summed strip trace of a general anisotropic distribution is assumed.

We also write $\mathfrak{D}_U^{(1)} = \mathfrak{D}_U$. This is the first level of the finite scale in Definition 4.1. The observable target class is its matching adjoint trace level.

Lemma (Flux-seed response completion is well posed). *The response domain in Definition 3.1 is a first-order flux-seed completion. More precisely, the smooth-core seminorms are finite, two finite homogeneous refinements give equivalent seminorms after passing to a common refinement, and the smooth core is dense by construction. If ρ_n is Cauchy in $\mathfrak{D}_U$, then for every stable test (W, ψ) the complete block pairings $\langle \mathcal{T}_{1,m,\theta}^{\text{phys,core}} \rho_n, \psi \rangle_W$ have a limit independent of the approximating sequence. The operators $\mathcal{T}_{1,m,\theta}^{\text{phys}}$ are therefore defined on the completion by this limiting pairing, and their boundedness $\mathfrak{D}_U \rightarrow \mathfrak{D}_U^w$ is the content of Theorem 3.2. This construction does not imply that the same point trace is bounded on the underlying $\mathcal{B}$, nor does it by itself prove transfer invariance of the graph domain.*

Proof. On the smooth core the weighted flux coefficients and resolved restrictions in Definition 2.18 have finite homogeneous support, so their integrals and variations are finite. A common refinement only subdivides those quantities and preserves physical face labels; artificial internal faces have opposite orientations. Hence the seed norms are equivalent under finite refinement. If ρ_n is seed-Cauchy, the estimate in Lemma 2.19 makes every complete physical block pairing Cauchy uniformly in (m) . Its limit defines the current on the completion independently of the approximating core sequence. $\square$

Theorem 3.2 (First-order physical response-current theorem). *Let $D_h(\theta)$ be a homogeneous branch domain and let $\Gamma_s \subset \partial D_h(\theta_s)$ be a moving singular curve with normal velocity V_Γ in the blown-up strip chart. For a stable test leaf W , a test function ψ , and a smooth core density ρ , cut W at Γ_0 , at artificial strip boundaries, and at terminal near-grazing cells. On the remaining central transversal pieces, the derivative of the branch characteristic function is the stable-leaf trace pairing*

$$(35) \quad \langle \mathcal{B}_\Gamma \rho, \psi \rangle_W = \sum_{y \in W \cap \Gamma_0} (V_\Gamma \cdot n_\Gamma)(y) \rho|_W(y) [\psi J_h]_\Gamma(y),$$

where $[\psi J_h]_\Gamma$ is the jump of the pulled-back test-Jacobian factor across adjacent homogeneous branches. This formula is a local smooth-core identity; no point trace of a general element of $\mathcal{B}$ is asserted. After summing adjacent branches, cancelling artificial strip faces, and combining the interior, Jacobian, flux, and physical-face terms in each grazing block, the complete first response current extends by flux-seed continuity and satisfies

$$(36) \quad \sum_{m \geq 0} |\langle \mathcal{T}_{1,m}^{\text{phys}} \rho, \psi \rangle| \leq C \|\rho\|_{\mathfrak{D}_U^{(3)}} \|\psi\|_{\mathfrak{A}_U^{(4)}}.$$

An isolated stripwise $\mathcal{B}_\Gamma$ need not have an absolutely summable weak norm.

Proof. Write the branch contribution in fixed coordinates as $1_{D_h(\theta_s)} J_{h_s}^{-1} \rho \circ h_s$. The ordinary derivative of the smooth factor gives branch-interior and Jacobian terms. The singular operation is the derivative of the moving-domain indicator. If Γ_s is written as $g_s = 0$ and if $\partial_s g_s|_0 = -V_\Gamma \cdot n_\Gamma$, then

$$(37) \quad \partial_s 1_{\{g_s > 0\}}|_{s=0} = (V_\Gamma \cdot n_\Gamma) \delta_{\Gamma_0}.$$

Pairing with a stable test curve gives the stated intersection sum. Central transversality gives a uniform intersection bound, while in $\mathbb{H}_k$ the impact equations and blown-up coordinate give polynomial bounds for V_Γ , crossing denominators, and branch Jacobians. Thus each stable-leaf pairing is bounded by

$$(38) \quad |\langle \mathcal{B}_\Gamma \rho, \psi \rangle_W| \leq C k^{M_1} \|\rho\|_{C_{\text{br}}^1} m_W \{y \in W : \text{the traced descendant of } y \text{ lies in } \mathbb{H}_k\}.$$

This is a local smooth-core estimate and is not summed in absolute value over k . After the physical terms and artificial-face cancellations are assembled, the renormalized block estimate gives

$$(39) \quad \sum_{m \geq 0} |\langle \mathcal{T}_{1,m}^{\text{phys}} \rho, \psi \rangle| \leq C \|\rho\|_{\mathfrak{D}_U^{(3)}} \|\psi\|_{\mathfrak{A}_U^{(4)}},$$

by Lemma 2.19. The flux-seed norm gives a uniform Cauchy estimate for smooth-core approximants and hence the extension to $\mathfrak{D}_U$; the unstable comparison follows by matching trace intersections under the local unstable holonomy between nearby stable leaves, with unmatched pieces retained in the same dyadic weak-terminal block. $\square$

Remark (Higher traces require resolved currents). Formal differentiation produces same-wall terms such as $\delta'(g)$ and, for two transverse walls in the two-dimensional collision section, a codimension-two coarea atom. Three or more independent delta constraints are not defined by declaring their intersection a finite vertex. The positive construction in Definition 4.13 and Theorem 4.14 therefore differentiates the resolved moving-domain current itself: repeated walls give normal derivatives, two clean walls give coarea atoms, and resolved vertices give parameter derivatives of an existing point current. The renormalized physical block identity then sums the resulting finite-order losses without an absolute strip moment.

Proposition 3.3 (First transfer derivative). *Let θ_s be an admissible C^1 parameter curve. On the smooth finite-strip core, differentiation of the transported transfer operator gives*

$$\nabla_s \mathcal{L}_{\theta_s}|_0 \rho = \sum_h \mathcal{L}_{0,h} (\mathcal{D}_h^{\text{int}} + \mathcal{D}_h^{\text{Jac}}) \rho + \sum_{\Gamma \subset S} \mathcal{B}_\Gamma \rho.$$

The branch-interior and Jacobian series and the physical-boundary trace series extend by continuity to a bounded map $\mathfrak{D}_U \rightarrow \mathfrak{D}_U^w$. Artificial partition boundaries cancel pairwise before the second sum is formed. For the transported invariant density, the difference quotient converges to this source in the target-class dual topology of Definition 4.1.

Proof. Differentiate each homogeneous inverse-branch formula. Differentiating its smooth weight gives $\mathcal{D}_h^{\text{int}} + \mathcal{D}_h^{\text{Jac}}$, with the impact, reflection, flux, and transport derivatives included. Differentiating the branch indicator gives the physical trace of Theorem 3.2. The branch estimates use the cited billiard strong-to-weak bounds. Adjacent artificial cutoffs form a partition of unity, so their indicator derivatives cancel. Completion of the core gives the asserted bounded complete physical current by Lemma 2.19, and the final convergence statement follows from Proposition 4.12. $\square$

Theorem 3.4 (Exact invariant-source Ward identity and strong primitive). *Work in one fixed transported collision fiber and write $\rho_s = d(\Phi_s^* \nu_s)/d\nu_0$. Then $\mathcal{L}_s \rho_s = \rho_s$. If the transported family is C^J on the smooth response core, then for every $1 \leq j \leq J$*

$$(40) \quad G_j := \sum_{\ell=1}^j \binom{j}{\ell} (\partial_s^\ell \mathcal{L}_s|_0) (\partial_s^{j-\ell} \rho_s|_0) = (I - \mathcal{L}_0) \partial_s^j \rho_s|_0$$

as an identity in the distributional source/test pairing. Consequently the fully assembled invariant source has the explicit strong primitive $\partial_s^j \rho_s|_0$. All artificial-face currents and all residual endpoint currents in that total source cancel exactly.

The theorem does not claim that an individual derivative letter $(\partial_s^\ell \mathcal{L}_s|_0)h$, with arbitrary h , has a strong primitive. That stronger operator statement is precisely the additional Ward or source-recovery condition needed for general dynamical susceptibilities.

Proof. The normalized collision flux is invariant for every table. Pulling it to the reference fiber gives $\mathcal{L}_s \rho_s = \rho_s$, with ρ_s the explicit transport Jacobian. Differentiate this identity j times and move the $\ell = 0$ term to the right; this is (40). The equality holds first against smooth finite-strip tests. The branch/interior, Jacobian, flux, and physical face terms are the complete distributional derivative of the left-hand side, so no endpoint term can be discarded separately. Adjacent artificial faces have opposite orientations and cancel before the equality is completed. Since the right-hand side is the difference of two strong transported densities, completion gives the asserted source identity. In particular every coefficient of a residual distribution supported only at $u = 0$ in the total invariant source is zero. $\square$

Proposition 3.5 (Represented strong recovery criterion). *Assume the period-one Demers–Zhang spectral gap on the centered strong density space. Let g be a centered complete assembled source in $\mathfrak{X}_U^{(m)}$, and write $\bar{g} = S_m \mathbf{Q}_m g$ for its total represented current. At the base strong level the following are equivalent:*

- (i) *there is a centered $h \in \mathfrak{D}_U$ such that $\bar{g} = (I - \mathcal{L}_0)h$ in the distributional pairing;*
- (ii) *the total represented current $\bar{g}$ belongs to the centered strong space $\mathfrak{D}_U$.*

When these conditions hold, the primitive is canonical and satisfies

$$(41) \quad h = (I - \mathcal{L}_0)^{-1} \bar{g} = \sum_{n \geq 0} \mathcal{L}_0^n \bar{g}, \quad \|h\|_{\mathfrak{D}_U} \leq C_{\text{gap}} \|\bar{g}\|_{\mathfrak{D}_U}.$$

Consequently, in a clean local decomposition with separated one-face and two-face point strata, a recovered represented current cannot retain a nonzero isolated singular coordinate. Those coordinates must cancel in the complete physical assembly before the ordinary density resolvent is used. If the response state also carries a separate terminal symbol, vanishing or closure of that component is an additional requirement not implied by the distributional equality. Intrinsic recovery as an element of the dynamical labelled module further requires

$$g = \iota_{\text{reg}} \bar{g} \quad \text{in } \mathfrak{X}_U^{(m)},$$

where ι_{reg} is the canonical regular embedding. Thus cross-label residues and $\pi_{\text{term}} g$ must vanish separately. For a specified complete block, the weaker certified scalar reset of Proposition 3.10 is an alternative: it proves that the terminal truncation vanishes in every subsequent fixed scalar pairing and then re-embeds $\bar{g}$, without asserting the displayed equality in $\mathfrak{X}_U^{(m)}$. The same assertion holds at a higher response level whenever the centered spectral-gap estimate is available in that level's strong graph norm.

Proof. If $\bar{g} = (I - \mathcal{L}_0)h$ and $h \in \mathfrak{D}_U$, boundedness of $\mathcal{L}_0$ on the strong space gives $\bar{g} \in \mathfrak{D}_U$. Conversely, the period-one spectral gap makes the Neumann series in (41) converge in the centered strong norm, and telescoping proves $(I - \mathcal{L}_0)h = \bar{g}$.

A clean point atom is not even in the ordinary Demers–Zhang weak completion, and a nonzero isolated transverse one-face current is, in general, weak but not strong by Proposition 4.10. Local tests supported near one separated stratum therefore detect any surviving coefficient. Hence membership of the total represented assembly in $\mathfrak{D}_U$ forces every such singular coordinate to cancel (or to combine with another term on the same total stratum into zero). The total-current map deliberately forgets cross-label and terminal information, so this argument makes no assertion about those coordinates. Equality with the regular embedding is exactly the additional dynamical-module requirement. The higher-level statement is the same Neumann-series argument in the stipulated graph norm. $\square$

Theorem 3.6 (A genuinely nonconjugate second-order recovered assembly). *Consider a finite-horizon table containing circular scatterers i and k , with radii R_i, R_k , and an isolated unobstructed normal period-two orbit between them. Keep all centers and all other scatterers fixed and set*

$R_i(s) = R_i + s$ for $|s|$ small. Assume the resulting family remains in the period-one/no-boundary-pair-birth geometry (R1)–(R2) of Assumption 2.6; no arbitrary-seed (R3) is used. Let P_0 be the total boundary length and put $a = 2\pi/P_0$. Under the transport which fixes the boundary label, polar parameter, and collision angle, the pulled-back normalized flux density is

$$(42) \quad \rho_s(j, \sigma, \varphi) = \frac{P_0}{P_0 + 2\pi s} \begin{cases} 1 + s/R_i, & j = i, \\ 1, & j \neq i. \end{cases}$$

In particular

$$(43) \quad \dot{\rho}_0 = \begin{cases} R_i^{-1} - a, & j = i, \\ -a, & j \neq i, \end{cases} \quad \ddot{\rho}_0 = \begin{cases} 2a^2 - 2a/R_i, & j = i, \\ 2a^2, & j \neq i. \end{cases}$$

These functions are centered strong densities. If $B_1 = \partial_s \mathcal{L}_s|_0$ and $B_2 = \partial_s^2 \mathcal{L}_s|_0$, then the complete first- and second-order invariant assemblies satisfy

$$(44) \quad G_1 = B_1 \rho_0 = (I - \mathcal{L}_0) \dot{\rho}_0,$$

$$(45) \quad G_2 = B_2 \rho_0 + 2B_1 \dot{\rho}_0 = (I - \mathcal{L}_0) \ddot{\rho}_0.$$

Thus the total represented one-face and point current in each complete assembly cancels to a strong density. The family verifies the represented base-level recovery criterion and the zero-waiting-time triangular step through order two, with explicit primitives (43).

This family is not a C^1 conjugate deformation for a generic such period-two orbit. The next theorem upgrades the complete fixed-length block to arbitrary positive waiting lengths. Neither result asserts eventwise or all-seed (S3), label-resolved terminal closure, or the all-seed operator bound (S1).

Proof. The normalized collision flux is $Z_s^{-1} \cos \varphi ds_s d\varphi$, with $Z_s = 2P_s$ and $P_s = P_0 + 2\pi s$. On component i , the arclength Jacobian is $(R_i + s)/R_i$; it is one on every other component. This proves (42). Two differentiations give (43); centeredness follows by differentiating $\int \rho_s d\nu_0 = 1$. Since the collision components are disjoint, these piecewise constant functions lie in every finite strong seed level used here.

Differentiate $\mathcal{L}_s \rho_s = \rho_s$ once and twice. This gives (44) and (45). Their right-hand sides are strong, so Proposition 3.5 forces the singular sector of each complete left-hand assembly to vanish. Notice that the individual summands $B_2 \rho_0$ and $B_1 \dot{\rho}_0$ are not separately declared strong.

For nonconjugacy, use paraxial Jacobi coordinates along the normal two-cycle. With

$$F_\tau = \begin{pmatrix} 1 & \tau \\ 0 & 1 \end{pmatrix}, \quad R_\kappa = \begin{pmatrix} 1 & 0 \\ 2\kappa & 1 \end{pmatrix},$$

the return derivative is conjugate to $R_{\kappa_i} F_\tau R_{\kappa_k} F_\tau$, whose trace is

$$2 + 4\tau(\kappa_i + \kappa_k) + 4\tau^2 \kappa_i \kappa_k.$$

Here $\tau'(0) = -1$, $\kappa'_i(0) = -\kappa_i^2$, and $\kappa'_k(0) = 0$, so the derivative of this trace is

$$-4(\kappa_i + \kappa_k) - 4\tau \kappa_i^2 - 8\tau \kappa_i \kappa_k - 4\tau^2 \kappa_i^2 \kappa_k < 0.$$

The periodic multiplier therefore changes. A C^1 conjugacy would preserve the multiplier spectrum of the isolated periodic orbit, which proves the claim. $\square$

Theorem 3.7 (Positive-waiting block recovery through second order). *Let $s \mapsto (\mathcal{L}_s, \rho_s)$ be any C^2 transported invariant family for which the complete first- and second-order finite-time current identities are defined, and write*

$$B_1 = \partial_s \mathcal{L}_s|_0, \quad B_2 = \partial_s^2 \mathcal{L}_s|_0, \quad \rho_j = \partial_s^j \rho_s|_0.$$

For $N \geq 1$, set

$$(46) \quad \mathcal{D}_N := \partial_s(\mathcal{L}_s^N)|_0 = \sum_{a=0}^{N-1} \mathcal{L}_0^{N-1-a} B_1 \mathcal{L}_0^a,$$

$$(47) \quad \begin{aligned} \mathcal{E}_N &:= \partial_s^2(\mathcal{L}_s^N)|_0 \\ &= \sum_{a=0}^{N-1} \mathcal{L}_0^{N-1-a} B_2 \mathcal{L}_0^a + 2 \sum_{0 \leq a < b \leq N-1} \mathcal{L}_0^{N-1-b} B_1 \mathcal{L}_0^{b-a-1} B_1 \mathcal{L}_0^a. \end{aligned}$$

Then the complete blocks containing all insertion positions and all positive waiting gaps satisfy

$$(48) \quad \mathcal{G}_{1,N} := \mathcal{D}_N \rho_0 = (I - \mathcal{L}_0^N) \rho_1 = (I - \mathcal{L}_0) H_{1,N},$$

$$(49) \quad \mathcal{G}_{2,N} := \mathcal{E}_N \rho_0 + 2 \mathcal{D}_N \rho_1 = (I - \mathcal{L}_0^N) \rho_2 = (I - \mathcal{L}_0) H_{2,N},$$

$$(50) \quad H_{j,N} := \sum_{r=0}^{N-1} \mathcal{L}_0^r \rho_j.$$

The flux jets ρ_j are centered. Hence the period-one spectral gap gives the uniform bound

$$(51) \quad \sup_{N \geq 1} \|H_{j,N}\|_{\mathfrak{D}_U} \leq C_{\text{gap}} \|\rho_j\|_{\mathfrak{D}_U}, \quad j = 1, 2.$$

Thus arbitrary positive waiting lengths are recovered after summing the complete fixed- N block; isolated insertion words are not asserted to be recovered. For the radial inflation of Theorem 3.6, all primitives are explicit through (43). The theorem proves total-current recovery. Label-resolved terminal closure requires, in addition, convergence to the regular embedding in $\mathfrak{X}_U^{(m)}$.

Proof. The product rule for the N -fold product gives (46) and (47), including the factor two for two distinct insertion positions. Differentiate $\mathcal{L}_s^N \rho_s = \rho_s$ once and twice. Moving the undifferentiated $\mathcal{L}_0^N \rho_j$ term to the right gives the first equalities in (48) and (49). The factorization

$$I - \mathcal{L}_0^N = (I - \mathcal{L}_0) \sum_{r=0}^{N-1} \mathcal{L}_0^r$$

gives the displayed primitives. Differentiation of $\int \rho_s d\nu_0 = 1$ centers ρ_1, ρ_2 , so the strong spectral gap makes their geometric sums uniformly bounded. Every equality is first a finite complete source/test identity, hence it includes all branch, Jacobian, flux, one-face, and clean point terms before the strong conclusion is taken. The final sentence follows from the distinction between the total-current map $\mathbf{S}_m$ and the label-resolved realization $\mathbf{Q}_m$. $\square$

Lemma 3.8 (Fixed-length scalar grazing-terminal truncation). *In the setting of Theorem 3.7, fix $N \geq 1$ and $j \in \{1, 2\}$. After forming the complete order- j , length- N physical block, cancelling artificial faces, and only then cutting the physical grazing coordinate into dyadic blocks, let*

$$\mathcal{G}_{j,N}^{[\geq M]}$$

denote the sum of the blocks with index $m \geq M$. On the smooth flux-seed/trace-test core, and hence in the scalar adjoint-trace completion, whenever the displayed response-level density norms are finite, there is a finite constant $C_{N,j}$ such that

$$(52) \quad |\langle \mathcal{G}_{j,N}^{[\geq M]}, f \rangle| \leq C_{N,j} 2^{-M} \max_{0 \leq \ell \leq j} \|\rho_\ell\|_{\mathfrak{D}_U^{(j+2)}} \|f\|_{\mathfrak{A}_U^{(j+3)}}.$$

Thus the grazing-terminal truncation of every fixed-length complete block vanishes in the scalar current topology. The assertion is deliberately not a label-resolved terminal-closure theorem: it neither proves $\|\pi_{\text{term}} \mathcal{G}_{j,N}\|_{\mathfrak{T}_U^{(m)}} = 0$ nor supplies a constant uniform in N .

Proof. For fixed N the product-rule formulae (46)–(47) contain finitely many insertion positions. On each intervening finite-time regular branch, the ordinary transfer and observable pullback have finite response-level bounds. Form the complete differentiated physical identity before estimating any

one-sided term. The renormalized block estimate (14) then gives $C2^{-m}$ for each resulting physical block. The finite number of insertion positions and finite-time operator constants are absorbed into $C_{N,j}$. Summing the geometric series for $m \geq M$ proves (52). Since this argument pairs the already assembled total current with f , it forgets branch labels and does not estimate the independent terminal coordinate of the ambient module, which proves exactly the stated limitation. $\square$

Theorem 3.9 (Uniform post-assembly grazing tail for radial blocks). *In the radial-inflation family of Theorem 3.6, let $j \in \{1, 2\}$, $N \geq 1$, and let $\overline{\mathcal{G}}_{j,N}$ denote the total current after complete assembly, rather than its historical one-sided expansion. Thus*

$$(53) \quad \overline{\mathcal{G}}_{j,N} = (I - \mathcal{L}_0^N) \rho_j.$$

Let ζ_M be a uniformly bounded smooth cutoff supported in the physical grazing collar $u \leq C2^{-M}$, equal to one on all dyadic blocks with index at least $M + O(1)$. Then

$$(54) \quad \sup_{N \geq 1} |\langle \zeta_M \overline{\mathcal{G}}_{j,N}, f \rangle| \leq C_j 2^{-4M} \|f\|_{C^0},$$

where C_j is independent of M and N . Moreover the canonical block-boundary state

$$(55) \quad x_{j,N}^{\text{asm}} := \iota_{\text{reg}} \overline{\mathcal{G}}_{j,N} \in \mathfrak{X}_U^{(0)}$$

is uniformly bounded in N and has zero terminal coordinate.

This theorem upgrades the post-assembly scalar tail from fixed N to uniform N . It does not say that the pre-assembly labelled history has zero π_{term} -coordinate: the passage from that history to (55) is precisely the certified assembly reset, not an identity in the historical module.

Proof. For the radial family the two flux jets in (43) are bounded functions, constant on each collision component. Since the billiard map is invertible and preserves the normalized collision flux,

$$\mathcal{L}_0^N h = h \circ \mathcal{P}_0^{-N} \quad \text{a.e.}, \quad \|\mathcal{L}_0^N h\|_{L^\infty(\nu_0)} = \|h\|_{L^\infty(\nu_0)}.$$

Hence

$$\sup_N \|\overline{\mathcal{G}}_{j,N}\|_\infty \leq 2\|\rho_j\|_\infty.$$

In the physical grazing coordinate the invariant flux is $d\nu_0 = 2u^3 a(u, s) du ds$, with a uniformly bounded. Therefore

$$\nu_0\{u \leq C2^{-M}\} \leq C' 2^{-4M},$$

and Hölder's elementary L^∞ - L^1 estimate gives (54). A bounded density defines an ordinary weak-density element with norm controlled by its L^∞ norm; the regular embedding is bounded by construction. This proves the uniform bound for (55), whose terminal coordinate is zero by definition. The final qualification follows because $\mathbf{S}_m$ forgets labels and terminal data, whereas $\mathbf{Q}_m$ does not. $\square$

Proposition 3.10 (Certified scalar assembly reset). *Let $g^{(<M)}$ be the truncations of a complete fixed-length physical source at a block boundary, and suppose that there is a centered $\bar{g} \in \mathfrak{D}_U^{(m+1)}$ and numbers $\eta_M \downarrow 0$ such that*

$$(56) \quad |\langle \mathbf{S}_m \mathbf{Q}_m g^{(<M)} - \bar{g}, f \rangle| \leq \eta_M \|f\|_{\mathfrak{A}_U^{(m+2)}}$$

for every trace test f . Then the block has the canonical assembly reset

$$\mathfrak{R}_m^{\text{asm}} g := \iota_{\text{reg}} \bar{g}.$$

For every fixed $q \geq 0$ for which $\mathcal{U}_0^q : \mathfrak{A}_U^{(m+2)} \rightarrow \mathfrak{A}_U^{(m+2)}$ is bounded,

$$(57) \quad \lim_{M \rightarrow \infty} \langle \mathbf{S}_m \mathbf{Q}_m g^{(<M)}, \mathcal{U}_0^q f \rangle = \langle \bar{g}, \mathcal{U}_0^q f \rangle.$$

Thus the labels required inside a singular event may be discarded after its complete scalar assembly has been certified. A finite succession of such certified blocks can be reset inductively.

If in addition $\bar{g} = (I - \mathcal{L}_0)h$ with centered strong h , the post-reset scalar resolvent telescopes from h . This is a blockwise scalar closure operation, not an identity $g = \iota_{\text{reg}} \bar{g}$ in the label-resolved

module and not a bounded projection on all of $\mathfrak{X}_U^{(m)}$. For the radial-inflation blocks, Theorems 3.7 and 3.9 and Lemma 3.8 verify the reset through order two for every fixed waiting length N , with strong primitives and post-assembly scalar grazing tails uniformly bounded in N .

Proof. Apply (56) with $\mathcal{U}_0^q f$. The fixed- q observable bound sends the right-hand side to zero and proves (57). After the limit, the input to the next block is the ordinary strong density $\bar{g}$, so the same argument applies inductively to any fixed finite chain of certified blocks. When $\bar{g} = (I - \mathcal{L}_0)h$, duality gives

$$\langle \bar{g}, \mathcal{U}_0^n f \rangle = \langle \mathcal{L}_0^n h, f \rangle - \langle \mathcal{L}_0^{n+1} h, f \rangle,$$

and the strong spectral gap controls the tail. The radial conclusion uses (48)–(51) for $\bar{g}$ and (52) for (56). None of these scalar identities identifies the pre-reset labelled history with the regular embedding. $\square$

Proposition 3.11 (Historical terminal accumulation dichotomy). *Let a complete first-order length- N block be expanded before assembly reset, and retain the insertion time as part of the terminal label. Thus its historical terminal space contains the isometric ℓ^1 -sum*

$$\mathfrak{T}_N^{\text{hist}} = \bigoplus_{a=0}^{N-1} \ell^1 \mathfrak{T}_a,$$

where a is the position of the differentiated event. Suppose the one-event derivative of the stationary input has terminal coordinate τ , and undifferentiated propagation merely transports it to the corresponding time-labelled copy τ_a . Then

$$(58) \quad \left\| \pi_{\text{term}}^{\text{hist}} \sum_{a=0}^{N-1} \mathcal{L}_0^{N-1-a} B_1 \mathcal{L}_0^a \rho_0 \right\|_{\mathfrak{T}_N^{\text{hist}}} = \sum_{a=0}^{N-1} \|\tau_a\| = N \|\tau\|.$$

The same conclusion holds for a high-grazing truncation $\tau^{[\geq M]}$. Consequently

$$\sup_{N \geq 1} \|\pi_{\text{term}}^{\text{hist}} \mathcal{G}_{1,N}^{[\geq M]}\| < \infty$$

can hold only if the one-event terminal germ at that truncation vanishes (or if the history quotient is changed so that different insertion times are assembled before taking the norm). In particular a uniform-in- N intrinsic historical terminal-closure theorem is not a consequence of a decaying one-event tail: for every nonzero tail, $\sup_N N \|\tau^{[\geq M]}\| = \infty$.

For the radial blocks this gives a precise dichotomy. Either the actual labelled coefficient calculation proves eventwise terminal cancellation, or the uniform statement available after complete assembly is necessarily the regular reset of Theorem 3.9, not convergence of the raw history in the label-resolved terminal norm.

Proof. The insertion positions are distinct coordinates of an ℓ^1 -sum. The terminal convention assigns no fictitious later billiard image to a nonextending germ, so undifferentiated letters do not identify two different time labels. The norm of the displayed sum is therefore the sum of the coordinate norms, which proves (58). The truncation statement is the same calculation in the closed subspace of indices with block number at least M . Taking the supremum over N proves the dichotomy. The last paragraph only applies this abstract fact to the pre-reset expansion in (46); the post-assembly identity $(I - \mathcal{L}_0^N)\rho_1$ lives in the regular coordinate and is a different object. $\square$

Theorem 3.12 (Countable assembly-first scalar susceptibility). *Let $r \geq 0$ index a countable family of complete physical blocks. Suppose the r th block has finite truncations $g_r^{(<M)}$, a certified assembly reset to a centered strong current*

$$\bar{g}_r = (I - \mathcal{L}_0)h_r, \quad h_r \in \mathfrak{D}_U,$$

and a scalar reset error $\eta_{r,M}$ as in (56). Let f_r be centered admissible tests and assume

$$(59) \quad \sum_{r \geq 0} \|h_r\|_{\mathfrak{D}_U} \|f_r\|_{\mathfrak{A}_U^{(1)}} < \infty, \quad \lim_{M \rightarrow \infty} \sum_{r \geq 0} \eta_{r,M} \|f_r\|_{\mathfrak{A}_U^{(1)}} = 0.$$

Then the assembly-first double susceptibility is absolutely convergent and independent of the finite terminal truncations:

$$(60) \quad \sum_{r \geq 0} \sum_{n \geq 0} \langle \bar{g}_r, \mathcal{U}_0^n f_r \rangle = \sum_{r \geq 0} \langle h_r, f_r \rangle,$$

$$(61) \quad \lim_{M \rightarrow \infty} \sum_{r \geq 0} \langle \mathbf{S}_m \mathbf{Q}_m g_r^{(<M)}, f_r \rangle = \sum_{r \geq 0} \langle \bar{g}_r, f_r \rangle.$$

Thus an infinite scalar response does not require a uniform intrinsic norm on the raw labelled terminal history. It requires summable strong primitives and summable scalar reset errors. This theorem does not create either estimate; it identifies the strictly weaker, physically sufficient target.

Proof. The strong spectral gap and the primitive identity give

$$|\langle \bar{g}_r, \mathcal{U}_0^n f_r \rangle| \leq C \vartheta^n \|h_r\|_{\mathfrak{D}_U} \|f_r\|_{\mathfrak{A}_U^{(1)}}.$$

The first condition in (59) and Tonelli's theorem give absolute convergence; summing in n telescopes to $\langle h_r, f_r \rangle$, and then summing in r proves the first identity. Summing the reset inequalities over r and using the second condition proves the second identity. $\square$

Theorem 3.13 (A non-invariant coboundary susceptibility class). *Let $s \mapsto \mathcal{L}_s$ be the nonconjugate radial-inflation family of Theorem 3.6. Let h_s be any centered C^J transported section of the strong density scale, uniformly bounded on a compact parameter interval, and set*

$$g_s = (I - \mathcal{L}_s)h_s.$$

For every centered C^J transported observable f_s in the corresponding strong dual class, the dynamical scalar susceptibility

$$(62) \quad \chi_s(g_s, f_s) := \sum_{n \geq 0} \langle g_s, \mathcal{U}_s^n f_s \rangle = \langle h_s, f_s \rangle$$

exists, is C^J , and satisfies

$$(63) \quad \partial_s^j \chi_s(g_s, f_s) = \sum_{\ell=0}^j \binom{j}{\ell} \langle \partial_s^\ell h_s, \partial_s^{j-\ell} f_s \rangle, \quad 0 \leq j \leq J,$$

with the transported density-observable pairing understood. Choosing a nonconstant h_s gives a non-invariant source on a genuinely nonconjugate billiard family.

The theorem is an exact-coboundary dynamical benchmark. It proves neither termwise differentiation of the infinite insertion expansion nor all-seed (S1)/(S3); those stronger conclusions require the vector-valued and source-specific estimates isolated above.

Proof. For every finite N , duality gives

$$\sum_{n=0}^{N-1} \langle g_s, \mathcal{U}_s^n f_s \rangle = \langle h_s, f_s \rangle - \langle \mathcal{L}_s^N h_s, f_s \rangle.$$

The uniform centered strong spectral gap sends the last term to zero geometrically, uniformly for s in the compact interval. This proves (62). Its right-hand side is a C^J transported pairing, and the ordinary Leibniz rule gives (63). No parameter derivative of $\mathcal{L}_s$ is used to obtain this identity, so no unproved all-branch completion is hidden in the argument. $\square$

Theorem 3.14 (Uniform correlation susceptibility for strong sources). *Let $s \mapsto \mathcal{L}_s$ be the radial-inflation family on a compact clean interval with the uniform period-one Demers–Zhang spectral gap. Let g_s be any centered strong source and f_s a centered admissible observable such that*

$$\sup_s \|g_s\|_{\mathfrak{D}_U} < \infty, \quad \sup_s \|f_s\|_{\mathfrak{A}_U^{(1)}} < \infty.$$

Then the genuinely correlation-generated series

$$(64) \quad \Xi_s(g_s, f_s) := \sum_{n \geq 0} \langle g_s, \mathcal{U}_s^n f_s \rangle$$

converges geometrically and uniformly in s . Its canonical primitive is

$$(65) \quad h_s = (I - \mathcal{L}_s)^{-1} g_s = \sum_{n \geq 0} \mathcal{L}_s^n g_s, \quad \Xi_s(g_s, f_s) = \langle h_s, f_s \rangle.$$

If the transported finite-time pairings are continuous in s , then Ξ_s is continuous. In particular this applies to

$$g_s = a_s \rho_s - \langle \rho_s, a_s \rangle \rho_s$$

for a uniformly strong transported correlation density $a_s \rho_s$.

Thus on a period-one strong density space there is no separate “non-coboundary strong sector”: every centered strong source has the canonical primitive (65). The new moving-singularity problem is differentiability of $s \mapsto h_s$, not existence of the fixed-table susceptibility. No parameter differentiability is asserted by this theorem.

Proof. The uniform centered spectral gap gives constants $C < \infty$ and $\vartheta < 1$ such that

$$|\langle \mathcal{L}_s^n g_s, f_s \rangle| \leq C \vartheta^n \|g_s\|_{\mathfrak{D}_U} \|f_s\|_{\mathfrak{A}_U^{(1)}}$$

uniformly in s . The Weierstrass test proves uniform convergence and the strong Neumann series in (65); duality gives the last equality there. Continuity follows by taking a finite partial sum, using continuity of each transported finite-time pairing, and then using the uniform geometric tail. Centering of the displayed correlation density is immediate. $\square$

Theorem 3.15 (Differentiated-invariance radial table-velocity benchmark). *Let $s \mapsto \mathcal{L}_s$ be the nonconjugate radial inflation of Theorem 3.6, on a compact clean interval on which the transported invariant density ρ_s is given by (42). Define the table-velocity current*

$$(66) \quad g_s^{\text{vel}} := (\partial_s \mathcal{L}_s) \rho_s,$$

where the derivative is the complete physical branch derivative and its represented total current is taken after assembly. Then g_s^{vel} is a nonzero centered strong source and

$$(67) \quad g_s^{\text{vel}} = (I - \mathcal{L}_s) \partial_s \rho_s.$$

Thus its canonical primitive is supplied by differentiated invariance.

At $s = 0$, put $B_j = \partial_s^j \mathcal{L}_s|_0$, $g_j = \partial_s^j g_s^{\text{vel}}|_0$, and $h_j = \partial_s^{j+1} \rho_s|_0$. The first two complete Poisson assemblies are

$$(68) \quad q_1 := g_1 + B_1 h_0 = (I - \mathcal{L}_0) h_1,$$

$$(69) \quad q_2 := g_2 + 2B_1 h_1 + B_2 h_0 = (I - \mathcal{L}_0) h_2.$$

Consequently the total represented currents in q_1 and q_2 have vanishing moving-face and clean point sectors, and their strong totals are the explicit second and third derivatives of (42). This is not a full certificate for Proposition 3.16: no separate label-resolved terminal reset or strong remainder (73) is asserted here.

For every centered C^2 transported observable f_s ,

$$(70) \quad \sum_{n \geq 0} \langle g_s^{\text{vel}}, \mathcal{U}_s^n f_s \rangle = \langle \partial_s \rho_s, f_s \rangle$$

converges uniformly and is C^2 , with

$$\begin{aligned} \partial_s \Xi_s &= \langle \partial_s^2 \rho_s, f_s \rangle + \langle \partial_s \rho_s, \partial_s f_s \rangle, \\ \partial_s^2 \Xi_s &= \langle \partial_s^3 \rho_s, f_s \rangle + 2 \langle \partial_s^2 \rho_s, \partial_s f_s \rangle + \langle \partial_s \rho_s, \partial_s^2 f_s \rangle. \end{aligned}$$

The source is invariance-generated: both its strongness and its primitive come from differentiating $\mathcal{L}_s \rho_s = \rho_s$. The theorem is therefore a direct telescoping benchmark, not an independently prescribed correlation density and not a source-module recovery certificate. It does not claim that an arbitrary local correlation density passes the proper closed gluing condition.

Proof. Differentiate $\mathcal{L}_s \rho_s = \rho_s$. This gives (67) as a complete physical current identity, so the represented source is centered and strong. The explicit first derivative in (43) is nonzero and

centered. If $g_s^{\text{vel}} = 0$, simplicity of the invariant density on the period-one component would force the centered invariant function $\partial_s \rho_s$ to vanish, a contradiction. Hence the source is nonzero.

Differentiate $g_s^{\text{vel}} = (I - \mathcal{L}_s) \partial_s \rho_s$ once and twice. Moving the terms hit by transfer derivatives to the left gives (68) and (69). The right sides are centered strong densities, so equality of the total represented currents removes the total face and point sectors in the complete left sides. It does not identify or reset the separate historical terminal labels. Finally, telescope the finite correlation sum using (67); the uniform period-one spectral gap removes the terminal term. The right side of (70) is an explicit transported pairing, and two ordinary differentiations give the displayed formulae. $\square$

Proposition 3.16 (Second-order Poisson recovery criterion). *In the setting of Theorem 3.14, suppose g_s has two transported strong jets g_1, g_2 at $s = 0$, and write*

$$B_1 = \partial_s \mathcal{L}_s|_0, \quad B_2 = \partial_s^2 \mathcal{L}_s|_0, \quad h_0 = (I - \mathcal{L}_0)^{-1} g_0.$$

Form the complete assembled first Poisson source

$$(71) \quad q_1 := g_1 + B_1 h_0.$$

If its represented current is centered strong and its terminal history has a certified assembly reset, put $h_1 = (I - \mathcal{L}_0)^{-1} q_1$. Next form

$$(72) \quad q_2 := g_2 + 2B_1 h_1 + B_2 h_0.$$

If this second complete source has the same recovery property, put $h_2 = (I - \mathcal{L}_0)^{-1} q_2$.

Assume, finally, that the complete source expansion has the strong remainder

$$(73) \quad \|(I - \mathcal{L}_s)(h_0 + sh_1 + \tfrac{1}{2}s^2 h_2) - g_s\|_{\mathfrak{D}_U} = o(s^2),$$

after the two certified resets. Then

$$(74) \quad h_s = h_0 + sh_1 + \tfrac{1}{2}s^2 h_2 + o_{\mathfrak{D}_U}(s^2),$$

and, for a C^2 transported observable f_s , the correlation susceptibility (64) is twice differentiable with

$$\begin{aligned} \Xi'_0 &= \langle h_1, f_0 \rangle + \langle h_0, f_1 \rangle, \\ \Xi''_0 &= \langle h_2, f_0 \rangle + 2\langle h_1, f_1 \rangle + \langle h_0, f_2 \rangle. \end{aligned}$$

Conversely, strong twice differentiability of the canonical primitive forces (71)–(72) to be centered strong identities. Hence these two assembled sources are the exact second-order triangular test for a non-invariant correlation density; they cannot be replaced by eventwise all-seed recovery.

Proof. Differentiate the Poisson equation $(I - \mathcal{L}_s)h_s = g_s$ once and twice. The resulting identities are

$$(I - \mathcal{L}_0)h_1 = q_1, \quad (I - \mathcal{L}_0)h_2 = q_2,$$

which proves necessity and the formulae for the canonical jets. For sufficiency let $\tilde{h}_s = h_0 + sh_1 + s^2 h_2/2$. The uniform centered strong resolvent bound and (73) give

$$\|h_s - \tilde{h}_s\|_{\mathfrak{D}_U} \leq C_{\text{gap}} \|(I - \mathcal{L}_s)\tilde{h}_s - g_s\|_{\mathfrak{D}_U} = o(s^2),$$

which is (74). Differentiate the transported pairing $\Xi_s = \langle h_s, f_s \rangle$. The qualification about complete assembly is essential: if the one-face or point sector of q_1 or q_2 survives, it is not a strong source and the displayed resolvent step is unavailable. $\square$

Proposition 3.17 (Moving-face gluing obstruction for Poisson response). *Let $s \mapsto g_s$ be a centered strong C^1 source family and let*

$$h_0 = (I - \mathcal{L}_0)^{-1} g_0.$$

Fix an isolated moving physical face $\Gamma_s = \{G_s = 0\}$ on which the two branch formulae are $\mathcal{A}_{+,s}$ and $\mathcal{A}_{-,s}$. Assume no other face current has the same labelled support on the chosen open subarc. The singular part of the first Poisson source $q_1 = g_1 + B_1 h_0$ on that subarc is

$$(75) \quad \dot{G}_0 [\mathcal{A}_{+,0} h_0 - \mathcal{A}_{-,0} h_0]_{\Gamma_0} \delta(G_0)$$

up to the fixed smooth coarea factor and orientation. Hence strong differentiability of the canonical primitive $h_s = (I - \mathcal{L}_s)^{-1} g_s$ forces the gluing condition

$$(76) \quad \dot{G}_0 [\mathcal{A}_{+,0} h_0 - \mathcal{A}_{-,0} h_0]_{\Gamma_0} = 0$$

on every isolated moving face. If it fails on one open subarc, then q_1 is not strong and the strong derivative $\partial_s h_s|_0$ does not exist.

Suppose at least one moving face has $\dot{G}_0 \neq 0$ and its two branch preimages are distinct. On any strong source space for which the reduced resolvent and the two branch trace maps are continuous, the sources satisfying all conditions (76) form a closed proper linear subspace. Thus first-order strong Poisson response is nongeneric among strong correlation sources, even though every centered strong source has a fixed-table primitive.

Proof. The strong jet g_1 has no transverse current. Differentiating the two-sided branch indicator gives exactly the jump term (81) with $h = h_0$, which is (75). An isolated nonzero transverse current is weak but not strong by Proposition 4.10; it therefore cannot be cancelled by the regular part of g_1 . The necessity of (76) follows also by differentiating the Poisson equation, which would require $(I - \mathcal{L}_0)\partial_s h_s|_0 = q_1$ in the strong space.

The gluing map

$$g_0 \mapsto \dot{G}_0 [\mathcal{A}_{+,0} - \mathcal{A}_{-,0}](I - \mathcal{L}_0)^{-1} g_0|_{\Gamma_0}$$

is continuous and linear on the stipulated trace domain, so its kernel is closed. It is not the zero map: choose a centered smooth strong h whose values at the two distinct branch preimages differ on a small subarc and put $g = (I - \mathcal{L}_0)h$. Its canonical primitive is h , and the displayed jump is nonzero. The joint kernel over all moving faces is therefore a proper closed subspace. A proper closed subspace has empty interior, proving the nongenericity assertion. $\square$

Corollary (The prescribed correlation-density test). *For the radial-inflation family, let*

$$g_s = a_s \rho_s - \langle \rho_s, a_s \rangle \rho_s$$

with $a_s \rho_s$ a transported strong correlation density, and let $h_0 = (I - \mathcal{L}_0)^{-1} g_0$. Before any second-order calculation is available, strong differentiability requires (76) on every radial moving face. If this test passes, the second-order problem is exactly the strong recovery of q_2 in (72), including its one-face and point-current coefficients. Therefore a natural radial correlation density cannot be certified merely from the strong regularity of a_s ; its Poisson primitive must satisfy the branch-gluing identities generated by the moving singularities.

Theorem 3.18 (Exact observable-current benchmark with complete finite-order recovery). *Let a_s be a transported C^J family of physical collision observables such that*

$$(77) \quad h_s := (a_s - \langle \rho_s, a_s \rangle) \rho_s$$

is a centered C^J strong density section. Define its one-collision observable current by

$$(78) \quad g_s := (I - \mathcal{L}_s) h_s.$$

Equivalently, on the smooth invertible core, g_s is the invariant-flux density of the discrete observable increment $a_s - a_s \circ T_s^{-1}$. It is nonzero whenever this increment is nonzero.

Write $B_\ell = \partial_s^\ell \mathcal{L}_s|_0$, $g_j = \partial_s^j g_s|_0$, and $h_j = \partial_s^j h_s|_0$. At every $1 \leq j \leq J$, the complete assembled Poisson source is the strong identity

$$(79) \quad q_j := g_j + \sum_{\ell=1}^j \binom{j}{\ell} B_\ell h_{j-\ell} = (I - \mathcal{L}_0) h_j.$$

Consequently every represented one-face and clean point-current coefficient in q_j cancels. In particular the moving-face gluing test and the full second-order source q_2 are satisfied, with canonical jets h_1, h_2 .

For every transported centered test observable f_s ,

$$(80) \quad \sum_{n \geq 0} \langle g_s, \mathcal{U}_s^n f_s \rangle = \langle h_s, f_s \rangle,$$

and the left side therefore has the full C^J Leibniz response of the right side. Applied to the nonconjugate radial-inflation family, this is a non-invariant exact correlation-current class which passes all finite-order represented recovery tests without eventwise all-seed (S3).

The theorem is not a prescribed-source assertion: the source is an exact one-collision current by construction. It supplies an explicit consistency class, while an independently specified correlation density remains subject to the gluing obstruction of Proposition 3.17.

Proof. Differentiate $(I - \mathcal{L}_s)h_s = g_s$ exactly j times. Moving all terms in which a derivative hits $\mathcal{L}_s$ to the right gives (79). Its right-hand side is a centered strong density. The strong-recovery criterion then forces all represented conormal and point coefficients of the complete left-hand assembly to cancel. For $j = 1, 2$ the identity is precisely

$$q_1 = g_1 + B_1 h_0 = (I - \mathcal{L}_0)h_1, \quad q_2 = g_2 + 2B_1 h_1 + B_2 h_0 = (I - \mathcal{L}_0)h_2.$$

For finite N , duality gives the telescoping identity

$$\sum_{n=0}^N \langle (I - \mathcal{L}_s)h_s, \mathcal{U}_s^n f_s \rangle = \langle h_s, f_s \rangle - \langle h_s, \mathcal{U}_s^{N+1} f_s \rangle.$$

The last term tends to zero geometrically because h_s and f_s are centered and the period-one strong spectral gap is uniform. This proves (80); differentiating the finite strong pairing gives the asserted C^J formula. $\square$

Prescribed-source boundary. On the centered strong space at a fixed table, the spectral gap makes $I - \mathcal{L}_s$ invertible, so every centered strong source is formally a coboundary with its canonical Poisson primitive. The meaningful distinction is therefore not “coboundary versus non-coboundary” at one parameter value, but whether the source section g_s is specified independently of that primitive and whether the canonical primitive varies strongly with s . For such a prescribed section, the first two necessary and sufficient tests are exactly q_1, q_2 in Proposition 3.16; the first test contains the proper closed gluing condition of Proposition 3.17. Theorem 3.15 supplies one nonzero invariance-generated benchmark. Differentiated invariance gives its primitive and the total-current q_1, q_2 identities, hence the direct telescoping C^2 susceptibility. It does not prove the label-resolved terminal reset or the strong remainder required for the full criterion, and it does not show that a generic independently prescribed local correlation density passes the gluing condition. The exact-current theorem above remains a separate benchmark.

Proposition 3.19 (Eventwise all-seed recovery already fails at first order). *Let $\Gamma_s = \{g_s = 0\}$ be a moving physical face separating two one-sided branch formulae $\mathcal{A}_{+,s}$ and $\mathcal{A}_{-,s}$. On the smooth core, the complete first derivative contains the transverse current*

$$(81) \quad \dot{g}_0 [\mathcal{A}_{+,0}h - \mathcal{A}_{-,0}h]_{\Gamma_0} \delta(g_0),$$

up to the fixed orientation and smooth coarea factor. If this coefficient is nonzero on an open subarc, then the total current represented by $B_1 h$ is not in the strong Demers–Zhang space and has no strong primitive.

Consequently, whenever $\dot{g}_0 \not\equiv 0$ and the two branch traces are distinct, there are smooth strong seeds h for which eventwise recovery fails. Thus no all-seed version of (S3) can hold for a genuinely moving branch face. The positive result in Theorem 3.7 works because all insertion positions and density-jet terms are assembled before recovery is tested.

Proof. Differentiate

$$H(g_s)\mathcal{A}_{+,s}h + H(-g_s)\mathcal{A}_{-,s}h.$$

The two indicator derivatives give (81); all smooth interior and Jacobian derivatives remain regular on the two sides and cannot cancel a nonzero isolated transverse current. Such a current is weak but not strong by Proposition 4.10. The represented recovery criterion Proposition 3.5 therefore excludes a strong primitive. If the two branch trace operators differ, a smooth seed supported where their difference is nonzero makes the coefficient nonzero. This proves the all-seed obstruction and the final comparison. $\square$

Proposition (Cubic-safe endpoint orders and the fourth-order obstruction). *Let a local moving cutoff contribution be $c_s(u, y)u^3H(u - \eta_s(y))$, with c_s smooth and $\eta_0 = 0$. For parameter orders*

$j \leq 3$, every endpoint delta derivative produced by the cutoff is annihilated by u^3 . At order four the leading endpoint term is

$$(82) \quad -6(\dot{\eta}_0)^4 c_0(0, y) \delta(u)$$

up to the fixed orientation convention; lower terms contain only derivatives of c_s multiplying $u^3 \delta^{(q)}$, $q \leq 2$, and vanish. Thus the cubic flux zero verifies the cutoff endpoint part through order three, whereas order four and above require an actual Ward identity or an additional zero of the numerator.

Proof. The highest distribution in $\partial_s^j H(u - \eta_s)$ is $(-\dot{\eta}_0)^j \delta^{(j-1)}(u)$; all other endpoint terms have lower delta order. The identities $u^3 \delta^{(q)} = 0$ for $q = 0, 1, 2$ prove the first claim. At $j = 4$, the top term is $(\dot{\eta}_0)^4 u^3 \delta^{(3)}(u) = -6(\dot{\eta}_0)^4 \delta(u)$. The product rule gives the stated lower terms. This is a complete local cutoff calculation and makes no claim about the separate regular coefficient $F_j^{\text{reg}}(0)$. $\square$

Theorem (First-order deterministic observable regularity). *Let f_s be a C^1 transported observable section whose centered value $f_0 - \langle \rho_0, f_0 \rangle$ belongs to the adjoint trace class in Definition 4.1. Then the first-order source $g = (\nabla_s \mathcal{L}_{\theta_s}|_0) \rho_0$ defines the scalar functional $\langle g, R_{\theta_0}^{\text{obs}} f_0 \rangle$ in (25). No element $(I - \mathcal{U}_{\theta_0})^{-1} \Pi_{\theta_0}^{\text{obs}, \perp} f_0$ in a positive trace space is introduced. The finite-order paired derivatives are proved separately in Proposition 4.17.*

Theorem 3.20 (First-order moving-singularity response). *Let θ_s be a C^1 curve in the finite-response parameter data, let ρ_0 be the invariant collision density at $s = 0$, and let f_s be a C^1 transported observable section satisfying the adjoint trace condition in Definition 4.1. Let the first transfer derivative from Proposition 3.3 be*

$$g = \nabla_s \mathcal{L}_{\theta_s}|_0 \rho_0$$

with branch/interior plus physical-trace decomposition

$$g = \sum_h \mathcal{L}_{0,h} (\mathcal{D}_h^{\text{int}} + \mathcal{D}_h^{\text{Jac}}) \rho_0 + \sum_{\Gamma \subset S} \mathcal{B}_\Gamma \rho_0$$

in the weak trace range. Then $I(s) = \langle \rho_s, f_s \rangle$ is differentiable at zero and

$$(83) \quad I'(0) = \langle \rho_0, \nabla_s f_s|_0 \rangle + \langle g, R_0^{\text{obs}} f_0 \rangle.$$

This value equals the first-order geometric flux derivative in (4); the transfer expression is a consistency representation of a static average. The second pairing is the trace pairing supplied by the target observable class. In particular, the theorem does not assert $(I - \mathcal{L}_0)^{-1} g \in \mathfrak{D}_U$. Nevertheless this complete invariant source is recovered in the precise sense $g = (I - \mathcal{L}_0) \partial_s \rho_s|_0$ by Theorem 3.4; no such statement is made for an isolated trace summand.

Proof. Direct differentiability follows already from Proposition 2.1. Put $h = \partial_s \rho_s|_0$. The exact invariant-source Ward identity gives $g = (I - \mathcal{L}_0)h$, with (h) centered and strong. Hence, for $f_0^\circ = \Pi_0^{\text{obs}, \perp} f_0$, duality gives the scalar identity

$$\sum_{n=0}^{N-1} \langle g, \mathcal{U}_0^n f_0^\circ \rangle = \langle h, f_0^\circ \rangle - \langle \mathcal{L}_0^N h, f_0^\circ \rangle.$$

The ordinary Demers–Zhang strong spectral gap sends the second term to zero geometrically. This proves existence of the scalar functional $\langle g, R_0^{\text{obs}} f_0 \rangle$, without convergence of the partial sums in any observable jet norm, and identifies it with $\langle h, f_0^\circ \rangle$. Adding the transported-observable derivative gives (83). Total mass of g is zero because $\langle 1, \mathcal{L}_s \rho_0 \rangle = \langle 1, \rho_0 \rangle$, so the centering is compatible with the cohomological equation. Both expressions were obtained by differentiating the same finite-time invariance identity, so their equality with (4) follows before the $N \rightarrow \infty$ limit. $\square$

Definition 3.21 (First-order weak derivative range). Let $\mathfrak{W}_U^{\text{der}} \subset \mathfrak{D}_U^w$ be the closure of finite sums of weak inputs generated from strong response densities by:

- (i) branch-interior, Jacobian, and roof derivatives of the homogeneous transfer family;
- (ii) first moving-boundary trace operators defined by (33);

(iii) finite linear combinations of such terms after zero-mean projection and finite homogeneous refinement.

Elements of $\mathfrak{W}_U^{\text{der}}$ carry their terminal-child decomposition, dyadic physical-block data, and unstable-holonomy matching data as part of the norm. More precisely, for a smooth-core generator g write $g = g_{\text{cen}} + g_{\text{term}}$, where g_{cen} is supported on central homogeneous descendants and g_{term} on high-strip, grazing, or multiple-impact terminal descendants. Set

$$\|g\|_{\mathfrak{W}_U^{\text{der}}} = \|g_{\text{cen}}\|_{\mathfrak{D}_U^w} + \|g_{\text{term}}\|_{\mathfrak{W}_U^{\text{term}}},$$

where the terminal seminorm is the first-order physical-block stable-leaf testing norm with the factor $2^{\epsilon_1 m}$. For a finite sum, take the infimum of the sum of these generator norms over all such decompositions, and complete. The response calculus only uses weak inputs in $\mathfrak{W}_U^{\text{der}}$; no weak-to-strong smoothing is asserted for an arbitrary element of $\mathfrak{D}_U^w$.

Proposition 3.22 (Observable-side trace insertion). *Let the complete first-order branch/interior plus physical-trace source satisfy $B\rho_0 \in \mathfrak{W}_U^{\text{rec}}$, and let f satisfy the adjoint trace hypothesis in Definition 4.1. Then*

$$\langle B\rho_0, R_0^{\text{obs}} f \rangle$$

is well defined and bounded by the first-order source norm times the target observable norm. The assertion is a dual pairing statement. It neither implies nor uses

$$(I - \mathcal{L}_0)^{-1} \Pi_0^{\text{den}, \perp} B\rho_0 \in \mathfrak{D}_U.$$

In particular, Kato's one-space resolvent identity is not applied to the strong-to-weak derivative $\partial_\theta \mathcal{L} : \mathfrak{D}_U \rightarrow \mathfrak{D}_U^w$, and no higher projector/resolvent derivative is asserted. An isolated trace which belongs only to $\mathfrak{W}_U^{\text{der}}$ is not covered; recovery is a property of the fully assembled physical source.

4. CERTIFIED FINITE-ORDER CONORMAL CALCULUS

Definition 4.1 (Response Banach scale). Let $\mathcal{C}_{\text{hom}}^r(U)$ be the transported branchwise smooth core. For $0 \leq m \leq J + 3$, define $\mathfrak{D}_U^{(m)}$ as the completion of this core for the independent flux-seed norm of order $m + 2$:

$$(84) \quad \|\rho\|_{\mathfrak{D}_U^{(m)}} = \mathcal{N}_{m+2}^{\text{flux}}(\rho).$$

Thus $\mathfrak{D}_U^{(1)} = \mathfrak{D}_U$. Conormal letters are defined on the core by the resolved indicator calculus in Theorem 4.14, combined into physical dyadic blocks, and bounded on every finite decorated core by Lemma 2.19; their completed countable-atlas extension is the independent mapping statement (S1), not a consequence of the definition. Their bounds are not inserted in (84). The weak density scale $\mathfrak{D}_U^{w, (m)}$ uses the corresponding Demers–Zhang stable-leaf tests and one fewer unstable comparison. It contains the regular and transverse one-face outputs for which those tests are defined; it is not enlarged by fiat to contain codimension-two point masses.

For a branchwise smooth observable define the independent one-sided jet/ holonomy seminorm

$$\mathcal{H}_m(f) = \sup_{\Gamma} \max_{0 \leq \ell \leq m} \|\partial_{n_\Gamma}^\ell f|_{\Gamma^\pm}\|_{C^\alpha} + \sup_{(W_1, W_2, \mathfrak{h}^u)} \frac{\|J_m f|_{W_1} - (J_m f|_{W_2}) \circ \mathfrak{h}^u\|_{C^0}}{d(W_1, W_2)^\beta},$$

where $J_m f$ is the vector of one-sided normal/tangential jets of order at most m , Γ ranges over the labelled physical faces of the clean stratification, and $\mathfrak{h}^u$ is the local unstable holonomy between matched stable test curves. The adjoint trace scale $\mathfrak{A}_U^{(m)}$ is the completion of physical branchwise C^r observables for the independent norm

$$(85) \quad \|f\|_{\mathfrak{A}_U^{(m)}} = \|f\|_{\mathcal{B}^\#} + \|f\|_{C_{\text{tr}}^m} + \mathcal{H}_m(f).$$

The norm contains no supremum over derivative-generated sources. Pairing with such sources is a theorem consequence of the physical block estimate. The embeddings are

$$\mathfrak{D}_U^{(m+1)} \hookrightarrow \mathfrak{D}_U^{(m)} \hookrightarrow \mathfrak{D}_U^{w, (m)}, \quad \mathfrak{A}_U^{(m+1)} \hookrightarrow \mathfrak{A}_U^{(m)}.$$

No embedding $\mathfrak{D}_U^{w, (m)} \hookrightarrow \mathfrak{D}_U^{(m)}$ is postulated.

Definition 4.2 (Ambient decorated conormal source scale). Let $\mathcal{G}_m^{\text{at}}$ be the countable set of *atomic* decorated supports in the fixed one-step response atlas. An index $\gamma \in \mathcal{G}_m^{\text{at}}$ records the homogeneous chart, dyadic block, physical face or central clean face pair, one-sided incident branch ω , and derivative order. There are three types:

$$\gamma = \text{reg}, \quad \gamma = (\Gamma, \omega, \ell), \quad \gamma = (z, \omega, \alpha).$$

Here Γ is a one-face germ; by (6), every clean pair point z lies in the central region $u \geq u_J$. A regular decorated point is retained only when it is the one-sided image of such a central point. Nonextending grazing and multiple-impact germs carry a terminal symbol and no fictitious map value.

There is one additional physically canonical presentation. For a physical face family Γ , an incident label ω , and a dyadic block b , put

$$\hat{\Gamma}_{\Gamma, \omega, b} = \bigcup_{2^b \leq k < 2^{b+1}} \Gamma_{k, \omega},$$

where the union is taken only across artificial homogeneity cuts, after the two artificial-face orientations have cancelled. Let $\hat{\mathcal{G}}_0$ contain the grouped indices $\hat{\gamma} = (\hat{\Gamma}_{\Gamma, \omega, b}, \omega, \ell)$ for which the coefficient jets have matching one-sided traces at every artificial cut. The coefficient space $E_{\hat{\gamma}, 0}$ is the completion of those matching tuples for the physical block-variation norm: the L^1 norms and total variations of the concatenated rescaled coefficient jets on $\hat{\Gamma}_{\Gamma, \omega, b}$. Its current map $q_{\hat{\gamma}, 0}$ integrates the concatenated coefficient against the one-sided trace jet on the whole physical block. A common refinement merely subdivides the same variation measure, so this grouped coefficient space does not depend on the artificial subdivision. At level m , the same notation $E_{\hat{\gamma}, m}$ uses concatenated jets through order $m - \ell$.

At a higher source level $m > 0$, a grouped index is admitted to $\hat{\mathcal{G}}_m$ only when its actual one-sided chain-rule image has a finite weighted column sum into level $m - 1$. A block whose later image undergoes an uncontrolled countable splitting is therefore available at base level but is not declared to be a propagating higher-level generator. Set

$$\mathcal{G}_m = \mathcal{G}_m^{\text{at}} \sqcup \hat{\mathcal{G}}_m.$$

This makes the one-event regrouping below an element of the Banach module, while leaving recursive regrouping as the separate content of (S3).

Let $E_{\text{reg}, m} = \mathfrak{D}_U^{w, (m)}$, let $E_{\Gamma, \omega, \ell} = C^{m-\ell}(\Gamma, \omega)$, and let $E_{z, \omega, \alpha} = \mathbb{C}$. The canonical current map $q_{\gamma, m} : E_{\gamma, m} \rightarrow (\mathfrak{A}_U^{(m+2)})^*$ is respectively the ordinary weak pairing, the one-sided conormal derivative, or the decorated point derivative; for $\gamma = \hat{\gamma}$ it is the grouped physical block current just defined. Write $A_{\eta\gamma}^{(m)} : E_{\gamma, m+1} \rightarrow E_{\eta, m}$ for the finite one-sided chain-rule coefficient from γ to an ordinary or terminal output index η . For $\gamma = \text{reg}$, this notation means the cited one-step Demers–Zhang weak operator; for a nonextending germ the only nonzero coefficient is its terminal copy. Each atomic nonregular index already contains its one-sided target-strip refinement, so its local chain-rule column has finitely many outputs. For an admitted grouped index the column is its actual, possibly countable, physical image column and is retained at level $(m+1)$ only when the weighted sum in (86) is finite. Summation over the full countable family of input columns is not used to prove (S1); that all-branch estimate remains the separate response-scale certificate.

Choose weights recursively. At level zero, for both atomic and grouped indices, put

$$w_{\gamma, 0} = 2^{\epsilon_* b(\gamma)} \max\{1, \|q_{\gamma, 0}\|\},$$

and give a terminal index the same base weight with $q_{\gamma, 0} = 0$. Terminal weights at higher levels are obtained by the same rule, with the identity copy as their only coefficient. After the ordinary and terminal weights at level m have been fixed, put

$$(86) \quad w_{\gamma, m+1} = 2^{\epsilon_* b(\gamma)} \max\{1, \|q_{\gamma, m+1}\|\} + \sum_{\eta} w_{\eta, m} \|A_{\eta\gamma}^{(m)}\|.$$

Give $E_{\gamma, m}$ the weighted norm

$$(87) \quad \|e\|_{\gamma, m}^{\text{src}} := w_{\gamma, m} \|e\|_{E_{\gamma, m}},$$

where $b(\gamma)$ is the dyadic grazing-block index (zero for central point indices). The norm of $q_{\gamma,m}$ includes the physical coarea Jacobian and the one-sided coordinate-change matrix. It is finite by the central determinant bound and the trace-test norm. By construction,

$$(88) \quad \sum_{\eta} w_{\eta,m} \|A_{\eta\gamma}^{(m)}\| \leq w_{\gamma,m+1}.$$

Form the Banach ℓ^1 -sum over both presentations,

$$(89) \quad \mathbb{E}_m = E_{\text{reg},m} \oplus_1 \bigoplus_{\gamma \in \mathcal{G}_m \setminus \{\text{reg}\}}^{\ell^1} E_{\gamma,m}, \quad e = (e_{\gamma}).$$

Let Ω_m be the countable set of one-sided branch labels, augmented by the regular label, and let $\mathcal{G}_m^{\text{term}}$ be the terminal indices. Put

$$\mathfrak{T}_U^{(m)} = \bigoplus_{\tau \in \mathcal{G}_m^{\text{term}}}^{\ell^1} E_{\tau,m}$$

with the weights (87). Define the label-resolved realization space

$$\mathbb{Y}_m = \bigoplus_{\omega \in \Omega_m}^{\ell^1} (\mathfrak{A}_U^{(m+2)})^* \oplus_1 \mathfrak{T}_U^{(m)}$$

and the joint realization map

$$(90) \quad \mathbf{Q}_m e = \left(\left(\sum_{\substack{\gamma: \omega(\gamma)=\omega \\ \gamma \notin \mathcal{G}_m^{\text{term}}}} q_{\gamma,m} e_{\gamma} \right)_{\omega \in \Omega_m}, (e_{\tau})_{\tau \in \mathcal{G}_m^{\text{term}}} \right).$$

Each series converges absolutely. Retaining the branch label is essential: two currents which cancel only after summing different one-sided branches can have different future images and are not identified in the dynamical source module. Terminal coefficients are likewise not sent to zero. Define

$$(91) \quad \mathfrak{X}_U^{(m)} = \mathbb{E}_m / \ker \mathbf{Q}_m, \quad \|[e]\|_{\mathfrak{X}_U^{(m)}} = \inf_{k \in \ker \mathbf{Q}_m} \|e + k\|_{\mathbb{E}_m}.$$

Since $\mathbf{Q}_m$ is continuous into the Hausdorff Banach space $\mathbb{Y}_m$, its kernel is closed; hence $\mathfrak{X}_U^{(m)}$ is a Banach space. The quotient removes representation ambiguity within each one-sided label without discarding dynamically relevant cross-label or terminal data. In particular, if a matching atomic strip tuple and its grouped block coefficient represent the same physical current, then

$$((c_{\Gamma,k,\omega})_k, -\widehat{c}_{\Gamma,\omega,b}) \in \ker \mathbf{Q}_m.$$

Thus it is the within-label quotient, rather than the scalar total-current map, which identifies the two presentations. We use the same symbol $\mathbf{Q}_m$ for the induced injective realization of the quotient in $\mathbb{Y}_m$. The bounded total-current map

$$S_m : \mathbb{Y}_m \longrightarrow (\mathfrak{A}_U^{(m+2)})^*, \quad S_m((y_{\omega})_{\omega}, t) = \sum_{\omega} y_{\omega},$$

forgets the terminal coordinate only at the final scalar pairing. The canonical regular embedding is

$$\iota_{\text{reg}} : \mathfrak{D}_U^{w,(m)} \longrightarrow \mathfrak{X}_U^{(m)}, \quad \iota_{\text{reg}} g = [(g, 0, 0, \dots)].$$

Call a face-preserving chart change *admissible* when its triangular jet matrices and their inverses satisfy the two weighted column-sum bounds obtained from (88). The uniform C^{J+2} rescaled physical chart changes and their common homogeneous refinements are admissible: their triangular jet matrices and inverses are uniformly bounded, the central coarea determinants stay bounded below, and the dyadic indices change by at most a fixed amount. On grouped indices a common refinement is an isometry of the physical block-variation measure before the uniformly bounded triangular jet change. The induced maps on the two coefficient ℓ^1 -sums are then bounded inverses, preserve every labelled current and terminal symbol, and descend to equivalent quotient norms.

Thus the construction is independent, up to equivalent norm, of the admissible response atlas and is not a supremum over derivative-generated sources. Write $\mathfrak{X}_{U,h}^{(m)}$ for the closed quotient image of coefficients carrying the one-sided branch label h , and write $\pi_{\text{term}} : \mathfrak{X}_U^{(m)} \rightarrow \mathfrak{T}_U^{(m)}$ for the bounded terminal-coordinate projection induced by $\mathbf{Q}_m$.

Proposition 4.3 (Point atoms require the ambient source scale). *On the finite simple atlas:*

(i) *the pairing extends continuously as*

$$\mathfrak{X}_U^{(m)} \times \mathfrak{A}_U^{(m+2)} \rightarrow \mathbb{C};$$

(ii) *a nonzero codimension-two point atom, and a fortiori any of its derivatives, is not in the ordinary Demers–Zhang weak stable-leaf completion;*

(iii) *each central decorated one-step branch on which the one-sided billiard map extends C^r has a bounded pushforward*

$$(92) \quad \mathcal{P}_{h*} : \mathfrak{X}_{U,h}^{(m+1)} \rightarrow \mathfrak{X}_U^{(m)},$$

The direct sum over the one-step central branches is bounded by (88). A terminal germ is recorded in the component detected by π_{term} and is not assigned a further image. Iteration along a fixed regular decorated itinerary is therefore bounded with one source-level loss per step, but no decay or global $\mathfrak{X}$ -semigroup is asserted;

(iv) *ordinary density mixing does not imply decay of centered point-source correlations. If z is a regular periodic point and $G_z = \delta_{z,\omega} - \rho_\theta$, then a smooth centered f can be chosen so that $\langle G_z, \mathcal{U}_\theta^{nq} f \rangle$ stays nonzero along the period q .*

Consequently a scalar resolvent containing point atoms requires an independently verified recovered/cancellation property; it cannot be deduced from the Demers–Zhang strong spectral gap. Moreover, no recovered class satisfying (S2) can contain every centered point atom, and (S3) cannot hold uniformly on the whole point-current sector of $\mathfrak{X}_U^{(m)}$.

Proof. The definition of the local weights makes every $q_{\gamma,m}$ contractive up to the fixed atlas constant. Absolute convergence in the ℓ^1 -sum and composition with the total-current map $\mathbf{S}_m$ prove (i), including independence of representative. Terminal coordinates do not enter this scalar pairing. For (ii), let $\delta_{z,\varepsilon}$ be a two-dimensional mollification of a point mass. Choose an admissible stable test leaf in one incident component crossing the $O(\varepsilon)$ ball about z . Then, for a test equal to one there,

$$\int_W \delta_{z,\varepsilon} dm_W \asymp \varepsilon^{-1}.$$

Hence the defining weak stable-leaf norm diverges; derivatives diverge faster. For (iii), a propagating decorated branch is a C^r diffeomorphism on its one-sided germ. The distributional chain rule sends face jets to face jets and point jets to point jets, with a loss of one test-jet level and coefficient matrices $A_{\eta\gamma}^{(m)}$. Therefore

$$\sum_{\eta} w_{\eta,m} \left\| \sum_{\gamma} A_{\eta\gamma}^{(m)} e_{\gamma} \right\| \leq \sum_{\gamma} \sum_{\eta} w_{\eta,m} \|A_{\eta\gamma}^{(m)}\| \|e_{\gamma}\| \leq \sum_{\gamma} w_{\gamma,m+1} \|e_{\gamma}\|.$$

This proves the direct one-step bound before passing to the quotient. Let $\mathbb{A}_m$ denote this coefficient map. On an extending labelled germ, the distributional chain rule is an intertwining identity between $\mathbb{A}_m$ and pushforward of the labelled current; on a nonextending germ it intertwines with the identity embedding into the corresponding terminal coordinate. Hence there is a bounded label-resolved map $\mathbf{P}_m$ on the realized range such that

$$\mathbf{Q}_m \mathbb{A}_m = \mathbf{P}_m \mathbf{Q}_{m+1}.$$

In particular $\mathbb{A}_m(\ker \mathbf{Q}_{m+1}) \subset \ker \mathbf{Q}_m$, which is the kernel-invariance needed to descend to the quotient. This proves (92) without inventing a billiard value on the singularity. Finally choose a regular periodic orbit disjoint from the singular set and a smooth f supported near one orbit point, then subtract its invariant mean. Evaluation on every q -th iterate proves (iv). If $G_z = (I - \mathcal{L}_\theta)h$ for a centered $h \in \mathfrak{D}_U^{(m+1)}$, the ordinary strong spectral gap and duality would force $\langle G_z, \mathcal{U}_\theta^n f \rangle \rightarrow 0$

exponentially for every centered smooth f , contradicting (iv). Hence G_z has no strong primitive of the form required in (S2), which proves the final assertion. $\square$

Proposition 4.4 (Scalar physical tails cannot imply all-branch (S1)). *The total-current map $S_m : \mathbb{Y}_m \rightarrow (\mathfrak{A}_U^{(m+2)})^*$ does not control the label-resolved source norm. More precisely, suppose a physical face has two incident labels $\omega_+ \neq \omega_-$ and, on infinitely many disjoint dyadic blocks b , the two labelled realization ranges contain the same nonzero current v_b . For every finite interval $M \leq b \leq K$, the labelled element*

$$y_{M,K} = \sum_{b=M}^K (v_b \mathbf{e}_{\omega_+} - v_b \mathbf{e}_{\omega_-})$$

satisfies $S_m y_{M,K} = 0$, although after normalizing $\|v_b\| = 1$,

$$\|y_{M,K}\|_{\mathbb{Y}_m} = 2(K - M + 1).$$

Likewise, S_m is identically zero on the retained terminal coordinate. Hence even a zero scalar physical tail gives no Cauchy estimate in $\mathfrak{X}_U^{(m)}$.

Consequently the scalar 2^{-b} estimate can prove neither the countable-atlas derivative map nor terminal-module closure. A sufficient all-branch input must instead be vector valued, for example a bound of the form

$$(93) \quad \sum_{\gamma: b(\gamma)=b} w_{\gamma,m} \|c_\gamma(B_{j,\theta}\rho)\|_{E_{\gamma,m}} \leq C 2^{-\delta b} \|\rho\|_{\mathfrak{D}_U^{(m+j+2)}}$$

after the admissible within-label quotient minimization, with $\delta > 0$. Establishing such a bound for the actual derivative coefficients is the unresolved analytic content of (S1).

Proof. The two label components are distinct coordinates in the ℓ^1 -sum $\mathbb{Y}_m$, so their norms add even though their images under S_m cancel. Finite partial sums therefore have the displayed norm and are not Cauchy as $K \rightarrow \infty$, whereas every total-current pairing is zero. The same conclusion for terminal data is immediate from the definition of S_m . Since the induced realization $\mathbf{Q}_m : \mathfrak{X}_U^{(m)} \rightarrow \mathbb{Y}_m$ is continuous, an $\mathfrak{X}_U^{(m)}$ -Cauchy estimate would imply a $\mathbb{Y}_m$ -Cauchy estimate, contradicting this construction. Summing (93) over b , on the other hand, does give an absolutely convergent coefficient lift and hence the desired source-module bound. This proves both claims. $\square$

Proposition (Dyadic entropy threshold for all-branch (S1)). *For one response level, let M_b be the number of nonzero labelled coefficient columns in grazing block b , let $W_b = \max_{\gamma: b(\gamma)=b} w_{\gamma,m}$, and suppose an actual derivative source has the within-label minimized coefficient bound*

$$\|c_\gamma(B_j \rho)\|_{E_{\gamma,m}} \leq A_b \|\rho\|_{\mathfrak{D}_U^{(m+j+2)}}.$$

Then the block contribution to (S1) is at most $M_b W_b A_b$. In particular, if

$$M_b \leq C 2^{\kappa b}, \quad W_b \leq C 2^{\mu b}, \quad A_b \leq C 2^{-\sigma b},$$

the vector-valued completion follows whenever $\sigma > \kappa + \mu$, with $\delta = \sigma - \kappa - \mu$ in (93).

For the standard quadratic homogeneity strips, one dyadic block $2^b \leq k < 2^{b+1}$ contains $\asymp 2^b$ strip indices before any within-label regrouping. Thus a proof based on separate strip columns has $\kappa = 1$. The scalar assembled 2^{-b} bound alone is at the borderline before the positive source weight μ is even paid and cannot verify (S1). A radial-inflation proof must therefore obtain either additional coefficient decay, or an explicit within-label physical regrouping that lowers the effective column entropy. This is the exact quantitative alternative left by (S1), rather than an unspecified “all-branch constant.”

Proof. The first assertion is the triangle inequality in the weighted ℓ^1 -sum after minimizing within each label. The three displayed dyadic estimates give

$$\sum_{\gamma: b(\gamma)=b} w_{\gamma,m} \|c_\gamma(B_j \rho)\| \leq C 2^{-(\sigma - \kappa - \mu)b} \|\rho\|.$$

The number of integers in $[2^b, 2^{b+1})$ is comparable with 2^b , which proves the quadratic-strip statement. No lower bound on an actual billiard coefficient is asserted; the proposition identifies the decay or regrouping that a positive proof must supply. $\square$

Proposition 4.5 (Within-label physical regrouping removes the artificial strip entropy). *Fix one response event and one of the finitely many physical one-step face families supplied by (R1)–(R2). For a face Γ , an incident label ω , and a dyadic block b , let $\{\Gamma_{k,\omega} : 2^b \leq k < 2^{b+1}\}$ be the restrictions produced only by the artificial homogeneity-strip subdivision. Suppose that, after the artificial-face partition-of-unity terms have been cancelled, the one-sided physical coefficients have matching traces on every common artificial cut. Then they concatenate, in the distributional sense, to one labelled block current*

$$(94) \quad \sum_{2^b \leq k < 2^{b+1}} q_{\Gamma_{k,\omega},m} c_{\Gamma_{k,\omega}} = \hat{q}_{\Gamma,\omega,b,m} \hat{c}_{\Gamma,\omega,b}.$$

At base level the right-hand side is the grouped generator $\hat{\gamma} \in \hat{\mathcal{G}}_0$ of Definition 4.2. The difference between the atomic tuple and this grouped representative lies in $\ker \mathbf{Q}_0$, and hence

$$(95) \quad \left\| \left[\sum_k c_{\Gamma_{k,\omega}} \right] \right\|_{\mathfrak{X}_U^{(0)}} \leq \hat{w}_{\Gamma,\omega,b,0} \|\hat{c}_{\Gamma,\omega,b}\|_{E_{\gamma,0}^{\sim}}.$$

At level $m > 0$ the same statement holds exactly when $\hat{\gamma} \in \hat{\mathcal{G}}_m$, namely when its actual recursive image column has the finite weighted sum required in the definition. This regrouping never identifies different incident labels and never removes a terminal coordinate.

Let F_* be the number of one-step physical face families, including their two incident labels. If the grouped currents in block b satisfy

$$(96) \quad \hat{w}_{\Gamma,\omega,b,m} \|\hat{c}_{\Gamma,\omega,b}\| \leq C 2^{-(\sigma-\mu)b} \|\rho\|_{\mathfrak{D}_U^{(m+j+2)}}$$

for all (Γ, ω) , with $\sigma > \mu$, then the one-event source-side part of (S1) follows by summing over b . In particular the effective column entropy of the grouped physical presentation is $\kappa_{\text{eff}} = 0$, rather than the artificial value $\kappa = 1$ from the separate quadratic-strip presentation.

For radial inflation, therefore, the unresolved estimate is no longer required to beat the 2^b number of artificial strips. It is enough to prove the same-label gluing in (94) and a strict grouped decay margin $\sigma > \mu$. Cross-label cancellation in the total physical current cannot be used for this purpose.

Proof. The restrictions of a distribution to adjacent artificial cells add to its restriction to their union. The derivative of the artificial partition of unity appears with opposite signs on the two cells, and the matching-trace hypothesis removes the only boundary term created by concatenation. This proves (94) within the fixed label. The matching atomic tuple and the grouped coefficient have the same component of the joint realization map. Their difference is therefore in $\ker \mathbf{Q}_0$, and choosing the grouped coefficient as the quotient representative proves (95). This is the step which a scalar total-current regrouping cannot supply. The no-boundary-birth atlas has finitely many one-step physical faces, so after this regrouping there are at most F_* labelled columns in every dyadic block. Consequently (96) gives

$$\sum_{\Gamma,\omega} \hat{w}_{\Gamma,\omega,b,m} \|\hat{c}_{\Gamma,\omega,b}\| \leq C F_* 2^{-(\sigma-\mu)b} \|\rho\|,$$

and the geometric sum over b converges. Since labels and terminal data were retained, this is a legitimate source-module lift rather than a scalar total-current cancellation. The statement concerns the initial complete event. A later propagation which splits a grouped current must satisfy the separate recursive column bound already built into (S3). $\square$

Definition (Level-compatible uniform block-flux graph domain). For a resolved radial grazing block b , let

$$F_{\theta,b}(\rho) = 1_{\mathcal{G}_b} u^3 a_{\theta}(u, s) \rho_{\theta}(u, s), \quad R_{\ell,\beta} = (u \partial_u)^\ell \partial_s^\beta.$$

Let $\mathcal{P}_b$ be a homogeneous subdivision, let $\mathcal{F}_b^{\text{phys}}$ be its labelled one-sided physical faces, and let $\mathcal{P}_b^{\text{cut}}$ be its artificial internal cuts. Define

$$\begin{aligned} \|H\|_{\widehat{\mathcal{V}}_b^p} := & \sum_{\ell+|\beta|\leq p} \left\{ \sum_{P\in\mathcal{P}_b} (\|R_{\ell,\beta}H\|_{L^1(P)} + |R_{\ell,\beta}H|_{BV(P)}) \right. \\ & + \sum_{\Gamma\in\mathcal{F}_b^{\text{phys}}} (\|J_\Gamma R_{\ell,\beta}H|_{\Gamma^\pm}\|_{L^1} + |J_\Gamma R_{\ell,\beta}H|_{BV}) \\ & \left. + \sum_{C\in\mathcal{P}_b^{\text{cut}}} \|R_{\ell,\beta}H|_{C^+} - R_{\ell,\beta}H|_{C^-}\|_{L^1(C)} \right\}. \end{aligned}$$

Here J_Γ is the physical coarea Jacobian and only same-label sides are compared at an artificial cut. The last line vanishes for the physically concatenated representative.

Let $\mathcal{W}_b^s$ be the standard stable test leaves in the block, $\mathcal{M}_b$ the matched triples $(W_1, W_2, \mathfrak{h}^u)$, and $\mathcal{U}_b$ the unmatched descendants from the standard coupling cut. With the same b_0, c_0, α, β and short-leaf scaling as in the Demers–Zhang norm, put

$$\begin{aligned} |H|_{w,b,p} &= \max_{\ell+|\beta|\leq p} \sup_{W\in\mathcal{W}_b^s, \|\psi\|_{C^\alpha(W)}\leq 1} \left| \int_W R_{\ell,\beta}H \psi \, dm_W \right|, \\ |H|_{hol,b,p} &= \max_{\ell+|\beta|\leq p} \sup_{(W_1, W_2, \mathfrak{h}^u)\in\mathcal{M}_b} \sup_{\|\psi_i\|_{C^\alpha}\leq 1} \frac{\left| \int_{W_1} R_{\ell,\beta}H \psi_1 \, dm - \int_{W_2} R_{\ell,\beta}H(\psi_2 \circ \mathfrak{h}^u) J_{\mathfrak{h}^u} \, dm \right|}{d(W_1, W_2)^\beta}, \\ |H|_{unm,b,p} &= \max_{\ell+|\beta|\leq p} \sum_{V\in\mathcal{U}_b} \sup_{\|\psi\|_{C^0(V)}\leq 1} \left| \int_V R_{\ell,\beta}H \psi \, dm_V \right|, \\ \|H\|_{\mathcal{W}_b^p} &= |H|_{w,b,p} + b_0 |H|_{hol,b,p} + c_0 |H|_{unm,b,p}. \end{aligned}$$

These are norms of the undifferentiated flux coefficient; neither contains $B_1\rho$.

For $0 \leq m \leq J+3$, set $p = m+2$ and define on the smooth finite-strip core

$$(97) \quad \|\rho\|_{\text{bfj}}^{(m)} := \mathcal{N}_{m+2}^{\text{flux}}(\rho) + \sup_{\theta,b} \left\{ 2^{3b} \|F_{\theta,b}(\rho)\|_{\widehat{\mathcal{V}}_b^p} + 2^b \|u^{-2} F_{\theta,b}(\rho)\|_{\mathcal{W}_b^p} \right\}.$$

Its completion is $\mathfrak{D}_{U,\text{bfj}}^{(m)}$. For two common refinements and two radial blown-up charts there is an atlas constant C_{at} , independent of b, θ , such that the corresponding norms differ by factors between C_{at}^{-1} and C_{at} . Indeed subdivision is additive in the variation measures, the artificial-cut jump occurs with both orientations, and the rescaled chart Jacobians and their inverses through order p are uniformly bounded; the standard two-way stable-leaf matching gives the weak/holonomy comparison.

Most importantly, the first term in (97) gives the level-correct embedding

$$(98) \quad \mathfrak{D}_{U,\text{bfj}}^{(m)} \hookrightarrow \mathfrak{D}_U^{(m)}, \quad \|\rho\|_{\mathfrak{D}_U^{(m)}} \leq \|\rho\|_{\text{bfj}}^{(m)}.$$

The graph domain contains the invariant flux jets (43). Its two extra block terms are essential: $\mathcal{N}_{m+2}^{\text{flux}}$ alone does not give either 2^{-3b} block-flux decay or 2^{-b} decay after multiplication by u^{-2} . No bounded action of $\mathcal{L}_\theta$ on this graph domain, and no return of a recovered first event to it, is asserted.

Theorem 4.6 (One-event radial grouped bound on the level-three graph domain). *Consider the radial-inflation family of Theorem 3.6. The base response letter has the bounded source-module realization*

$$B_1 : \mathfrak{D}_{U,\text{bfj}}^{(3)} \longrightarrow \mathfrak{X}_U^{(0)}.$$

After the same-label physical regrouping of Proposition 4.5, the singular one-face coefficient in block b satisfies

$$(99) \quad \widehat{w}_{\Gamma,\omega,b,0} \|\widehat{c}_{\Gamma,\omega,b}(B_1\rho)\|_{E_{\gamma,0}^\wedge} \leq C 2^{-(3-\epsilon_*)b} \|\rho\|_{\text{bfj}}^{(3)},$$

and the regular interior/Jacobian block satisfies

$$(100) \quad \|(B_1\rho)_b^{\text{reg}}\|_{\mathfrak{D}_U^{w,(0)}} \leq C2^{-(1-\epsilon_*)b}\|\rho\|_{\text{bfj}}^{(3)}.$$

The central clean point sector is finite, and the order-one grazing endpoint is zero because $u^3\delta(u) = 0$. Hence the complete base-level label-resolved first derivative is a bounded one-event map on the displayed graph domain, with summability margin $\delta = 1 - \epsilon_* > 0$.

By (98) the domain lies at the correct formal input level $\mathfrak{D}_U^{(3)} = \{\mathcal{N}_5^{\text{flux}} < \infty\}$. Nevertheless this theorem is not the source-side clause (S1): it proves neither invariance of $\mathfrak{D}_{U,\text{bfj}}^{(3)}$ under $\mathcal{L}_\theta$ nor return of a recovered output to that domain for a next response event.

Proof. For a circular target with centre c_i , use

$$G_{j \rightarrow i}(z, s) = |(q_j(z) - c_i) \times v(z)|^2 - (R_i + s)^2.$$

On $G_{j \rightarrow i} = 0$, $\partial_s G = -2R_i$, whereas differentiation in the line angle gives $|\partial_\varphi G| = 2R_i\tau$. Uniform separation and finite horizon therefore give a uniform coarea bound. The physical face velocity, coarea multiplier, and all their rescaled derivatives through the required order act boundedly on $\widehat{\mathcal{V}}_b^5$. The block term in (97) beyond $\mathcal{N}_5^{\text{flux}}$ consequently gives

$$\|\widehat{c}_{\Gamma,\omega,b}(B_1\rho)\|_{E_{\gamma,0}^\wedge} \leq C\|F_{\theta,b}(\rho)\|_{\widehat{\mathcal{V}}_b^5} \leq C2^{-3b}\|\rho\|_{\text{bfj}}^{(3)}.$$

Artificial partition derivatives cancel, and matching traces create no jump variation, so this is the grouped coefficient in the original quotient module, not merely a scalar distributional estimate. Multiplication by the source weight 2^{ϵ_*b} , together with the uniformly bounded current map, proves (99).

The regular interior/Jacobian derivative has at most one impact denominator:

$$|\dot{\tau}| + \|\partial_s T_s|_0\| \leq C|\langle v, n \rangle|^{-1} \leq Cu^{-2}.$$

All remaining reflection, chart, stable-leaf, and holonomy multipliers are uniformly bounded in the rescaled atlas. The second supremum in (97), including its unmatched weak pieces, therefore gives

$$\|(B_1\rho)_b^{\text{reg}}\|_{\mathfrak{D}_U^{w,(0)}} \leq C2^{\epsilon_*b}\|u^{-2}F_{\theta,b}(\rho)\|_{\mathcal{W}_b^5} \leq C2^{-(1-\epsilon_*)b}\|\rho\|_{\text{bfj}}^{(3)}.$$

Both block sums are geometric. Clean pair points lie in the central region by (R2), and $u^3\delta(u) = 0$ removes the order-one endpoint. The resulting uniform Cauchy estimate extends the smooth-core map to the stated completion. Nothing in this argument replaces the two blockwise suprema by the weaker integrated flux-seed norm. $\square$

Proposition 4.7 (The second radial letter is not an all-seed (S1) map). *For the radial-inflation family, the isolated second transfer letter $B_2 = \partial_s^2 \mathcal{L}_s|_0$ does not satisfy the base label-resolved source estimate in (S1) on every strong seed. More precisely, on a circular target let $r = R + s$, fix an incoming oriented line at perpendicular distance $h < r$ from the centre, and put*

$$d(r) = \sqrt{1 - h^2/r^2} = |v \cdot n| = u^2.$$

In the transported target arclength chart the impact angle $\alpha(r) = \arcsin(h/r)$ satisfies

$$(101) \quad \alpha''(r) = \frac{h(2r^2 - h^2)}{r^5 d(r)^3} \asymp d(r)^{-3} = u^{-6}$$

as the outgoing collision becomes grazing. Since the collision flux is $u^3 a(u, y) du dy$, a locally affine target observable detects in the interior second derivative the nonintegrable coefficient

$$(102) \quad u^3 \alpha''(r) \asymp u^{-3}.$$

Consequently no within-label regrouping of artificial homogeneity strips can give a decaying dyadic ℓ^1 source bound for $B_2\rho$ uniformly over strong ρ . The obstruction already occurs for a smooth seed which is nonzero on the chosen branch.

This does not contradict the second-order invariant or exact-current theorems. Their complete sources contain the additional terms $2B_1\rho_1$, respectively $g_2 + 2B_1h_1$, and the exact differentiated

invariance/Poisson identities cancel (102) before the source is placed in the strong regular embedding. Thus second and higher radial response must be formulated for specified complete assemblies; extending the first-order all-seed theorem to B_2 is impossible on the present source scale.

Proof. Write the target intersection in coordinates in which the incoming line is horizontal. Its perpendicular coordinate is h , while the longitudinal distance from tangency is $a(r) = \sqrt{r^2 - h^2} = rd(r)$. Differentiating $\alpha(r) = \arcsin(h/r)$ twice gives (101).

Choose a sequence of smooth test functions supported in the interiors of dyadic target blocks $u \asymp 2^{-b}$, away from the physical tangency face and from artificial strip cuts, and make each test affine in α on a smaller block. Their rescaled trace-test norms are uniform. On those supports the dual identity

$$\langle B_2 \rho, f_b \rangle = \int \rho \partial_s^2 (f_b \circ \mathcal{P}_s)|_0 d\nu_0$$

has no face term. The quadratic observable term vanishes on the affine region, while the linear term is $(\partial_\alpha f_b) \alpha''(r)$. Smooth transport corrections have strictly lower grazing order. Taking ρ nonzero with fixed sign there, flux invariance gives the fixed-sign block coefficient $u^3 \alpha''(r) \asymp u^{-3}$. It grows rather than decays after multiplication by the positive source weight $2^{\epsilon_* b}$. Because the tests avoid every block boundary, neither artificial-face cancellation nor a conormal endpoint current can remove this interior term. This violates the vector-valued dyadic estimate required in (S1). $\square$

Assumption 4.8 (Collision response-scale and triangular recovery certificates). The independent completions in Definition 4.1 satisfy, uniformly on the clean chart, the following structural estimates at every level used up to order J .

(S1) $\mathcal{L}_\theta : \mathfrak{D}_U^{(m)} \rightarrow \mathfrak{D}_U^{(m)}$ is bounded. Let $\mathfrak{W}_U^{\text{der},(m)}$ be the closure in $\mathfrak{X}_U^{(m)}$ of the complete order- m derivative sources. Then $B_{j,\theta} : \mathfrak{D}_U^{(m+j+2)} \rightarrow \mathfrak{W}_U^{\text{der},(m)} \subset \mathfrak{X}_U^{(m)}$. By Proposition 4.4, this clause requires a label-resolved vector-valued block estimate such as (93); it is not implied by the scalar physical 2^{-b} tail.

(S2) There is a distinguished closed recovered source class $\mathfrak{W}_U^{\text{rec},(m)} \subset \mathfrak{W}_U^{\text{der},(m)}$ and a bounded strong-primitive map

$$(103) \quad \mathcal{K}_{\theta,m} : \mathfrak{W}_U^{\text{rec},(m)} \longrightarrow \mathfrak{D}_U^{(m+1)}, \quad \bar{g} := \mathbf{S}_m \mathbf{Q}_m g = (I - \mathcal{L}_\theta) \mathcal{K}_{\theta,m} g,$$

with centered primitive and $\|\mathcal{K}_{\theta,m} g\|_{\mathfrak{D}_U^{(m+1)}} \leq C_m^{\text{prim}} \|g\|_{\mathfrak{W}_U^{\text{der},(m)}}$. At base strong level, Proposition 3.5 shows that this clause is equivalent to proving that the complete assembled source is centered and strong; the primitive is then the ordinary reduced density resolvent. Thus the genuinely new estimate is cancellation of the represented conormal/point sector together with a strong norm bound, not a second resolvent construction. To pass a block boundary, the certificate additionally supplies either intrinsic equality with the regular embedding or the scalar assembly-reset condition of Proposition 3.10; raw terminal data are never discarded merely because $\mathbf{S}_m$ forgets them. The fully assembled invariant sources in Theorem 3.4 belong to this class with the explicit primitive $\partial_s^j \rho_s$. The radial inflation in Theorem 3.6 verifies this cancellation at zero waiting time, while Theorem 3.7 verifies the complete fixed-length block uniformly for arbitrary positive gaps through order two, and Lemma 3.8, Theorem 3.9, and Proposition 3.10 close every fixed-length block in the scalar reset sense.

(S3) A *certified recovered response tree* starts from a specified centered source in $\mathfrak{W}_U^{\text{rec},(m)}$. It is triangular in the following sense: if a recovered centered source is evolved for a finite time and the next branch/Jacobian/roof/conormal event is inserted, the *complete assembled source at that event* again belongs to the next recovered level, either intrinsically or after a certified assembly reset. Its strong primitive is bounded by the product of the intervening transfer-letter norms and the preceding primitive norm. Difference quotients of these assembled primitives converge in the stated strong level. This clause does not assert that $B_{j,\theta} \rho$ is recovered for an arbitrary $\rho \in \mathfrak{D}_U^{(m+j+2)}$; the initial recovery and every subsequent closure step are part of the certificate. By Proposition 3.19, this

clause cannot hold for every strong seed at a genuinely moving face; it must be either source-specific or formulated for complete time blocks.

These are genuine structural hypotheses on the *assembled* physical source tree. They are not consequences of the ordinary density spectral gap or of the definition of the norm. Unlike the former (S2)–(S3), they do not assume the desired correlation decay: once (103) is available, that decay is a theorem consequence of the ordinary Demers–Zhang strong spectral gap. A raw isolated trace need not be recovered and is not placed in $\mathfrak{W}_U^{\text{rec},(m)}$. The exact invariant sources in Theorem 3.4 are the only response tree verified to satisfy (S2) in full finite order in this paper. The nonconjugate radial inflation verifies the complete invariant assembly through second order, but no nonconjugate moving-scatterer family is claimed here to satisfy the eventwise repeated closure (S3). It does verify source-specific blockwise positive-waiting reset through order two for every fixed block length; the infinite dynamical tree remains open. A uniform intrinsic historical terminal estimate is not an independent soft bound: by Proposition 3.11 it requires the one-event terminal germ to vanish.

Proposition 4.9 (No adjoint Lasota–Yorke contraction in a positive jet norm). *Let T be any C^2 uniformly hyperbolic diffeomorphism on a regular branch, with an expanding covector direction of factor $\lambda > 1$. A norm which dominates a nonzero first derivative of f in that direction cannot satisfy*

$$\|f \circ T^N\|_{\text{jet}} \leq a\|f\|_{\text{jet}} + C\|f\|_{\text{lower}}, \quad a < 1,$$

on all centered smooth f , when the lower norm is compact with respect to the jet norm. In particular the former adjoint trace Lasota–Yorke assertion for $\mathfrak{A}_U^{(m)}$ cannot be used to prove source-series convergence.

Proof. Choose a regular chart and smooth functions $f_k(x) = k^{-1}\chi(x)e^{ik\xi(x)}$, subtracting their means, where $d\xi$ is the expanding covector and χ is supported away from the branch boundary. Their first-jet norms stay bounded below and above, while the compact lower norm tends to zero along a subsequence. The chain rule gives $d(f_k \circ T^N) = DT^N*(df_k) \circ T^N$, so the controlled derivative is multiplied by at least $c\lambda^N$. Taking N with $c\lambda^N > a$ and then $k \rightarrow \infty$ contradicts the displayed inequality. Centering changes only a constant and hence no derivative. $\square$

Proposition 4.10 (Exact Demers–Zhang location of a transverse trace). *In the notation of Demers–Zhang, a nonzero codimension-one current supported on a C^2 curve uniformly transverse to the stable test leaves is, in general, an element of the weak completion $\mathcal{B}^w$ but not of the strong stable space $\mathcal{B}$. Therefore the Demers–Zhang strong spectral gap does not by itself produce a strong primitive for an isolated trace source.*

Proof. Let W_ℓ be a stable test segment of length ℓ meeting the trace curve once and transversely. In the weak norm the test normalization is $\|\psi\|_{C^p(W_\ell)} \leq 1$, so the point trace is bounded by the inverse crossing angle. In the Demers–Zhang strong stable norm the allowed tests satisfy $|W_\ell|^\alpha \|\psi\|_{C^q(W_\ell)} \leq 1$. Taking a test which is constant near the crossing with value $\ell^{-\alpha}$ makes the trace pairing of size $c\ell^{-\alpha}$, which diverges as $\ell \downarrow 0$. Thus the current need not lie in the strong space. The fixed-table Lasota–Yorke theorem contracts the strong norm and only bounds the weak norm; it supplies no exponential decay for an arbitrary weak trace source. A source-specific recovery/coupling estimate is genuinely additional. $\square$

Theorem 4.11 (Strong-primitive conormal coupling). *Assume the map spectral gap of Theorem 2.24. Let $g \in \mathfrak{W}_U^{\text{rec},(m)}$ be centered and put $\bar{g} = \mathbf{S}_m \mathbf{Q}_m g$ and $h = \mathbf{K}_{\theta,m} g$, so that $\bar{g} = (I - \mathcal{L}_\theta)h$. All pairings with g below mean the total-current pairing with $\bar{g}$. Then for every centered $f \in \mathfrak{A}_U^{(m+1)}$,*

$$(104) \quad |\langle g, \mathcal{U}_\theta^n f \rangle| \leq C_m \vartheta^n \|h\|_{\mathfrak{D}_U^{(m+1)}} \|f\|_{\mathfrak{A}_U^{(m+1)}},$$

and

$$(105) \quad \sum_{n=0}^{N-1} \langle g, \mathcal{U}_\theta^n f \rangle = \langle h, f \rangle - \langle \mathcal{L}_\theta^N h, f \rangle.$$

In particular the series is Cauchy with an explicit $O(\vartheta^N)$ tail. For a certified finite response tree satisfying (S3), applying this theorem after every complete assembled event gives the product decay and absolute summability of the scalar words in that tree.

Proof. Duality and $\bar{g} = (I - \mathcal{L}_\theta)h$ give

$$\langle g, \mathcal{U}_\theta^n f \rangle = \langle \mathcal{L}_\theta^n h, f \rangle - \langle \mathcal{L}_\theta^{n+1} h, f \rangle.$$

The two terms are bounded by the ordinary zero-mass Demers–Zhang spectral gap on the strong density space; the continuous embedding of the response density level into that space and the restricted observable dual give (104). Summing the displayed difference telescopes and proves (105). The triangular statement is an induction: (S3) supplies the next strong primitive, and the same argument supplies the next intervening geometric factor. Thus correlation decay is derived from the spectral gap and the primitive identity rather than assumed as a Koopman jet contraction. $\square$

Proposition 4.12 (Nontriviality and consequences of scale compatibility). *For each $m \leq J + 3$, the spaces in Definition 4.1 are nonzero Banach spaces independent, up to equivalent norm, of finite homogeneous refinement. The transported invariant density belongs to every required density level, and*

$$\mathcal{L}_\theta : \mathfrak{D}_U^{(m)} \longrightarrow \mathfrak{D}_U^{(m)}$$

is bounded under (S1). For $1 \leq j \leq J$, the j th transported transfer derivative extends as

$$(106) \quad B_{j,\theta} := \partial_\theta^j \mathcal{L}_\theta : \mathfrak{D}_U^{(m+j+2)} \longrightarrow \mathfrak{W}_U^{\text{der},(m)} \subset \mathfrak{X}_U^{(m)}.$$

Every physical branchwise C^r observable with finite displayed jets belongs to $\mathfrak{A}_U^{(m)}$. If (S2) holds, then for a centered recovered source $g \in \mathfrak{W}_U^{\text{rec},(m)}$, the scalar functional

$$(107) \quad \mathcal{R}_{\theta,g}^{\text{scal}}(f) := \sum_{n \geq 0} \langle g, \mathcal{U}_\theta^n f \rangle$$

is bounded on centered $\mathfrak{A}_U^{(m+1)}$, with norm at most $C_m(1 - \vartheta)^{-1} C_m^{\text{prim}} \|g\|_{\mathfrak{W}_U^{\text{der},(m)}}$. Neither R_θ^{obs} nor $B_{j,\theta}^*$ is asserted to be a bounded operator on the positive jet completion $\mathfrak{A}_U^{(m)}$.

Proof. Branchwise C^r functions with support compactly contained in one regular homogeneous chart have finite flux-seed norms, so every level is nonzero. The collision invariant density is 1 in the moving flux fiber; after normal-graph transport it is a C^r Jacobian and has the stated finite norms. Two finite refinements admit a common refinement. Central pieces are merely subdivided, artificial internal faces occur with opposite orientations, and terminal pieces retain the same strip index; hence both seed norms dominate the common-refinement norm by a finite-complexity constant.

The transfer and derivative maps are (S1), written in (106). The strong primitive from (S2) and Theorem 4.11 prove (107); (S3) supplies the corresponding primitive after a complete re-insertion event. This proposition deliberately gives a scalar resolvent functional and not a Koopman resolvent on the positive jet completion; the latter is ruled out by Proposition 4.9. $\square$

Definition 4.13 (Resolved indicator current). Let

$$D_s = \{z : g_{1,s}(z) > 0, \dots, g_{\ell,s}(z) > 0\}$$

be a moving branch domain in a two-dimensional collision chart. Resolve its finite singular stratification into the parameter-uniform C^J manifold with corners supplied by Assumption 2.6, $\tilde{D}_s$ whose codimension-one faces are labelled by one generator and whose codimension-two corners are labelled by an ordered pair of distinct faces. The order- m moving-boundary current is defined before any formal delta multiplication by

$$(108) \quad \mathcal{T}_D^{(m)} := \partial_s^m (\iota_{s*} [\tilde{D}_s])|_{s=0}$$

as a distribution on compactly supported branch tests. This definition is intrinsic under changes of the resolving chart and fixes the meaning of every higher trace used below.

Theorem 4.14 (Second and multiple moving traces). *For every $1 \leq m \leq J$, the current $\mathcal{T}_D^{(m)}$ in Definition 4.13 is a finite sum of:*

- (i) *branch-interior derivatives;*
- (ii) *one-face currents $a_{m,\ell}\delta^{(\ell)}(g_\Gamma)$, $0 \leq \ell < m$, representing repeated derivatives of the same moving wall;*
- (iii) *derivatives of codimension-two coarea atoms at clean intersections $g_\Gamma = g_\Lambda = 0$, with coefficient*

$$\frac{a_{\Gamma\Lambda}}{|\det(Dg_\Gamma, Dg_\Lambda)|};$$

- (iv) *parameter derivatives of the preceding point currents on the labelled clean pair-intersection vertices of the fixed simple stratified isotopy.*

All point-current terms lie in the central region $u \geq u_J$; the grazing collar contributes only one-face currents. Thus a point atom is never born from or lost into the grazing boundary inside the theorem. No term is defined as $\delta(g_1)\delta(g_2)\delta(g_3)$ in the two-dimensional collision section. Triple-labelled meetings are excluded by the simple-incidence hypothesis; they cannot be hidden inside a “resolved vertex.” For each $m \leq J$ and each finite decorated core subfamily $\mathcal{F}$,

$$(109) \quad \|\mathcal{T}_{D,\mathcal{F}}^{(m)}\rho\|_{\mathfrak{X}_U^{(m)}} \leq C_{m,\mathcal{F}}\|\rho\|_{\mathfrak{D}_U^{(m+2)}}.$$

The completed countable physical branch sum belongs to $\mathfrak{X}_U^{(m)}$, with a uniform version of this estimate, only under the all-branch mapping certificate (S1). No isolated trace component is asserted to have a uniform completion before the complete physical event is assembled.

Proof. Differentiate the pushforward current in (108). On a smooth face, the one-dimensional distributional chain rule gives

$$\partial_s^m H(g_s)|_0 = \sum_{\ell=0}^{m-1} P_{m,\ell}(\dot{g}, \dots, \partial_s^{m-\ell} g) \delta^{(\ell)}(g_0).$$

In normal coordinates $y = g_0$, the action of $\delta^{(\ell)}(y)$ moves ℓ normal derivatives onto the smooth coefficient, density, test, and chart Jacobian. This is a standard conormal distribution and requires no self-product.

At two distinct clean faces, the coarea formula turns the intersection into the displayed decorated point atom in $\mathfrak{X}_U^{(m)}$. Repeated parameter differentiation moves the atom and differentiates its coarea coefficient; it does not add a third normal equation. The clean lower bound in Assumption 2.6 and Lemma 4.15 controls its inverse determinant. A tangency of two distinct faces is outside the response chart; it is not reclassified as a vertex and no dyadic $1/\Delta$ summation is used.

For a finite decorated core, the coarea and trace-test bounds give (109) directly in the quotient coefficient norm. For the countable atlas, the local coefficient of an order- m current can have polynomial strip and propagation growth. The complete physical event is assembled before summation, and Lemma 2.19 supplies its basic 2^{-b} block tail. Membership in the stronger recursively weighted completion is exactly the separate all-branch statement (S1), not a consequence of that scalar tail. The ambient source norm and its corner pairing are Definition 4.2 and Proposition 4.3. Internal artificial faces appear twice with opposite induced orientations, so they cancel before completion. The local cases and determinant estimates are recorded in Lemmas C.3 to C.5 and Propositions C.6 and C.7. $\square$

Remark (Wavefront interpretation). For two distinct faces the wavefront conormals are linearly independent on a clean cell, which is exactly the Hörmander condition for the coarea product. For one repeated face the theorem uses a normal derivative of one conormal current, not a product with itself. Pair-current derivatives remain derivatives of that same point current. These cases exhaust the finite branch derivative tree on the stipulated simple chart; a triple unfolding is a different local problem and is not covered.

Lemma (Chart independence of resolved conormal currents). *Let $\beta_s : \tilde{D}_s \rightarrow D_s$ and $\beta'_s : \tilde{D}'_s \rightarrow D_s$ be two finite resolutions of the same moving branch domain, compatible with the physical face labels. Then for every $m \leq J$,*

$$\partial_s^m \beta_{s*}[\tilde{D}_s]|_0 = \partial_s^m \beta'_{s*}[\tilde{D}'_s]|_0$$

as distributions. Hence $\mathcal{T}_D^{(m)}$ is independent of the order in which strip, branch-change, and admitted multiple-impact cells are refined inside the one-step clean stratified isotopy of Theorem 2.9.

Proof. Take a common resolution $\widehat{D}_s$ obtained by the fiber product of the two finite stratifications followed by subdivision along every face preimage. There are proper face-preserving maps

$$p_s : \widehat{D}_s \rightarrow \widetilde{D}_s, \quad p'_s : \widehat{D}_s \rightarrow \widetilde{D}'_s$$

of degree one on the interior, and $\beta_s p_s = \beta'_s p'_s$. Integration of a compactly supported test function over the three interior currents therefore gives the same scalar function of s . It is C^J because the resolved charts and their Jacobians are C^J up to each face. Differentiate this scalar identity m times. Oppositely oriented artificial faces cancel in the common resolution, while physical faces and corners retain their labels. The resulting distributional derivatives agree. $\square$

Lemma 4.15 (Clean central two-face determinant bound). *Let $\Delta(z) = |\det(Dg_\Gamma(z), Dg_\Lambda(z))|$ on an admitted two-face trace cell. Under Assumption 2.6, that cell lies in $u \geq u_J$ and $\Delta \geq c_J$. Hence for every stable test segment W ,*

$$\int_{W \cap \{g_\Gamma = g_\Lambda = 0\}} \Delta^{-1} d\mathcal{H}^0 \leq C_J \#(W \cap \{g_\Gamma = g_\Lambda = 0\}).$$

The intersection multiplicity is part of the finite-word clean-atlas bound. Repeated faces carry one conormal support with normal derivatives and do not use this estimate. No claim is made that a sublevel estimate alone makes $1/\Delta$ integrable through a simple tangency.

Proof. The clean determinant lower bound gives $\Delta^{-1} \leq c_J^{-1}$ pointwise at every distinct-face intersection. Summing over the uniformly finite intersection set gives the displayed inequality. The counterexample $\Delta(t) = |t|$ shows why the excluded tangential case cannot be recovered by summing dyadic determinant shells: every shell contributes a nonvanishing constant. The boundary-fold example in Proposition 2.10 shows separately why a determinant jet vanishing at $u = 0$ cannot replace (6). $\square$

Proposition (Scalar limit interchange at finite order). *Let $g_s^{(j)}$ be a recovered order- j assembled source produced by a difference quotient of the transported transfer family, assume its strong primitives converge in the level stated in (S3), and let*

$$u_{f,N} = \sum_{n=0}^{N-1} \mathcal{U}_0^n \Pi_0^{\text{obs}, \perp} f.$$

For $j \leq J$, the two limits

$$\lim_{s \rightarrow 0} \lim_{N \rightarrow \infty} \langle g_s^{(j)}, u_{f,N} \rangle, \quad \lim_{N \rightarrow \infty} \lim_{s \rightarrow 0} \langle g_s^{(j)}, u_{f,N} \rangle$$

exist and agree as scalar source/test pairings. The common value is the observable-side word specified by Proposition 4.17.

Proof. Write $S_N(g, f) = \sum_{n=0}^{N-1} \langle g, \mathcal{U}_0^n f \rangle$. The strong-primitive identity gives

$$S_N(g, f) = \langle \mathcal{K}g, f \rangle - \langle \mathcal{L}_0^N \mathcal{K}g, f \rangle.$$

Hence

$$|S_N(g_s^{(j)}, f) - S_\infty(g_0^{(j)}, f)| \leq C \|\mathcal{K}g_s^{(j)} - \mathcal{K}g_0^{(j)}\|_{\mathfrak{A}_U^{(j+1)}} \|f\|_{\mathfrak{A}_U^{(j+1)}} + C \vartheta^N \|\mathcal{K}g_0^{(j)}\| \|f\|.$$

The first term tends to zero by (S3) and the second uniformly by the ordinary strong spectral gap. The same scalar estimate applies after each complete earlier event in an order- j word. No element $R_0^{\text{obs}} f$ in the positive jet completion is introduced. This proves joint Cauchy convergence and the interchange. $\square$

Theorem 4.16 (Observable-side scalar insertion for recovered sources). *For $0 \leq m \leq J$, let*

$$g \in \mathfrak{W}_U^{\text{rec}, (m)}, \quad \langle g, 1 \rangle = 0,$$

and let $f \in \mathfrak{A}_U^{(m+1)}$ be centered. Then

$$(110) \quad \langle g, R_\theta^{\text{obs}} f \rangle := \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \langle g, \mathcal{U}_\theta^n f \rangle$$

exists and

$$|\langle g, R_\theta^{\text{obs}} f \rangle| \leq C_m C_m^{\text{prim}} \|g\|_{\mathfrak{W}_U^{\text{der},(m)}} \|f\|_{\mathfrak{A}_U^{(m+1)}}.$$

More precisely, for $M > N$,

$$(111) \quad \left| \sum_{n=N}^M \langle g, \mathcal{U}_\theta^n f \rangle \right| \leq \frac{C_m C_m^{\text{prim}} \vartheta^N}{1 - \vartheta} \|g\|_{\mathfrak{W}_U^{\text{der},(m)}} \|f\|_{\mathfrak{A}_U^{(m+1)}}.$$

More generally, every alternating scalar word in a certified recovered tree, with total response order at most J , defines a bounded pairing between the corresponding ambient source and observable levels. No statement is made for a word whose first derivative source has not been certified recovered.

This theorem makes no assertion that

$$\sum_{n \geq 0} \mathcal{L}_\theta^n g \in \mathfrak{D}_U^{(m)}.$$

In particular, no finite prefix $g, \mathcal{L}g, \dots, \mathcal{L}^{N-1}g$ is assigned to the strong density space merely because the complete assembled source has a separate strong primitive.

Proof. For a smooth finite-strip source, duality gives

$$\sum_{n=0}^{N-1} \langle g, \mathcal{U}_\theta^n f \rangle = \sum_{n=0}^{N-1} \langle \mathcal{L}_\theta^n g, f \rangle.$$

The complete physical block estimate and the explicit one-sided jet/holonomy norm give the time-zero pairing bound. Applying Theorem 4.11 to every summand and summing the geometric tail gives (111). This proves Cauchy convergence uniformly on the smooth core.

Completion gives (110). Notice that the finite prefix was estimated only as a dual pairing. It was not inserted into a strong-density norm. Longer words are the scalar re-insertion blocks of (S3): evolve a centered recovered block, assemble the next complete event, take its strong primitive, and apply Theorem 4.11 again. No transpose derivative operator on $\mathfrak{A}_U^{(m)}$ is introduced. Each insertion consumes the finite number of source/test levels recorded in its order, so total order at most J closes before the primitive reserve is exhausted. $\square$

Proposition 4.17 (Certified scalar multi-space differentiation). *Let a centered input source and all of its successive complete derivative events form a certified recovered response tree in the sense of (S3). After pairing that tree at level $m + j + 3$ with an observable at level $m + j + 3$, the j th parameter derivative of its centered scalar susceptibility is a finite sum, over ordered compositions $\pi_1 + \dots + \pi_q = j$, of absolutely convergent scalar series*

$$(112) \quad \sum_{n_0, \dots, n_q \geq 0} \mathcal{C}_{\pi, \mathbf{n}, \theta}(\rho, f).$$

Here $\mathcal{C}_{\pi, \mathbf{n}, \theta}$ is the finite scalar correlation block obtained by differentiating the corresponding certified finite-time identity identity, centering after each insertion, and retaining every branch, Jacobian, flux, roof, and physical-face term. It is not notation for a composition of operators $R^{\text{obs}} B^* R^{\text{obs}}$ on a jet space.

Proof. Differentiate a finite partial sum. The ordinary product rule gives finitely many branchwise correlation blocks indexed by an ordered composition and by the intervening times $\mathbf{n}$. The initial source is recovered by the certificate, and the assembled strong primitives converge by the last clause of (S3). Estimate every intervening block by Theorem 4.11; the triangular primitive bound in (S3) makes the product of geometric factors summable over $\mathbf{n}$. Dominated convergence therefore passes the parameter limit through the multiple series. Induction gives the Faà di Bruno coefficients

and consumes exactly the displayed source/test levels. At no stage is a strong-to-weak derivative inserted into a one-space Kato identity. $\square$

Proposition 4.18 (Uniform certified finite-order response words). *For each $0 \leq j \leq J$, every ordered composition $\pi = (\pi_1, \dots, \pi_q)$ of j , and every certified recovered response tree whose initial primitive $h_{\pi, \rho, \theta}$ satisfies*

$$\|h_{\pi, \rho, \theta}\|_{\mathfrak{D}_U^{(j+3)}} \leq C_{\text{init}}(\pi) \|\rho\|_{\mathfrak{D}_U^{(j+3)}},$$

there is a constant $C_{\text{word}}(j, q)$, independent of θ , such that

$$(113) \quad \sum_{n_0, \dots, n_q \geq 0} |\mathcal{C}_{\pi, \mathbf{n}, \theta}(\rho, f)| \leq C_{\text{word}}(j, q) \|\rho\|_{\mathfrak{D}_U^{(j+3)}} \|f\|_{\mathfrak{A}_U^{(j+3)}}.$$

Proof. Apply the ordinary strong decay to the certified initial primitive, and then apply the triangular strong-primitive estimate (S3) after every subsequent complete event. Merely knowing (106) does not certify the first derivative source; that extra implication is not used here. The branch number, strip complexity, conormal multiplicity, and terminal pieces are already included in the operator norm of $B_{\pi_i, \theta}$; no artificial factor ϑ_0^n is inserted into a series. The only geometric decay is the Demers–Zhang strong decay applied to the constructed primitive in Theorem 4.11. $\square$

Theorem 4.19 (Certified finite-order dynamical susceptibility). *Assume Assumption 4.8. Let $g_\theta \in \mathfrak{W}_U^{\text{rec}, (m)}$ be centered, put $h_\theta = \mathcal{K}_{\theta, m} g_\theta$, and let $f_\theta \in \mathfrak{A}_U^{(m+J+3)}$ be a centered transported observable. Assume that h_θ and f_θ are C^J in the displayed strong source and observable levels and that every derivative event in their finite tree carries the (S3) certificate. Then the scalar*

$$(114) \quad \chi_\theta(g_\theta, f_\theta) = \sum_{n \geq 0} \langle g_\theta, \mathcal{U}_\theta^n \Pi_\theta^{\text{obs}, \perp} f_\theta \rangle = \langle g_\theta, R_\theta^{\text{obs}} f_\theta \rangle$$

is C^J . Its derivatives are the finite ordered observable-side words in Propositions 4.17 and 4.18; no density-side resolvent is applied to g_θ .

Proof. The explicit Cauchy tail (111) is uniform on the clean compact chart. Differentiate finite partial sums and use the assumed C^J strong primitive, the certified triangular recovery bounds (S3), followed by Theorem 4.11, to pass to the limit with the same uniform tail at each consumed source/test level. The product rule gives precisely the ordered words of Proposition 4.17; their uniform bounds are Proposition 4.18. $\square$

Lemma 4.20 (Collision-map response loss accounting). *The collision-map construction consumes primitive derivatives according to*

operation	input level	output
j th branch/Jacobian/conormal letter	$\mathfrak{D}_U^{(j+2)}$	$\mathfrak{X}_U^{(j)}$
j th projector or map-resolvent pairing	$\mathfrak{D}_U^{(j+3)} \times \mathfrak{A}_U^{(j+3)}$	$\mathbb{C}$
exported collision-map pairing	$0 \leq j \leq J, \quad r \geq J+4$	$\mathbb{C}$.

No continuous-time loss count is included: it would require the unconstructed weighted graph-domain and symbol mappings isolated in Proposition 2.15.

Proof. One derivative is spent on the impact/grazing coordinate and one on the unstable-holonomy comparison. Centering plus one observable trace-resolvent level gives the second row. The last row is the resulting finite collision budget. Every loss is attached to a displayed collision-map mapping statement. $\square$

Theorem 4.21 (Finite-order moving-singularity response). *Assume Assumption 4.8. Let θ_s be a C^J finite-response curve, let ρ_s be the transported invariant collision density, and let f_s be a C^J observable section in the required source/test levels. Then*

$$I(s) = \langle \rho_s, f_s \rangle$$

is C^J by the direct flux formula (4). The transfer expression (83) equals that direct derivative. For $2 \leq j \leq J$, differentiating the invariance equation gives an equivalent finite Faà di Bruno sum obtained by:

- (i) differentiating f_s , the flux transport, and the invariant projection;
- (ii) replacing every transfer derivative by its branch/interior, Jacobian, and resolved conormal decomposition from Theorem 4.14;
- (iii) placing the resulting conormal source in $\mathfrak{X}_U^{(j)}$; if its complete tree is certified recovered, insert it through the scalar observable-side words of Proposition 4.17.

If Φ_s is the fixed collision-section transport and $J_s = d(\Phi_s^* \nu_s)/d\nu_0$, then the non-circular bound is

$$|I^{(j)}(0)| \leq C_j \max_{\ell \leq j} \|\partial_s^\ell(f_s \circ \Phi_s)|_0\|_{C^0} \max_{\ell \leq j} \|\partial_s^\ell J_s|_0\|_{C^0}.$$

Proof. Pulling the invariant flux integral back by Φ_s proves differentiability and the displayed bound directly, with no invariant-density derivative on its right-hand side. To obtain the transfer representation, differentiate the finite-time invariance identity $\langle \rho_s, f_s \rangle = \langle \rho_s, \mathcal{U}_s^N f_s \rangle$, center after every derivative letter, and move all transfer derivatives to the observable side. At first order, the limit $N \rightarrow \infty$ is (83); its Cauchy remainder follows from the explicit invariant strong primitive. At order j , the product rule produces the resolved conormal letters of Theorem 4.14 and the successive-scale words of Proposition 4.17. When the assembled tree is certified, their uniform Cauchy limits and bounds are Proposition 4.18. Without that certificate the finite-time conormal identity remains valid, but no infinite scalar word is asserted. Since both sides are derivatives of the same finite-time identity, the singular trace/resolvent terms combine exactly to the geometric Leibniz formula (4). $\square$

4.1. Spectral separation and infinite scalar words.

Definition (Period-one response convention). The entire response theorem is formulated on the fixed period-one mixing component required in Assumption 2.6. The period is a dynamical hypothesis, not the period of an arbitrary finite transition graph and not a consequence of connectedness of the undirected scatterer graph. A family with a genuine $q > 1$ cyclic decomposition requires a separate formulation using $\mathcal{L}_\theta^q$, cyclic projectors, and periodic centering; it is not reassembled in this paper.

Proposition (Peripheral spectrum on the period-one component). *On the stipulated period-one mixing component, 1 is a simple eigenvalue of $\mathcal{L}_\theta$ and there is no other unit-modulus eigenvalue. This is the peripheral-spectrum input used in Theorem 2.24; no conclusion for a $q > 1$ cyclic component is claimed.*

Proof. Quasi-compactness is supplied by the billiard-specific Demers–Zhang Lasota–Yorke and compact-embedding theorem. Simplicity and exclusion of the remaining peripheral roots use the stipulated period-one mixing property and the classical ergodic/mixing theorem for the selected finite-horizon dispersing component [16, 6, 7]. Connectedness alone is not used as an aperiodicity proof. $\square$

Lemma (Uniform period-one spectral separation). *On the compact clean response chart, the selected component remains period one and the simple eigenvalue 1 stays uniformly separated from the rest of the spectrum.*

Proof. Period one is part of Assumption 2.6. The Demers–Zhang perturbation theorem gives continuous spectral projectors and spectral stability in the strong-to-weak perturbation metric. The clean chart is compact and does not cross a cyclic bifurcation, so the pointwise spectral separations have a positive common minimum. $\square$

Definition (Expanded observable resolvent word). For an ordered derivative composition $\pi = (\pi_1, \dots, \pi_q)$, differentiate the corresponding finite centered correlation block and retain all physical terms. Denote the resulting scalar at intervening times $\mathbf{n} = (n_0, \dots, n_q)$ by $\mathcal{C}_{\pi, \mathbf{n}, \theta}(\rho, f)$. The

expanded scalar word is

$$(115) \quad \sum_{n_0, \dots, n_q \geq 0} \mathcal{C}_{\pi, \mathbf{n}, \theta}(\rho, f).$$

Centering is inserted after every derivative event. This definition uses the finite branch calculus and scalar pairing only; it does not presuppose that a formal transpose B_j^* maps a positive observable jet space into itself. No extra numerical weight is attached to the summand.

Lemma 4.22 (Actual decay of a recovered derivative block). *Let $g \in \mathfrak{W}_U^{\text{rec}, (m)}$ be centered and let $\psi \in \mathfrak{A}_U^{(m+1)}$ be centered. There are $C_m > 0$ and $\vartheta \in (0, 1)$ such that*

$$(116) \quad |\langle g, \mathcal{W}_\theta^n \psi \rangle| \leq C_m C_m^{\text{prim}} \vartheta^n \|g\|_{\mathfrak{W}_U^{\text{der}, (m)}} \|\psi\|_{\mathfrak{A}_U^{(m+1)}}.$$

The constants are uniform on a compact response chart.

Proof. This is Theorem 4.11 together with the primitive bound in (103). The decay is therefore a consequence of the ordinary Demers–Zhang strong spectral gap for the primitive, not an assumption about Koopman contraction on a positive jet norm. $\square$

Proposition (Certified infinite scalar-word summability). *Every expanded word (115) of total derivative order at most J that belongs to a certified recovered response tree and has the initial primitive bound in Proposition 4.18 is absolutely summable after grouping into adapted blocks, and*

$$\sum_{n_0, \dots, n_q \geq 0} |\text{the } (n_0, \dots, n_q) \text{ summand}| \leq C_\pi \|\rho\|_{\mathfrak{D}_U^{(J+3)}} \|f\|_{\mathfrak{A}_U^{(J+3)}}.$$

The constant C_π contains the real geometric factors $(1-\vartheta)^{-(q+1)}$, the finite strong-primitive constants, finite branch multiplicities, and the proved physical grazing-block sums. It contains no postulated $\vartheta_0^{n_0+\dots+n_q}$ multiplier.

Proof. Apply the actual decay from Lemma 4.22 to the certified initial primitive. The triangular estimate (S3) then gives a strong primitive at every subsequent complete event, followed again by the actual decay from Lemma 4.22. Repeat for $n_{q-1}, \dots, n_0$. This bounds the absolute value of the summand by

$$C_\pi \prod_{a=0}^q \vartheta^{n_a} \|\rho\|_{\mathfrak{D}_U^{(J+3)}} \|f\|_{\mathfrak{A}_U^{(J+3)}}.$$

Now the geometric factors are consequences of (116). Summing them proves the result. Branch number and trace-intersection number are already included in the finite derivative-letter norms and in the complete physical block estimate. $\square$

No differentiated suspension-word theorem. The fixed-table contour above is not differentiated with respect to the table. Such a step would require the common weighted graph domains and negative-order symbol estimate described after Proposition 2.15; it is left for future work.

Proposition (Rigid translation through second order). *Translate one scatterer by $q_i(\sigma, s) = q_i(\sigma, 0) + sw$, and allow the incoming section coordinates to be transported, $x = x(s)$, $v = v(s)$. At a regular impact $x(s) + \tau(s)v(s) = q_i(\sigma(s), 0) + sw$, the first two impact-time derivatives are*

$$\dot{\tau} = \frac{n_i \cdot (w - \dot{x} - \tau \dot{v})}{n_i \cdot v}, \quad \ddot{\tau} = \frac{\dot{n}_i \cdot (w - \dot{x} - \tau \dot{v}) - n_i \cdot (\ddot{x} + 2\dot{\tau} \dot{v} + \tau \ddot{v})}{n_i \cdot v},$$

where $\dot{n}_i = -S_i(\dot{x} + \dot{\tau}v + \tau \dot{v} - w)^\top$. For fixed incoming coordinates this reduces to $\dot{\tau} = (n_i \cdot w)/(n_i \cdot v)$ and the corresponding fixed- (x, v) second derivative. The order-two transfer source is

$$B_{2,w}\rho = B_{2,w}^{\text{int}}\rho + B_{2,w}^{\text{Jac}}\rho + \sum_{\Gamma} (a_{\Gamma,w}\delta(g_\Gamma) + b_{\Gamma,w}\delta'(g_\Gamma))\rho + \sum_{\Gamma \cap \Lambda} c_{\Gamma \cap \Lambda,w}\delta_{\Gamma \cap \Lambda}\rho.$$

Its invariant response is the observable-side pairing prescribed by Proposition 4.17; no density-side resolvent is applied to $B_{2,w}\rho$.

Proof. The first derivative of the impact equation is

$$\dot{x} + \dot{\tau}v + \tau\dot{v} - w = q_{i,\sigma}\dot{\sigma}.$$

Its normal projection gives the displayed formula for $\dot{\tau}$; its tangential projection and the Weingarten identity give the formula for $\dot{n}_i$. Differentiating the normal equation a second time produces the terms $2\dot{\tau}\dot{v} + \tau\ddot{v}$ and gives the displayed formula for $\ddot{\tau}$. Differentiate the reflection law and branch Jacobian to obtain the interior and Jacobian terms. For a single moving branch face, the second derivative of $H(g_s)$ is

$$\ddot{g}\delta(g) + \dot{g}^2\delta'(g).$$

Two distinct clean faces give the coarea atom with determinant denominator. These are exactly the three cases in Theorem 4.14. The renormalized physical block identity sums their combined grazing contributions, and Theorem 4.16 supplies the final observable pairing. $\square$

5. FIVE-RESULT SYNTHESIS AND LOGICAL BOUNDARY

Theorem 5.1 (Actual static and source-specific collision-map response). *Let $J \geq 1$, let $r \geq J + 4$, let Θ be a compact finite-response parameter set, and assume Assumptions 2.3 and 2.4. Static collision averages are C^J and satisfy (4) without a response-scale hypothesis. For the genuinely nonconjugate radial inflation in Theorem 3.6, the complete invariant source returns to the centered strong density space through order two, with the explicit primitives (43). Moreover Theorem 3.7 recovers the complete fixed-length block uniformly over arbitrary positive waiting lengths, while Lemma 3.8 gives its fixed- N historical scalar grazing-terminal tail and Theorem 3.9 gives a sharper uniform-in- N tail for the total current after assembly. These source-specific conclusions use neither all-seed (S1) nor eventwise (S3), and they do not imply label-resolved terminal closure. They do satisfy the certified scalar assembly-reset criterion of Proposition 3.10, so any subsequent fixed scalar waiting block sees the recovered strong density rather than the historical one-sided labels. In addition, Theorem 3.13 gives a non-invariant C^J scalar susceptibility class on the same nonconjugate family by taking $g_s = (I - \mathcal{L}_s)h_s$ with a prescribed strong transported section h_s . This exact-coboundary class requires no all-branch response derivative. More generally Theorem 3.14 proves uniform convergence for every centered strong correlation source, while Proposition 3.16 identifies the two complete assembled source tests needed for twice differentiable correlation response. The explicit exact-current benchmark Theorem 3.18 passes every represented finite-order Poisson test. The differentiated-invariance benchmark in Theorem 3.15 is nonzero and has an infinite twice differentiable scalar susceptibility by direct telescoping; its total-current q_1, q_2 identities are not presented as a complete recovery certificate. Finally, Theorem 4.6 proves a base-level one-event graph-domain bound on $\mathfrak{D}_{U,\text{bf}}^{(3)} \hookrightarrow \mathfrak{D}_U^{(3)}$. Because transfer invariance and next-event closure are unproved, this result is explicitly not called (S1). There is also no all-seed second-order extension: Proposition 4.7 gives the sharp circular impact obstruction, so every positive higher-order radial result must use the complete assembled source.*

These are the unconditional or source-specific billiard conclusions of this theorem. They do not use the abstract finite-order certificates below.

Proof. The direct static statement is Proposition 2.1. The nonconjugate invariant assembly and its waiting-time and terminal statements are Theorems 3.6, 3.7 and 3.9, Lemma 3.8, and Proposition 3.10. The non-invariant scalar benchmarks, differentiated-invariance example, and finite-order exact current are Theorems 3.13 to 3.15 and 3.18 and Proposition 3.16. Finally, the restricted first radial source estimate and the sharp all-seed second-order obstruction are Theorem 4.6 and Proposition 4.7. $\square$

Theorem 5.2 (Abstract certified finite-order conormal calculus). *Let $J \geq 1$, let $r \geq J + 4$, and assume Assumptions 2.3 and 2.4.*

Assume in addition the period-one/no-boundary-birth/Ward chart Assumption 2.6 and the response-scale mapping certificate (S1) of Assumption 4.8. Then:

- (i) the collision maps have the billiard-specific Demers–Zhang anisotropic spectral gap on the stipulated period-one component;
- (ii) the standard strips and complete flux/trace identity give summable renormalized physical grazing blocks through order J , without an absolute polynomial strip moment;
- (iii) the transfer derivatives $B_{j,\theta}$, $j \leq J$, have finite branch/interior, Jacobian, and resolved-conormal decompositions on the response Banach scale;
- (iv) if a specified response tree additionally carries the independent (S2)–(S3) certificate, its reduced collision-map susceptibility is C^J , with scalar scale-resolvent words bounded as in Proposition 4.18 and Theorem 4.19 and map-scale loss as in Lemma 4.20.

In particular, for every admissible target observable f_s , the first derivative is

$$\left. \frac{d}{ds} \right|_{s=0} \langle \rho_s, f_s \rangle = \langle \rho_0, \nabla_s f_s|_0 \rangle + \left\langle (\nabla_s \mathcal{L}_s|_0) \rho_0, R_0^{\text{obs}} f_0 \right\rangle = \left. \frac{d}{ds} \right|_{s=0} \int_{\Sigma_0} (f_s \circ \Phi_s) J_s d\nu_0.$$

All higher static derivatives exist by direct differentiation of the flux measure. Their finite-time transfer identities have the conormal expansion of Theorem 4.21; an infinite observable-side word is asserted only when its assembled source tree has the (S3) recovery certificate. The construction uses neither an undefined product of three independent delta distributions nor a weak-to-strong finite-prefix inference. The theorem does not construct full finite-order all-seed (S1) or eventwise (S3) for a genuinely nonconjugate moving-scatterer family. The radial inflation gives only the source-specific blockwise invariant assembly stated above. The flux-conjugate class verifies (R3) only and is a symmetry class by Proposition 2.8. Moreover, by Proposition 4.4, the scalar physical tail used in that assembly cannot by itself prove the label-resolved all-branch map (S1).

Proof. The collision geometry, growth, and physical block summation are Theorems 2.20 and 2.21 and Lemma 2.19. The anisotropic map theorem is Theorem 2.24. The response scale and conormal calculus are Proposition 4.12 and Theorem 4.14. The first response identity is Theorem 3.20; all higher observable-side insertions and scale-resolvent derivatives are Theorems 4.16, 4.19 and 4.21 and Propositions 4.17 and 4.18, under the displayed recovery certificate. $\square$

Flow is not an exported theorem. The paper retains only the corrected fixed-table BDL correspondence and the obstructions in Propositions 2.14 and 2.15. A branch/preimage-weighted common graph domain, its weighted Lasota–Yorke and resolvent invariance, a negative-order dyadic symbol bound, and the zero-waiting endpoint cancellations remain unconstructed. Accordingly no moving-family flow implication or regularity budget is asserted.

Proposition (Response consistency checks). *The response formula has the following invariances. (C1) If $f_s \equiv 1$, then $I'(0) = 0$, because $\Pi_0^{\text{obs},\perp} 1 = 0$ and the transported derivative of the constant observable vanishes.*

(C2) *The source has zero total mass: $\langle g, 1 \rangle = 0$. Thus the total-mass derivative is zero.*

(C3) *A purely tangential reparametrization of a fixed physical wall has $V_\Gamma \cdot n_\Gamma = 0$. Its physical trace vanishes, while the branch-interior and transport-Jacobian terms cancel as the derivative of a conjugate coordinate representation.*

(C4) *Changing only the auxiliary transport trivialization changes the two terms in (83) by opposite commutator pairings; their sum is unchanged.*

(C5) *Artificial homogeneity-strip boundaries occur in adjacent partition functions $\chi_k + \chi_{k+1} = 1$; hence their central first derivatives cancel before the physical trace sum is formed.*

(C6) *Suppose one scatterer is translated rigidly, $q_i(\sigma, s) = q_i(\sigma, 0) + sw$. At a regular impact $x + \tau v = q_i(\sigma, 0)$, differentiation of the impact equation at fixed incoming coordinates gives*

$$\dot{\tau} = \frac{n_i \cdot w}{n_i \cdot v}.$$

If the corresponding moving branch boundary in collision coordinates is $g_{i,h,s}(z) = 0$, its section-normal velocity is

$$V_{\Gamma_{i,h}} \cdot n_{\Gamma_{i,h}} = - \frac{\partial_s g_{i,h,s}|_0}{|\nabla_z g_{i,h,0}|},$$

where $\partial_s g_{i,h,s}|_0$ is obtained from the displayed impact-time derivative and the differentiated reflection law. Thus the complete first-order source and its trace letter are

$$(117) \quad g_w = \sum_h \mathcal{L}_{0,h}(\mathcal{D}_{h,w}^{\text{int}} + \mathcal{D}_{h,w}^{\text{Jac}})\rho_0 + \sum_{i,h} \mathcal{B}_{\Gamma_{i,h},w}\rho_0,$$

$$(118) \quad \langle \mathcal{B}_{\Gamma_{i,h},w}\rho, \psi \rangle_W = \sum_{y \in W \cap \Gamma_{i,h}} (V_{\Gamma_{i,h}} \cdot n_{\Gamma_{i,h}})(y) \cdot \rho|_W(y) [\psi J_h]_{\Gamma_{i,h}}(y).$$

Inserting g_w into (83) gives the rigid-translation response. In particular, a physical vector w is not identified directly with a normal vector in collision-section coordinates.

(C7) For every transported smooth observable, the singular transfer expression is exactly the direct geometric derivative:

$$\langle \rho_0, \nabla_s f_s|_0 \rangle + \langle (\nabla_s \mathcal{L}_s|_0)\rho_0, R_0^{\text{obs}} f_0 \rangle = \int_{\Sigma_0} [\partial_s(f_s \circ \Phi_s)|_0 + f_0 \partial_s J_s|_0] d\nu_0.$$

Thus the trace and resolvent terms do not create a spurious dynamical correction to a static collision average.

Proof. Items (C1)–(C2) follow from centering and preservation of total mass. Items (C3)–(C4) are differentiation of a conjugacy and invariance of the density-observable pairing under the transport Jacobian. Item (C5) is the derivative of a partition of unity. Item (C6) follows by differentiating the impact equation, the reflection law, and then the section defining function; the trace formula is Theorem 3.2. Item (C7) is the first-order case of the finite-time invariance argument in Theorem 4.21, compared with Proposition 2.1. $\square$

5.1. Exported interface with homogenization. The response package is meant to be used as a theorem interface rather than as an informal regularity assumption. A later argument may invoke Theorem 5.1 only for the stated actual static, source-specific radial, and exact-current conclusions. The general finite-order conormal implication, including differentiated map susceptibilities for a certified recovered tree, is instead Theorem 5.2 and requires its displayed R/S certificates through the chosen J . A later argument may invoke a moving-family suspension response only after separately constructing the common weighted graph domains and negative-order symbol estimates identified after Proposition 2.15; no such theorem is exported here. The source-side scalar formulation is part of the exported interface: downstream arguments may pair conormal sources with the stated source/test levels, but may not replace them by an unproved strong density resolvent.

In this separation, the present paper supplies a conormal source framework, a source-specific nonconjugate block-recovery theorem through order two, and a conditional scalar interface. Apart from the proved base first radial letter, it does not supply a full higher-order label-resolved all-branch recovery theorem for a general dynamical response tree. A downstream macroscopic equation may use the certified interface, but is not part of the present argument.

6. CONCLUSION

The actual collision-map results and the certified abstract calculus are now separate. The source module contains ordinary weak terms, one-sided face currents, central point atoms, terminal symbols, and canonical same-label dyadic block generators. The latter make physical regrouping an identity in the quotient norm, not merely after the scalar labels have been forgotten. Under the finite Ward and triangular recovery certificates, this module gives the finite-order scalar calculus of Theorem 5.2; those certificates are not claimed for a generic moving table.

For the nonconjugate radial inflation, the complete invariant assembly recovers through order two, including arbitrary fixed positive waiting lengths and assembly-first terminal reset. On the level-three uniform block-flux jet graph domain, the first transfer letter has a one-event bound

$2^{-(1-\epsilon_*)b}$, with the physical face column itself bounded by $2^{-(3-\epsilon_*)b}$. The domain embeds in $\mathfrak{D}_U^{(3)}$, but the result is not (S1) because neither transfer invariance nor next-event closure is proved. This is sharp in order: the isolated second letter contains the nonintegrable interior coefficient u^{-3} , so every higher radial theorem must concern a specified complete assembly. The exact observable-current class gives an all-order telescoping benchmark. The radial table-velocity current is a nonzero differentiated-invariance benchmark and yields an infinite susceptibility. Its total-current q_1, q_2 identities do not establish label-resolved reset or the strong Taylor remainder; a generic local correlation source therefore remains open. The moving-face gluing condition shows why such a result is nongeneric.

The remaining collision problem is therefore precise: control recursive splitting of the block-grouped columns for a specified complete response tree and verify strong recovery for a source chosen independently of its Poisson primitive. Uniform intrinsic historical terminal closure is not a target; a nonzero time-labelled terminal germ accumulates linearly. Physical infinite response instead uses the assembly-first scalar theorem with summable primitives and reset errors.

Continuous time is retained only as fixed-table input plus obstruction, not as a moving-family criterion or theorem. A generic parameter letter costs one transverse level and one u^{-2} grazing level, while the pointwise weight u^2 is not comparable at consecutive collisions. A future proof must construct a dynamical branch/preimage-weighted corrected BDL completion, prove its weighted Lasota–Yorke and common-resolvent bounds, cancel the zero-waiting endpoint Fourier polynomial, and establish the regular $W^{p,1}$ time-kernel estimates. The present flow statement records only the obstructions to the simpler shortcuts.

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APPENDIX A. CITATION AND DEPENDENCY CONVENTIONS

The references used in the main text provide mechanisms at specific layers of the proof. Sinai billiard hyperbolicity, singular strata, homogeneity strips, and standard-family growth enter through [16, 15, 6, 7]. The strong/weak anisotropic transfer architecture for the grazing billiard map, compact embedding, and quasi-compactness enter through [10, 11, 14, 1]. The suspension high-frequency estimate uses the Dolgopyat cancellation mechanism and its finite-horizon billiard-flow implementation [12, 2]. This paper constructs the one-step clean atlas and flux-seed response scale, proves their finite-refinement properties and the finite-order resolved moving-boundary currents. The cited works supply the fixed-table map/flow mechanisms at the interfaces identified above; they are not cited for the new moving-singularity response calculus. For comparison, the specialized small-hole linear-response theorem [4] obtains decay by representing its singular source as a weak-star limit of regular standard families. Its crucial one-step vertical-line-to-unstable-family geometry is not shared by a terminal physical grazing trace, so that result is not cited as a proof of (S1)–(S3).

Definition (Venue-facing citation route for the suspension estimate). The high-frequency suspension estimate is cited in three separate layers. The cancellation mechanism is the Dolgopyat oscillatory two-branch argument [12]. The finite-horizon Sinai-flow implementation of the analytic part is taken from [2, arXiv v3, including its correction note]: fake foliations, standard-family decomposition, the Dolgopyat bound of BDL Lemma 8.1, the resolvent contraction of BDL Proposition 9.1, and the flow consequences of BDL Theorems 1.2 and 1.4. The Inventiones text is arXiv v2; arXiv v3 appends the two-page corrections to Lemmas 3.2, 4.6, and 7.5, and those corrected forms are the ones used here. The uniform geometric contact and pair-selection data are verified in Proposition 2.14; the parameter-uniform flow spaces and graph domains are the separate hypothesis Assumption 2.11. Neither layer is replaced by a short collision-curvature heuristic.

Definition (Dependency convention). The proof dependencies are read in the following order: primitive billiard geometry, singular map and growth estimates, anisotropic transfer spaces, spectral and suspension resolvents, the resolved conormal trace scale, and finally the observable-side scale-resolvent pairing. No weak-to-strong finite-prefix recovery is used.

APPENDIX B. TRACE AND FREQUENCY BOOKKEEPING CORRECTIONS

Lemma (First-order strip tail). *Let M_1 be the strip power obtained by differentiating the impact, Jacobian, flux, and crossing factors once. On the smooth core and with a finite strip cutoff,*

$$|\langle \mathcal{T}_{1,k,\theta}^{\text{core}} \rho, \psi \rangle_W| \leq C k^{M_1} \|\rho\|_{C_{\text{br}}^1} m_W(E_{1,k,\theta}(W)).$$

This local inequality is not summed absolutely over k . Instead the complete physical current is grouped into dyadic blocks and summed by Lemma 2.19. Neither the quadratic-strip width $O(k^{-3})$ nor the unweighted growth lemma supplies an arbitrary moment. Completion in the flux-seed norm gives the combined first response current; no pointwise trace bound on $\mathcal{B}$ is asserted.

Lemma (Resonant-window counting correction). *Suppose $\Phi_b(s) = b(\tau_\alpha^{(N)}(s) - \tau_\beta^{(N)}(s))$ and $|\Phi'_b(s)| \geq |b|\Delta$ on a fixed interval. For fixed angular width ω , each component of*

$$\{s : \text{dist}(\Phi_b(s), 2\pi\mathbb{Z}) < \omega\}$$

has length $O(|b|^{-1})$, but there are $O(|b|)$ relevant resonances. Consequently the union has only an $O(1)$ bound and generally occupies a fixed positive proportion; no $O(|b|^{-1})$ estimate for the whole union is valid. The high-frequency estimate used in this paper is therefore the BDL generator-resolvent theorem from Propositions 2.14 and 2.23, whose gain comes from local pairing, amplitude cutoffs, and iteration.

Remark (No weak-to-strong prefix inference). From $\mathcal{L}^{N*}g \in \mathfrak{D}_U$ one cannot infer $\sum_{n \geq 0} \mathcal{L}^n g \in \mathfrak{D}_U$ when the finite prefix contains the original weak distribution $g \notin \mathfrak{D}_U$. Accordingly the paper uses the observable-side pairing in Equation (83) and makes no density-side recovery claim.

APPENDIX C. LOCAL PROOFS FOR THE SINGULAR RESPONSE CALCULUS

This appendix contains the local calculations used by the response theorem. It has deliberately been reduced to eight load-bearing statements. In particular, it contains no independent constant records, no density-side recovery assertion, and no high-frequency contour argument based only on a polynomial resolvent bound. Every estimate below states its input, output, and dependence on the hypotheses in the main text.

C.1. Impact, reflection, and the complete physical block.

Proposition C.1 (Local impact and reflection differentiation). *Let h_θ be a regular one-step collision branch contained in a homogeneous chart, and write the next impact as*

$$G_\theta(z, \tau) = F_\theta(q(z) + \tau v(z)) = 0.$$

If $c_\theta = n_\theta \cdot v \neq 0$ at the impact, then every mixed derivative of the flight time of total order $a \leq J+2$ is a finite sum

$$(119) \quad \partial^a \tau_\theta = \sum_{\pi} c_\theta^{-1-|\pi|} P_{a,\pi}(\partial^{\leq a} F_\theta, \partial^{\leq a} q, \partial^{\leq a} v, \partial^{< a} \tau_\theta),$$

where the $P_{a,\pi}$ are universal polynomials. Derivatives of the reflection law

$$R_\theta(q, v) = (q, v - 2(v \cdot n_\theta(q))n_\theta(q))$$

and of the branch Jacobian have the same form: a finite sum of products of normal-frame, curvature, flight-time, and crossing-factor derivatives. On $\mathbb{H}_k$ all coefficients are bounded by a polynomial in k , with degree depending only on a and the primitive C^{J+4} bounds.

Proof. Differentiate $G_\theta(z, \tau_\theta(z)) = 0$. At each order, isolate the unique term $\partial_\tau G_\theta \partial^a \tau_\theta = c_\theta \partial^a \tau_\theta$; all remaining terms contain lower flight-time derivatives. This gives (119) by induction. The reflection formula is algebraic in v and n_θ , while the branch Jacobian is a finite product of flight and reflection factors. The homogeneous strip bounds $|c_\theta|^{-1} \leq Ck^2$ and the finite derivative induction give the last claim. No summation over k is made in this proposition. $\square$

Theorem C.2 (Verified Ward identities for complete invariant sources). *Let ρ_s be the transported invariant collision density and let*

$$G_j = \sum_{\ell=1}^j \binom{j}{\ell} (\partial_s^\ell \mathcal{L}_s|_0) \rho_0^{(j-\ell)}.$$

Then, as a distributional identity on the smooth flux-seed/test core,

$$(120) \quad G_j = (I - \mathcal{L}_0) \rho_0^{(j)}, \quad \langle G_j, 1 \rangle = 0.$$

For the local cutoff model $c_s(u, y) u^3 H(u - \eta_s(y))$, the singular endpoint terms vanish identically for $j \leq 3$, whereas the leading fourth-order endpoint is

$$(121) \quad -6 \dot{\eta}_0^4 c_0(0, y) \delta(u).$$

Consequently the cubic flux zero alone proves no order- j endpoint cancellation for $j \geq 4$. Under the additional operator Ward identities (8), the complete physical order- j coefficient satisfies

$$F_{j,m}(u, y; \rho, \psi) = u \tilde{F}_{j,m}(u, y; \rho, \psi)$$

and its dyadic block has the uniform tail

$$(122) \quad |\langle \mathcal{T}_{j,m}^{\text{phys}} \rho, \psi \rangle| \leq C_j 2^{-m} \|\rho\|_{\mathfrak{D}_U^{(j+2)}} \|\psi\|_{\mathfrak{D}_U^{(j+3)}}.$$

Proof. Differentiate $\mathcal{L}_s \rho_s = \rho_s$ exactly j times and move the $\ell = 0$ term to the left. This gives (120); pairing with 1 gives zero because $\mathcal{L}_s$ preserves total mass.

For the cutoff model, repeated differentiation gives sums of $u^3 \delta^{(n)}(u)$ multiplied by parameter jets of η_s . The distributional identity

$$u^3 \delta^{(n)}(u) = \begin{cases} 0, & n < 3, \\ -n(n-1)(n-2) \delta^{(n-3)}(u), & n \geq 3 \end{cases}$$

proves the first three assertions and gives (121). Thus the remaining operator-level cancellation is precisely the finite family in (8); it is not inferred from the cubic zero. If $\dot{\eta}_0(y_0) \neq 0$ and the localized complete coefficient at that germ is nonzero, the fourth-order atom cannot vanish for every localized seed/test. Hence arbitrary-seed R3 is not a generic property of a moving grazing face; only an exact assembled-source identity can remove this obstruction.

Once these residual endpoint coefficients vanish, $F_j^{\text{reg}}(0, y) = 0$, and the fundamental theorem of calculus gives

$$F_j^{\text{reg}}(u, y) = u \int_0^1 \partial_u F_j^{\text{reg}}(tu, y) dt.$$

On the m -th blown-up dyadic block, $u \asymp 2^{-m}$. The derivative and holonomy bounds in (R3) therefore give (122). Artificial faces have already cancelled pairwise before this estimate, so no unsigned strip moment is used. $\square$

C.2. One-wall and clean two-wall currents.

Lemma C.3 (Second derivative of one moving wall). *Let*

$$g_s = g_0 + s \dot{g}_0 + \frac{1}{2} s^2 \ddot{g}_0 + o(s^2)$$

in a C^2 chart, with $Dg_0 \neq 0$. Then

$$(123) \quad \partial_s^2 H(g_s)|_0 = \ddot{g}_0 \delta(g_0) + \dot{g}_0^2 \delta'(g_0).$$

Equivalently, if $g_s = g_0 - sh_1 + \frac{1}{2} s^2 h_2 + o(s^2)$, the coefficient of $\delta(g_0)$ is $+h_2$. In normal coordinates $y = g_0$, its action on a compactly supported amplitude a is

$$\langle \partial_s^2 H(g_s)|_0, a \rangle = \int_{y=0} \ddot{g}_0 a J d\sigma - \int_{y=0} \partial_y (\dot{g}_0^2 a J) d\sigma.$$

Proof. Apply the one-variable distributional chain rule twice: $\partial_s H(g_s) = \dot{g}_s \delta(g_s)$. A second derivative gives $\ddot{g}_s \delta(g_s) + \dot{g}_s^2 \delta'(g_s)$. The final formula is the definition of δ' after the coarea change of variables. $\square$

Lemma C.4 (Repeated derivatives of one generator). *Let g_s be a single C^J face generator with $Dg_0 \neq 0$. For $1 \leq m \leq J$,*

$$(124) \quad \partial_s^m H(g_s)|_0 = \sum_{\ell=0}^{m-1} P_{m,\ell}(\dot{g}_0, \dots, \partial_s^{m-\ell} g_0) \delta^{(\ell)}(g_0),$$

where $P_{m,\ell}$ is the corresponding Bell polynomial. This is a normal derivative of one conormal current, not a product $\delta(g_0)^q$. For a smooth core density and test, its weak norm is bounded by the order- $(m+2)$ flux-seed norm times the order- $(m+3)$ one-sided trace-test norm.

Proof. Faà di Bruno's formula applied to the distribution H gives (124). In the normal coordinate $y = g_0$, move the ℓ normal derivatives from $\delta^{(\ell)}(y)$ to the coefficient, density, test, and chart Jacobian. These are exactly the finite normal jets controlled by the stated seed/test norms. Completion follows after physical block assembly by Theorem C.2. $\square$

Lemma C.5 (Clean double-trace coarea formula). *Let g_Γ, g_Λ be two distinct C^J face generators in a two-dimensional collision chart and suppose*

$$\Delta_{\Gamma\Lambda}(z) = |\det(Dg_\Gamma(z), Dg_\Lambda(z))| \geq \delta > 0$$

on their common zero set. Then for every compactly supported continuous amplitude a ,

$$(125) \quad \langle \delta(g_\Gamma) \delta(g_\Lambda), a \rangle = \sum_{z \in \{g_\Gamma = g_\Lambda = 0\}} \frac{a(z)}{\Delta_{\Gamma\Lambda}(z)}.$$

Parameter derivatives of this current are finite sums of derivatives of the moving point, of a , and of $\Delta_{\Gamma\Lambda}^{-1}$. They remain point-current derivatives associated with the same labelled pair and do not create a third delta factor.

Proof. The map $z \mapsto (g_\Gamma(z), g_\Lambda(z))$ is a local diffeomorphism at each common zero. A partition of unity and the two-dimensional change of variables formula give (125). The implicit function theorem makes each labelled zero C^J in the parameter. Differentiation of the finite point-evaluation formula proves the final claim. $\square$

Proposition C.6 (Determinant lower-bound alternatives). *Under (R2), every labelled distinct-face pair lies in exactly one of the following cases.*

- (i) *On the central region $u \geq u_J$, the pair is admitted and $\Delta_{\Gamma\Lambda} \geq c_J > 0$.*
 - (ii) *In the grazing collar $0 \leq u < u_J$, distinct physical faces have no common zero; only one-face grazing currents are admitted.*
 - (iii) *The pair has a boundary birth/death, is tangent, has a triple unfolding, or changes incidence. It is outside the clean response chart and contributes no two-wall atom to the theorem.*
- There is no fourth case in which infinitely many determinant shells are summed merely from a sublevel-width estimate.*

Proof. The central bound and the empty grazing collar are the two quantitative parts of (R2). The simple normal-crossings hypothesis excludes triple-labelled vertices and incidence changes. A tangency or boundary birth is outside the admitted atlas. Finally, the example $\Delta(t) = |t|$ has

$$\int_{2^{-p-1}}^{2^{-p}} \frac{dt}{\Delta(t)} = \log 2$$

for every p ; this proves that sublevel width alone cannot justify an infinite-shell alternative. The half-chart fold $g_1 = y$, $g_2 = y - u^2 + s$, $u \geq 0$, shows that a boundary determinant jet $\det(Dg_1, Dg_2) = 2u$ would not suffice: an interior intersection is born for $s > 0$. This is why the empty-collar condition is explicit. $\square$

Proposition C.7 (Finite-order clean two-wall current calculus). *Assume (R2)–(R3), and let $m \leq J$. The order- m derivative of a resolved branch indicator on the simple two-dimensional atlas is a finite sum of:*

- (a) *branch-interior derivatives;*

- (b) the one-wall terms in Lemma C.4;
- (c) parameter derivatives of the clean point atoms in Lemma C.5.

The point atoms in (c) occur only in the central region $u \geq u_J$; the grazing collar has no distinct-face common zero. For every finite decorated core $\mathcal{F}$, after artificial faces are cancelled and physical grazing terms are assembled,

$$\|\mathcal{T}_{D,\mathcal{F}}^{(m)}\rho\|_{\mathfrak{X}_U^{(m)}} \leq C_{m,\mathcal{F}}\|\rho\|_{\mathfrak{D}_U^{(m+2)}}.$$

The constant depends on the finite one-step atlas, the determinant lower bound, the order- m face jets, and the Ward bound, but not on a determinant shell cutoff. A uniform countable-atlas completion is asserted only under (S1).

Proof. Differentiate the pushforward of the resolved indicator current. A point with no active face gives (a), one active face gives (b), and two active faces give (c). Simple normal crossings exclude a fourth local incidence type. The one-face and two-face formulas are the preceding lemmas. Point atoms are placed in the decorated ambient source scale of Definition 4.2; they are not asserted to belong to the ordinary Demers–Zhang weak stable-leaf completion. Determinants are controlled by Proposition C.6; their polynomial strip loss is combined with all other physical terms before summation. Theorem C.2 then supplies the dyadic factor 2^{-m} , while adjacent artificial faces cancel with opposite orientation. On a finite core this proves the displayed estimate. The recursively weighted all-branch sum includes additional propagation weights; its completion is precisely the independent mapping certificate (S1), and is not inferred from the scalar 2^{-m} tail. $\square$

C.3. Continuous-time boundary. No differentiated flow theorem is part of this proof ledger. The retained fixed-table BDL input and the transverse/grazing/global-weight obstructions are Propositions 2.14 and 2.15. A future proof must first construct the dynamical branch/preimage-weighted common graph domain and its negative-order inverse-Laplace symbol estimate; only then would a graph-domain Neumann argument and dyadic integration by parts become applicable.

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